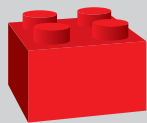
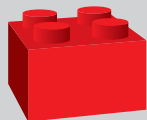


with, existing uses or rights.¹⁵⁴ For example, in the case of sustainable aquaculture for ecosystem enhancement, tidal energy generation, or the deployment of ecological infrastructure, rights could be compulsorily acquired and compensated for. Furthermore, the law could provide that rights could be extinguished – and reallocated – for repeated breaches of regulatory conditions, which is not currently possible under the RMA.¹⁵⁵

A future system will need mechanisms by which rights can be reallocated over time. Market mechanisms make this reasonably straightforward, but regulatory tools pose challenges. Some activities may require more security of tenure (eg wind farms may require a longer duration of rights), while others may need a more agile mechanism for reallocation to alternatives.



A formal forum could be established whereby new entrants or sectors wishing to use the marine space in a way that conflicts with existing uses could have some legal pathway to negotiate access rather than being excluded.



The Public Works Act or minerals-type access arrangements could be used to accommodate publicly important uses of the marine environment.

Reforms might contemplate more novel tools for reallocating rights, not just between different uses, but also between different locations. In particular, we may need to rethink what “occupation” rights mean in an age of climate and other environmental change. Some of the sites consented in the early days of marine farming have, for example, already proved unsuitable for the activity, particularly for the cold-water salmon species farmed in the Marlborough Sounds. The seawater in shallow and enclosed sites has been warming, fish have been dying and eutrophication of the seabed has been occurring. It has become clear that both environmental quality and farm productivity might be improved if they were relocated. However, the RMA does not envisage farms moving, with any relocation being treated as a new application and with no certainty of success. In such a context, marine farmers are understandably reluctant to give up existing rights to space.

One solution might be for existing aquaculture space to be tradeable for new areas, or for “backup” sites to be planned for and consented in advance. Another option would be to support a transition (where possible) towards aquaculture operations that are mobile (eg nets that could be towed rather than permanently fixed to the seabed),¹⁵⁶ and to provide for tools that allow those operations to shift between approved areas (a permit based on biomass rather than location). For example, in Norway licenses are attached to biomass which can be moved between different aquaculture areas depending on environmental conditions and market requirements, helping to avoid the risk that operators become trapped in marginal or unsuitable sites, as well as maximising productivity.¹⁵⁷

A mechanism could even be developed to shift permits granted under the RMA/NBA to ones granted in the EEZ, if open ocean aquaculture was to be encouraged. (This might be one benefit of merging these statutes together – see Chapter 11). As mentioned earlier in the context of property rights, one option would be for such “mobile” occupation rights to be perpetual and tradeable (since concerns about the environmental impacts of the right on a particular location would be less important).¹⁵⁸



Rights in a future system could be made more spatially agile, especially when it comes to fixed occupation rights. That is particularly relevant to aquaculture operations, which may need to shift or become more operationally mobile, but it could also apply in the future to other activities as environmental conditions change (eg floating wind farms¹⁵⁹ or tidal energy facilities).

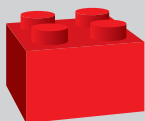
Environmental human rights

While rights to use resources, as discussed above, are commonly used to allocate scarce resources and to provide commercial certainty for investment, they can also be used in other, more novel, ways. For example, all New Zealanders could be given rights to the protection (rather than use) of the moana. There has been some experimentation with this idea, internationally, through the grant of environmental human rights (the right to a healthy environment, a right to clean water and so forth). It has even been enshrined in some countries’ constitutions, and in the Stockholm Declaration (1972).¹⁶⁰ In October 2021, the United Nations Human Rights Council went so far as to recognise that a clean, healthy and sustainable environment is a human right, with the High Commissioner describing the triple planetary threats of climate change, pollution and nature loss as the single greatest human rights challenge of our era.¹⁶¹

The grant of environmental human rights, of course, reflects an anthropocentric perspective; if using a resource would harm people's rights, and people choose to exercise those rights, then the system would gladly defend them against incursion. Some have warned that this has risks. It can lead to a dilution of sustainability, a justification for development, and an individual and property-focused "anthropocentric reductionism".¹⁶² In other words, it can be dangerous and unrealistic – even wishful thinking – to provide even a general right to a healthy environment if this is not accompanied by related environmental obligations and duties.¹⁶³ For one, people may simply choose not to exercise a right if it doesn't affect them personally or if they are "bought out" (ie externalities are internalised).

Environmental rights might potentially also clash with other rights, such as property rights in ITQ or Te Tiriti rights, requiring legal resolution. A high-level right would probably require extensive litigation to interpret in context (what degree of harm *would* be acceptable?). And it might even be used in improper ways to mask anti-competitive behaviour, as protections under the RMA have done in the past (eg the "supermarket wars").

But a general human right to a healthy marine environment (in the Bill of Rights Act or a new Oceans Act) could at least be helpful in highlighting when high-level government policy was in breach of such rights. This could be framed at a relatively general level, such as a right for environmental degradation not to create a substantial risk of illness, localised food depletion or ecosystem collapse, reflecting the principle of environmental justice and intergenerational equity. In other words, it would provide a legal mechanism to prompt government action and improve overall transparency, even if the only source of relief was a declaration in the courts. Indeed, that is the main point of the Bill of Rights Act. At a more granular level, environmental human rights could clarify things like the extent to which management of fish stocks and other marine activities is a matter in which the broader public has an interest (and a right to participate).



A future system could enshrine human rights to a healthy marine environment (eg in the Bill of Rights Act). However, that would have challenges in practice, and may not be a silver bullet solution to addressing environmental issues.

Rights for nature and legal personhood

There is a more unconventional way in which we could use the concept of "rights" in a future system. This is to give rights not to people, but rather to nature itself. Here, nature (or an aspect of it) is treated as an entity capable of having rights. This could be achieved through creating legal personhood. Such an approach would reflect the principle of ecological justice and an ecocentric ethic (see Chapter 7).

A rights for nature approach is "fundamentally subversive of economic orthodoxy".¹⁶⁴ In fact, it is much more than just a "tool" like a regulation, plan or property right. It is a fundamental reimagining of the system that cuts across multiple themes, including institutional design. It turns many things on its head. For that reason, we consider it in Chapter 13 as one starting point for a future system and it reappears across other chapters.

A spotlight on rights for nature

Legal personhood is a status that is granted to non-human entities to confer on them certain rights and obligations. It is most commonly used in the case of corporations (we are comfortable saying that "Google" has done something, even though it doesn't really exist). However, legal personhood can be applied to a wide range of things.

When applied to nature, legal personhood gives an entity standing in its own right, to go to court to defend its rights. It also allows the entity to advocate for its rights or interests. Of course, non-speaking entities require humans to act on their behalf in order to do these things. Various different statutory and institutional setups can be used to effect legal personhood (see Chapters 11 and 12). But the purpose of legal personhood is to recognise the intrinsic rights of nature and reformulate the way we think about the relationship between humans and the environment. Rather than a resource to be exploited, nature is a living system with needs and interests – an equal of humanity.

If we were to confer rights on nature in the marine environment, what would that mean? First and foremost, a decision would need to be made about what worldview(s) should underpin it. In much of the literature, the basic ethic driving personhood is an ecocentric one, but implementation of legal personhood for nature in Aotearoa New Zealand (so far, only on land

in the context of Te Urewera and the Whanganui River) has been heavily based on te Ao Māori conceptions of the world.¹⁶⁵

If we were to extend a rights for nature approach to the moana, there is a rich body of knowledge and understanding that comes from te ao Māori that we could draw on. For example, conferring rights on Tangaroa could bring with it a body of history, knowledge and practices that may help us determine his rights and interests. In a general sense, his mana and mauri could mean we need to cease degrading activities, including on land.

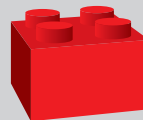
The ocean might even be regarded as a legal *community* rather than a legal *person*. There is not just one ancestor in the marine space. Hinemoana, the physical embodiment of the Pacific Ocean, Kiwa (an ocean guardian), Parawhenuamea (ancestor of freshwater), Rakahore (solid rock), and countless others representing sea creatures are present. While some may say it would be unwieldy and impractical to establish rights for all of these ancestors' interests, is this not the role already played by kaitiaki? And don't we already have complex structures that represent communities (councils) and shareholders (parent companies and subsidiaries)?

However, basing a marine legal personhood system exclusively on te ao Māori imports all of the implications of that normative system too. They cannot easily be separated. For example, in te ao Māori there is no recognition or explanation of Pākehā and tauīwi New Zealanders' obligations and responsibilities towards the ocean. It would arguably be inappropriate to apply te ao Māori obligations and responsibilities, because Pākehā and tauīwi do not have whakapapa to ocean ancestors, and thus have a very different relationship with the sea.

There are also other challenges with defining tikanga Māori through a Western judicial system, such as how to deal with metaphysical concepts alongside physical facts, and how to explain terms in English without isolating them from the cloak of their worldview.¹⁶⁶ This is not to say we shouldn't look to te ao Māori to inform a system of legal personhood; rather, we should ensure we think critically about its implications and identify nodes of agreement with other perspectives.

A system based on ecocentrism would have the benefit of applying to all New Zealanders, while sharing sentiments with a te ao Māori approach, such as around the inherent value of nature and the reliance of humans on it. However, a purely ecocentric approach may not be appropriate either. It does not recognise the special relationship Māori have with the environment and the ocean, and our Treaty obligations in this regard. Ultimately it will depend on whether we see the tool of legal personhood

as a mechanism purely for environmental protection, with Māori interests facilitated in some other manner; or whether it is a tool to empower Māori, with environmental protection only achieved as a secondary purpose through the exercise of tikanga in their relationship with the environment.



A future system could recognise that the moana itself has legally enforceable rights. The normative basis of recognising personhood for nature will be important, but potentially difficult to establish given the different worldviews of te ao Māori and te ao Pākehā.

Irrespective of its normative underpinnings, several design features of a rights for nature model would need to be considered. One would be the scale at which rights would be conferred. At one end of the scale, personhood could be conferred on the Ocean as a whole, potentially in the person of Tangaroa or Hinemoana. This would be akin to the constitutional-level protections for elements of nature seen in countries like Ecuador and Bolivia.



Inner Hauraki Gulf

Raewyn Peart

A spotlight on rights for nature in Ecuador and Bolivia

In 2008, Ecuador enshrined the rights of nature into its constitution, giving its mountains, rivers, forests, air and islands legally enforceable rights to exist, flourish and evolve. It is influenced by the indigenous concepts of *sumac kawsay* (good living) and Pachamama (Mother Earth). Article 71 reads:¹⁶⁷

Nature, or Pacha Mama, where life is reproduced and occurs, has the right to integral respect for its existence and for the maintenance and regeneration of its life cycles, structure, functions and evolutionary processes. All persons, communities, peoples and nations can call upon public authorities to enforce the rights of nature. To enforce and interpret these rights, the principles set forth in the Constitution shall be observed, as appropriate. The State shall give incentives to natural persons and legal entities and to communities to protect nature and to promote respect for all the elements comprising an ecosystem.

Bolivia has two pieces national legislation known as the Mother Earth Laws. The 2010 law sets out ten articles which form a principled framework. Article 7 sets out the rights of Mother Earth:¹⁶⁸

1. To life: The right to maintain the integrity of living systems and natural processes that sustain them, and capacities and conditions for regeneration.
2. To the diversity of life: It is the right to preservation of differentiation and variety of beings that make up Mother Earth, without being genetically altered or structurally modified in an artificial way, so that their existence, functioning or future potential would be threatened.
3. To water: The right to preserve the functionality of the water cycle, its existence in the quantity and quality needed to sustain living systems, and its protection from pollution for the reproduction of the life of Mother Earth and all its components.
4. To clean air: The right to preserve the quality and composition of air for sustaining living systems and its protection from pollution, for the reproduction of the life of Mother Earth and all its components.

5. To equilibrium: The right to maintenance or restoration of the interrelationship, interdependence, complementarity and functionality of the components of Mother Earth in a balanced way for the continuation of their cycles and reproduction of their vital processes.
6. To restoration: The right to timely and effective restoration of living systems affected by human activities directly or indirectly.
7. To pollution-free living: The right to the preservation of any of Mother Earth's components from contamination, as well as toxic and radioactive waste generated by human activities.

Other statements of principle are made, such as:¹⁶⁹

- Human activities, within the framework of plurality and diversity, should achieve a dynamic balance with the cycles and processes inherent in Mother Earth.
- Mother Earth is a dynamic living system comprising an indivisible community of all living systems and living organisms, interrelated, interdependent and complementary, which share a common destiny.
- The exercise of individual rights is limited by the exercise of collective rights in the living systems of Mother Earth. Any conflict of rights must be resolved in ways that do not irreversibly affect the functionality of living systems.

These approaches are quite different to the Aotearoa New Zealand application of legal personhood to date, in that they operate at a constitutional level, and place an onus on all citizens to uphold and enforce Mother Earth's rights. This is an approach that we could consider – for example, a “Tangaroa’s Law” which does not alter the nitty gritty of the existing ocean management system but allows citizens to hold the government to account (eg through recourse to declaration proceedings) when that system breaches the fundamental rights of the ocean.¹⁷⁰ This would have the advantage of creating a common set of ethics and principles across all the moana, recognising the interconnectedness and primacy of ecosystem health rather than just being focused sectoral objectives. However, its generality may make it only of symbolic value if it had to be interpreted in light of more specific and measurable objectives (eg MSY for fisheries management).

At the other end of the scale, rights and personhood could be conferred on more granular features of the marine environment. This could be particular populations or species (eg recognising the sentience and human-like characteristics of whales and dolphins, or the need to give agency to particularly vulnerable or threatened species that need it most). For example, a right to survive may be a more effective basis for setting limits for the mortality of threatened species than controls under a statute like the Fisheries Act (based on sustainable utilisation).

Some advocates propose that dolphins should be granted a non-human legal personhood status due to their high intelligence.¹⁷¹ Research on dolphins has shown that dolphins have distinct personalities from one another and can recognise themselves in the mirror. They have complex social structures and can work together to solve problems or teach each other tricks for fun. We already have legislation to specially protect marine mammals¹⁷² and our animal welfare legislation recognises that animals are sentient.¹⁷³ Whales and dolphins are often particularly significant to certain iwi and hapū. They are recognised as taonga (treasured) species in pūrākau (legends) such as that of Paikea, or Tinirau and Kae,¹⁷⁴ the popular consciousness (think of *Whale Rider*) and in settlement legislation.¹⁷⁵

However, there is a danger that prioritising whales and dolphins over all other marine life is inappropriate. Some warn that this encourages us to value species to the extent they are similar to humans.¹⁷⁶ We risk drawing wobbly moral lines between different species, especially when those lines are based on human's propensity to eat or make money from the death of certain species. Nonetheless, legal personhood could be an innovative way to ensure that important or taonga species are protected and able to thrive.

Alternatively, particular areas could become a legal “person”. That could include individual biogeographic regions or rohe moana, providing an opportunity to strengthen Māori connections with the moana. It would also recognise that te ao Māori places importance on metaphysical and non-living components of the marine environment, not just particular species or even ecosystems.¹⁷⁷ An area-based approach to personhood could also provide an opportunity to rethink MPAs and to tailor rights to the values and characteristics of a particular place. In short, each MPA could be a legal person which is an approach proposed by The Earth Law movement.¹⁷⁸ This approach has the advantage of drawing on the model already in existence for Te Urewera, which used to be a protected area (a national park). However, it could risk MPAs being treated in a more fragmented way rather than as a single network (if the rights of one could trump the overall impact of the whole).¹⁷⁹

A spotlight on the Wadden Sea

Tineke Lambooy, Jan van de Venis and Christiaan Stokkermans have proposed that a legal personhood model be applied to the Dutch part of the Wadden Sea.¹⁸⁰ The Wadden sea is a large tidal wetland which is highly valued by the people of the Netherlands and has significant environmental values. However, it is suffering from economic use, splintered governance and poor policy decisions. The proposal is to establish a “natureship”. This would grant the Wadden sea the rights and ability to act as a legal person.



Dusky dolphin, Kaikōura

Raewyn Peart

A spotlight on legal personhood in Aotearoa New Zealand

Aotearoa New Zealand currently has three natural area/features that have been granted legal personhood: the Whanganui River, Te Urewera, and Taranaki Maunga (under finalisation).

Legislation for the Whanganui River recognises that:¹⁸¹

- Te Awa Tupua is a legal person and has all the rights, powers, duties, and liabilities of a legal person.
- Te Awa Tupua is an indivisible and living whole, comprising the Whanganui River from the mountains to the sea, incorporating all its physical and metaphysical elements.

Section 13 lays out Tupua te Kawa, which “comprises the intrinsic values that represent the essence of Te Awa Tupua”. These are expressed in both te reo Māori and English with an accompanying English explanation. The four values are:¹⁸²

- *Ko te Awa te mātāpuna o te ora*: the River is the source of spiritual and physical sustenance
- *E rere kau mai i te Awa nui mai i te Kahui Maunga ki Tangaroa*: the great River flows from the mountains to the sea
- *Ko au te Awa, ko te Awa ko au*: I am the River and the River is me
- *Ngā manga iti, ngā manga nui e honohono kau ana, ka tupu hei Awa Tupua*: the small and large streams that flow into one another form one River

The Te Urewera Act also confers personhood on Te Urewera, but is designed to:¹⁸³

establish and preserve in perpetuity a legal identity and protected status for Te Urewera for its intrinsic worth, its distinctive natural and cultural values, the integrity of those values, and for its national importance, and in particular to—

- (a) strengthen and maintain the connection between Tūhoe and Te Urewera; and
- (b) preserve as far as possible the natural features and beauty of Te Urewera, the integrity of its indigenous ecological systems and biodiversity, and its historical and cultural heritage; and

- (c) provide for Te Urewera as a place for public use and enjoyment, for recreation, learning, and spiritual reflection, and as an inspiration for all.

This purpose reflects the special significance of the iwi relationship to the area, but also draws on ecocentric principles and to an extent, other public interest factors.

The principles for implementing the Act are that:¹⁸⁴

- (a) Te Urewera is preserved in its natural state:
- (b) the indigenous ecological systems and biodiversity of Te Urewera are preserved, and introduced plants and animals are exterminated:
- (c) Tūhoetanga, which gives expression to Te Urewera, is valued and respected:
- (d) the relationship of other iwi and hapū with parts of Te Urewera is recognised, valued, and respected:
- (e) the historical and cultural heritage of Te Urewera is preserved:
- (f) the value of Te Urewera for soil, water, and forest conservation is maintained:
- (g) the contribution that Te Urewera can make to conservation nationally is recognised.

Additionally, public access to Te Urewera must be maintained. The Board must act in a manner that achieves these principles as far as possible. The Act also sets out a framework for granting activity permits and concessions. Decision-making on activities therefore happens under the provisions of this Act, rather than piggybacking on other legislation such as the Conservation Act

This model provides an intriguing possibility when it comes to giving personhood and rights to individual MPAs (eg marine reserves, recreational fishing areas, ahu moana areas, seabed reserves etc). These could usefully contain their own regulatory frameworks rather than just piggybacking on tools in other frameworks (eg the Fisheries Act, RMA or EEZ Act), while recognising that there might be some shared jurisdiction or role (eg Fisheries New Zealand advising on recreational fishing limits). We discuss MPAs in Chapter 9.

Personhood could even be conferred at multiple scales – threatened species, protected areas, sentient creatures, mixed fish stocks, and the moana as a whole. After all, overlapping personhood is familiar in other areas of the law (eg parent companies and subsidiaries, trusts and charities, natural persons and partnerships).



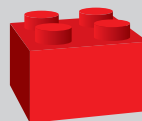
The scale at which the moana is granted rights is important. At one end of the scale, the ocean as a whole could be recognised as a person with rights. At the other end, particular places, species or features could be given personhood. Layers of personhood could even be established, just as we have layers of personhood for companies.

The exact nature of the legal rights conferred on aspects of the moana would also need to be determined. Any legal right requires specificity. This is where the novelty of a rights for nature approach could be most apparent, because it goes well beyond the familiar regulatory tools contained in frameworks like the RMA, Fisheries Act or conservation laws. By treating nature as a person, it opens up horizons for rights as broad as those enjoyed by human persons. For example, the ocean (or aspects of it) could have the right (or even the duty) to:

- make submissions on RMA-style processes when notified (eg plan changes, resource consent applications, NPSSs, NESs) and other tools like taiāpure, catch limits, and MPAs.
- propose marine reserves and other MPAs.
- make submissions on conservation processes including development of general policies, conservation management strategies and plans, and concession applications.
- engage in government policy decisions and legislative reform proposals (eg submit on discussion documents and bills).
- lodge appeals with the Environment Court where its rights were infringed and take declaration proceedings to determine the exact nature of its rights in particular situations.
- Undertake novel forms of enforcement action more akin to common law action (eg trespass, nuisance, negligence) rather than relying only on traditional mechanisms under the RMA.

- Make claims for personal injury.
- Own and defend property from others.

Under the status quo, individuals can submit and advocate for positive environmental outcomes. However, there are two distinct benefits of granting the moana the ability to do so. First, its advocacy will likely have more weight or mana, because it comes directly from the entity (albeit through human representatives) and therefore more accurately express its interests. Secondly, the entity may have far more resources and capability to advocate than individuals or volunteer organisations, meaning that advocacy is more consistently present and can be pursued to a larger degree. For example, its guardian body would likely contain or have access to experts such as scientists, lawyers, etc, and in some cases it could have its own budget – such as the Whanganui River Te Korotere.



If the moana was granted legal rights, the system would need to be clear as to what those rights involve. They could be much wider than current environmental protections in the RMA or Fisheries Act. They could be as broad as the rights enjoyed by humans.



Pakiri Beach

Craig Potton

8.7 Other regulatory tools

New forms of regulatory tools could also be developed in a strengthened oceans management system. Some might be created within the framing of the NBA. For example, an intriguing model to build on is provided by water conservation orders (see the spotlight below). Although water conservation orders themselves are not applicable to the marine area (and require significant improvements),¹⁸⁵ they could provide a template for other “order-based” tools that could be deployed in our seas. We could, for example, see oceans conservation orders included in the NBA as a mechanism to respond quickly to adverse environmental changes.

Raewyn Peart



Ponui Island

A spotlight on water conservation orders

Water conservation orders are a hangover from pre-RMA legislation,¹⁸⁶ and seek to provide targeted protection to water bodies having high intrinsic value (eg wild rivers).

Once an order is in place, councils must ensure that their policies and plans are not inconsistent with it, and resource consents must not be granted if they are contrary to it. Any party can apply for an order.¹⁸⁷ The bulk of applications have been by the New Zealand Fish and Game Council (with the Department of Conservation only having made one application). Uniquely, they are not subject to Part 2 of the RMA, and have their own highly protective purpose nested within the Act,¹⁸⁸ in a way not dissimilar to separate conservation legislation (eg national parks or reserves). They could, in theory, be equally at home in other, more protective, conservation legislation.

These orders are a direct mechanism by which the “normal” RMA planning process can be bypassed, where there are outstanding values to be protected. But why should that mechanism be limited to freshwater, and only then where outstanding values remain? For example, rather than just protecting examples of relatively pristine elements of the environment (wild and scenic rivers), could a similar process be used to impose directive environmental “emergency” orders at the other end of the spectrum – where bottom lines have been infringed and a part of the ocean is unacceptably degraded (eg in a polluted estuary)?

Could they be used as a mechanism by which untouched but vulnerable elements of the marine environment could be protected in a faster way, to recognise their intrinsic value (eg biogenic reefs, hydrothermal vents, or critical habitat for threatened species)? And could they be used to prevent or revoke land use or discharge rights where these are impacting the equivalent of “wild and scenic rivers” in the oceans, such as marine reserves at the mouths of sediment polluted rivers? We are becoming more used to emergency responses to threats like Covid-19, but slower burn environmental degradation of the oceans is of a similar magnitude and urgency.

The concept of an emergency order is not dissimilar to the Randerson Panel's suggestion that a general regulation-making power should remain outside of national direction (NPSs and NESs) where there is an immediate risk of environmental damage.¹⁸⁹ It could be a temporary measure to allow breathing room in order to learn more about the pressures facing an area, and could even be accompanied by support and compensation for lost rights (eg a temporary restriction on harvesting plantation forests in a catchment, reducing stocking rates, or closing a fishery).¹⁹⁰ We see a similar thing in the context of biosecurity incursions, where emergency restrictions can be draconian (eg the removal of shellfish and disestablishment of marine farms, or closure of harbours to anchoring) where the economy is imperilled.¹⁹¹ Yet environmental issues can be similarly urgent, and warrant a similarly firm response where limits are at risk.

This “order” style tool could potentially also form the framing for rāhui as a formal tool under the NBA (or other legislation like the Fisheries Act or even MPA legislation). However, there would be questions to address as to who was able to apply for such an order, what could trigger it (eg environmental harm or broader cultural factors) and who would be responsible for granting it. Further questions would arise about the relationship between these orders and the tikanga practice of rāhui: would such orders supersede customary cultural practices, or would they offer an opportunity to enhance and deploy them?

A spotlight on rāhui

It has been noted that Māori had long-established protected areas within their marine environments, either for conservation or other reasons. One such practice used to prevent degradation is rāhui. Rāhui were the first form of MPAs to exist in this country, and continue to be set today under tikanga.¹⁹²

Rāhui is a tikanga practice which restricts and sets aside an area in order to give time for it to physically and spiritually rebalance.¹⁹³ It can be used for several purposes. A permanent rāhui may be imposed on food gathering at a battle site or place specifically associated with death, such as a drowning. It pays respect to the area and to the people who may have been affected by the event.

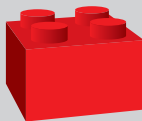
Another purpose of a rāhui is to replenish depleted resources and restore their mauri.¹⁹⁴ Rāhui are usually conceived of as temporary,¹⁹⁵ but they may persist over a long period of time if a problem is not resolved.¹⁹⁶ They can also differ in application: they may be a complete prohibition on entry and taking resources; be a ban on taking only one resource;¹⁹⁷ be a ban on a specific human activity;¹⁹⁸ be seasonal only;¹⁹⁹ or allow for the taking of resources but only with permission and in a specified way.²⁰⁰ Other various practices have been noted by the Waitangi Tribunal and have been summarised in *Hui-te-ana-nui: Understanding kaitiakitanga in our marine environment*.²⁰¹

Rāhui has no direct recognition in statute, although mechanisms such as temporary fisheries closures can be used as a means of enforcing aspects of a rāhui concerned with the harvest of kai moana (and they are often referred to in this way). Other mechanisms, like marine reserves, are less consistent with rāhui in that they have indefinite duration. This reflects a Western tradition of spatial separation of wilderness and human activity. Rāhui is also a tool to secure cultural practices, not just “environmental” ones as understood in the Western sense.²⁰² As such, the concept and practice of rāhui could in some cases present some tensions between Western and Māori ways of thinking.

Raewyn Peart



Rāhui sign, Umupuia



Emergency orders could be utilised in a future system where environmental limits were imperilled. That could be one basis on which legally binding rāhui could be deployed.

Regulatory tools under “non-marine” frameworks also warrant consideration. For example, how we address plastic waste is of enormous importance to our oceans, but (although much more could be done through the Fisheries Act, RMA and EEZ Act)²⁰³ a lot of this must be achieved largely outside “marine” statutes through product stewardship schemes, prohibitions on manufacture under the Waste Minimisation Act and measures under the Litter Act.²⁰⁴ The former could be amended so there is a *duty*, not just a *power*, for ministers to progress waste minimisation measures in order to meet targets (eg the volume of plastic entering the marine environment or being consumed by marine organisms). A similar accountability framework to that for climate change, could be established around this duty, whereby an independent agency measures progress and makes independent recommendations that must be considered (and reasons given for departure from them).

Similarly, the performance of our wastewater and stormwater infrastructure and urban centres can impact on the moana, meaning that a broad range of tools for funding and maintaining infrastructure, bylaws under the Local Government Act, and regulatory mechanisms like the Building Code have potential to make a difference.²⁰⁵ So too do vehicle emission standards and design requirements, given the cocktail of chemicals that runs into our stormwater drains from conventional vehicles.

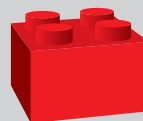
For example, the Building Code could have a refresh to focus on the marine impacts (and benefits) of construction materials and building design. Local Government Act infrastructure strategies could be strengthened to require impacts on the sea to be factored in. We have looked at such tools in our previous work on resource management

reform in the urban context, and the options presented there could contribute to better marine outcomes too (eg new build requirements for onsite stormwater management and pollution filtering mechanisms like rain gardens; connecting downpipes from older industrial roofs to wastewater systems to divert heavy metal runoff to treatment facilities; and avoiding investment in hard coastal protection structures where managed retreat is possible).

Some regulatory tools can also be found in unexpected places, such as the curiously broad regulation-making powers under the Territorial Sea, Contiguous Zone and Continental Shelf Act.²⁰⁶ That is hardly a fit for purpose tool for marine management, but if we are trying to squeeze possibilities out of the current system then things like this are worth a second thought.



The Waste Minimisation Act could contain a duty for ministers to progress regulatory tools like prohibitions and product stewardship schemes to meet mandatory targets for the reduction or elimination of plastic dangerous to marine life.



The Building Code, wastewater and stormwater infrastructure standards, and vehicle emissions standards could be strengthened to reduce the impacts of buildings and runoff on the marine environment.



Mangere wastewater treatment works

8.8 From regulatory to non-regulatory tools

Regulatory tools are essential for defending environmental limits, and can be used to make trade-offs (eg through planning and consenting frameworks). They can be used to frame the allocation of resources, by creating property rights or conferring regulatory permits. They are vital for recognising and protecting the interests of mana whenua (eg by safeguarding wāhi tapu and allowing for the exercise of customary rights). However, they have limitations when it comes to another key role the oceans management system might be expected to perform – pursuing positive outcomes.

That is not to suggest regulatory interventions are irrelevant when it comes to improving outcomes. They can be used in a number of innovative ways to drive positive change.

First, public authorities can have legal *duties* to take action. While that often relates to the delivery of goods and services (eg navigation infrastructure)²⁰⁷ it can also extend to specific obligations to enhance the natural environment. Such duties could be expanded in a future system. For instance, councils (or others) could be legally required to undertake wastewater and stormwater repairs and upgrades to ensure minimum standards are met for discharges to the marine environment by a certain date.²⁰⁸ Specific duties to restore particularly degraded marine ecosystems (as contemplated in the Seachange Tai Timu Tai Pari initiative) could be enshrined in statute, rather than just falling within an agency's general mandate to enhance biodiversity.

There could also be an active duty imposed upon authorities to conduct a large-scale joint operation to map marine habitats in all inshore areas. Indeed, it is incongruous that such effort has gone into meticulously tracking development capacity around high growth cities,²⁰⁹ when so little is known about the state of the marine environment which provides essential services to communities. As described earlier, habitat mapping has occurred in the United States, and has enabled the imposition of clear environmental limits.

A mapping and research effort could be accompanied by duties to actively monitor, not just agreed biophysical indicators (eg extent of kelp cover and expansion of activities like aquaculture), but also to evaluate and report on the effect that public interventions are having on those indicators (from practical things like kina removal to regulatory interventions like fisheries restrictions). Duties could be cross-agency ones, to ensure that indicators were measured and reported on in an integrated fashion.

A spotlight on environmental monitoring

Dr Marie Doole is of the view that “due to patchiness and poor utility” monitoring “is likely to underestimate the gravity of environmental problems”.²¹⁰ That may be the case in general terms, but the deficiency is exacerbated in the marine space, where monitoring is conducted for different purposes (eg fishing and conservation), and where what is happening in a highly fluid and connected environment is hard to see, measure and understand. Regional councils can struggle in the coastal marine area due to their size and capacity constraints.²¹¹ There are variations in collection techniques, the scale at which monitoring occurs, and data quality.²¹²

The introduction in 2015 of a national reporting framework with six-monthly domain-based reporting cycles was a major step forward.²¹³ The marine environment was specifically identified as a domain to be reported on. However, as the Parliamentary Commissioner for the Environment has pointed out, reporting on the data we happen to have does not remedy the problem of deficiencies and gaps in the data itself. Consistent time-series data is missing, habitats remain unmapped and poorly understood even when of vital importance to resources like fish stocks, and there is no real sense of what marine indicators should be chosen. Fisheries data is not integrated well, in terms of impacts on broader marine environments, or vice versa. And reporting does not take the next step of evaluating the effectiveness (or otherwise) of public interventions intended to influence indicators.

Nor is reporting a substitute for substantive duties to take corrective action in response.²¹⁴ No such duties are linked to environmental reporting at the moment. One option for a future system would be to locate environmental monitoring and reporting responsibilities within a centralised body that casts its eye over the whole marine space (eg an Oceans Agency or an Oceans Ministry), which is common overseas. Indicators – at least those relating to threatened marine species – could trigger least an obligation to commence a policy process if a certain status was breached..

To be meaningful, duties for public action need to be accompanied by funding (it costs a lot of money to *do* things), which can be hard to legislate for. However, a baseline level of funding could be provided by ringfenced

and independently managed funds (eg by directing it to specific purposes rather than general pots of money). We explore some options for doing so when we discuss funding tools further below.

Another way in which regulation could be used to create positive outcomes is through the use of accords, or negotiated regulation. These represent rules that are negotiated between a regulator and a regulated community, that are then made binding. That can encourage buy-in (or even championing) by sectors, and provide stepping-stones towards a new and improved status quo. For example, an accord might be reached on bottom trawling and other methods in exchange for assistance to transition to other methods and gear, or to manage forestry harvests in ways that do not involve clear-felling. However, it is crucial that environmental limits are not weakened in practice through negotiation. The public interest, and the objectives of the system, are more than just a bargain struck between sectoral or stakeholder interests.²¹⁵

Regulatory *restrictions* can also be used as an opportunity to enhance the natural environment, not just to mitigate adverse effects. This can be seen in the concept of a biobank, where harm to the environment can be used as a trigger to provide a net benefit.

A spotlight on offsetting

Environmental offsetting, generally speaking, allows a person to cause harm in one area or context in exchange for improvements in another. The original harm remains, so it is not the same thing as mitigation. In Aotearoa New Zealand, biodiversity offsets have been those discussed most.

Offsetting can be a disastrous road to environmental degradation if offsets are not proportionate to the harm caused. It is not an excuse to “purchase” rights to cause environmental harm, and offsets must be subservient to regulatory limits. Case law has provided useful constraints on the use of offsets under the RMA (eg the extent to which we compensate like for like),²¹⁶ and valuable work has been done on the topic as part of the collaborative development of an NPS for Indigenous Biodiversity.²¹⁷ Yet some have seen offsets as potentially inconsistent with strong ecocentrism, as trading off the wellbeing of one ecological system for that of another has ethical implications.²¹⁸ It can commodify biodiversity, which has intrinsic value as well as value through provision of ecosystem services.

However, using the principle of net gain when offsetting would mean that, in theory, harm in one place can be more than compensated for in another. Subject to careful constraints, this could be an extremely useful tool, as potential harm can be used as a trigger for overall enhancement. A positive example on this front can be seen in the United Kingdom’s Environment Act 2021, where some planning permissions are conditional on a net gain in biodiversity.²¹⁹

There remain issues with offsetting in the current system, not least the inconsistent approach across regions and lack of national direction. But one way forward could be the creation of a biobank. This is a scheme that, effectively, allows the measurement and trading of biodiversity values. Enhancement projects are purchased from a biobank by those who are required to provide compensation for biodiversity loss elsewhere. Under a biobank, offsetting requirements could be deployed in a more coordinated way to create, for example, functional ecological corridors rather than random islands of improvement. These could be used to supplement, or support, corresponding public funding for ecological enhancement – combining public and private resources to common aims in a coordinated way that maximises their overall positive impact.

A marine biobank is a potentially interesting tool when it comes to the deployment of MPAs (see Chapter 9). The externalities caused by activities like fishing, mining, tourism or coastal development (or even forestry and agriculture) could potentially be calculated, and used as a metric to enhance “like for like” in other marine areas (eg to pay for active restoration of kelp forests, the eradication of pests, or “buying out” other interests like ITQ or occupation rights in an area). That could see an MPA network grow and be actively enhanced over time in a way that is spatially planned.

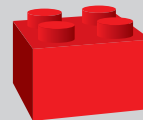
However, careful safeguards would be needed. Offsets would not be acceptable where impacts are on threatened species and ecosystems, and would be used only to address residual adverse effects on common and widespread species. Thorny ethical questions would also arise. Is it acceptable to exacerbate the decline of biodiversity in some places – eg beneath finfish farming operations close to shore – by providing offsets in MPAs elsewhere? Or to erode the protection of an MPA (eg by

continuing to allow sedimentation from catchments) if net benefits could be gained on the other side of the country or way out in the EEZ? It is difficult to know how much value we should assign to site or location-specific indicators versus regional or national level indicators (“overall” biodiversity gains). Moreover, why would policy makers not simply restrict damaging activities *as well as* creating MPAs? It is by no means obvious that one should be treated as a trade-off or “payment” for the other.

Marine biobanking would also face challenges not seen on land. It may be extremely difficult to estimate the impacts of some activities on biodiversity beyond the footprint of a fixed activity. We often simply don’t know what harm is being done. It would be even harder to *value* such things and provide “like for like”. And it may be more difficult to ensure that gains (eg through the removal of kina and planting of kelp) are maintained over long periods of time and in the face of systemic pressures like climate change. Maintaining kelp or coral gardens in the face of continued fishing and land-based pressures might prove much harder than purchasing and planting land with indigenous trees.

A future system could also provide a regulatory environment that is friendly to the deployment of desirable activities relative to undesirable ones (eg by cutting red tape). The Minister for the Environment has recently expressed an interest in embedding the Covid-19 fast-track consent process in a future resource management system, and that could be used to make activities like seaweed farming, sustainable shellfish farming or green infrastructure easier to progress than others. Some could even be given formal controlled activity status for which consent must be granted (relative to non-complying status for others) or given express weight in tendering processes for the allocation of marine space). However, all of this can be seen as the system picking winners, the legitimacy of which may depend on what the proper roles of the system are (see Chapter 6). Moreover, which activities are seen as “desirable” depends on one’s objectives, which are not necessarily clear. For example, should a future system be aiming for a large expansion of offshore aquaculture to create jobs, generate export earnings and potentially reduce pressure on wild fisheries and land-based protein sources? Should it make it easier for such things to occur?

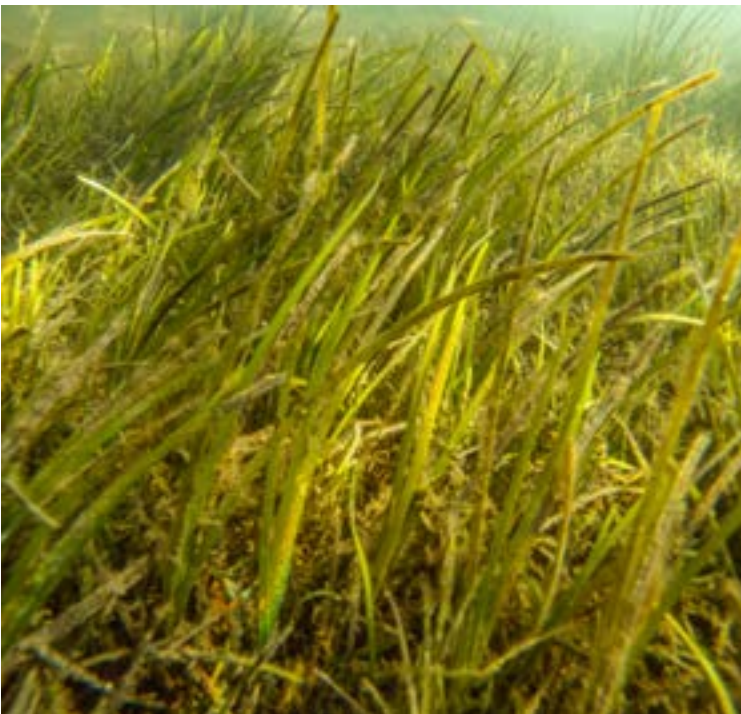
Regulatory tools could be used to drive positive outcomes in the marine environment. Duties on public authorities to pursue restoration and enhancement could be strengthened; sectoral accords could be reached which industries to provide improvements; a marine biobanking framework could be deployed; and fast track processes (or less stringent activity status) could provide an incentive for projects having public interest/environmental benefits.



There are limits to how much regulation can drive positive change in the marine environment. Arguably more is required than just stopping people doing things, letting nature regenerate itself, making public authorities do the heavy lifting, or providing offsets for harm. To improve outcomes, the system may need to start speaking the language of incentives, not coercion.

Funding tools

Non-regulatory tools are also essential to allow (and incentivise) public authorities to perform their roles. Earlier, we highlighted that a regulatory duty on public authorities to pursue positive outcomes requires consistent and meaningful funding. It can be extremely expensive to do anything in the marine environment.



Seagrass, Whangaparapara harbour

But funding is required for almost anything to happen in a system reliant on the intervention of public institutions. Money is, alongside information, like the circulatory system of the oceans management system. Without it, institutions cannot make policies and regulations, undertake enforcement action, or provide advocacy. Money breathes life into all these things.

Some policy and regulatory roles are necessary because of the actions of specific people, such as consent holders, who should therefore bear the burden of associated processing, monitoring and enforcement costs.²²⁰ This is consistent with the polluter-pays, user-pays, and beneficiary-pays principles.²²¹ It also costs a great deal of money to develop policy and regulation, such as strategies, policy statements, and plans. Associated state of the environment monitoring is expensive too, as is the research needed to underpin regulatory tools in the marine environment (notably habitat protection).

Much of this work has application throughout the country or a region, and is in the public interest. As such, it is appropriate for it to be funded through general channels that reflect the whole regulated (and monitored) community (ie a general pool of funds distributed through central and local government budgeting processes).

However, the public cost of undertaking such functions can be considerable, especially for regional councils where core funding relies on a small (and sometimes shrinking) population and rating base. These councils have the same basic planning responsibilities as larger ones, but population is not always an indicator of the size of the marine area to be managed,²²² the complexity of activities being undertaken in it, or the degree of land-based pressures being faced.²²³

Funding needs for territorial authority functions are also significant, particularly when it comes to repairing, replacing and maintaining wastewater and stormwater infrastructure. Some have questioned whether the incentives and constraints provided by current funding and financing mechanisms will be sufficient for councils to address significant historical spending deficits on such infrastructure, as well as spiralling future costs. Wastewater upgrades needed to give effect to the NPS for Freshwater Management have been estimated to cost \$1.4–\$2.1 billion, with ongoing operating costs at \$60–\$90 million.²²⁴ This is a complex area, and one that we discussed in our work on urban resource management reform, and we refer readers there for more information.²²⁵ However, it is highly relevant to water quality in the moana, given the risks that wastewater and stormwater pose in estuaries and beaches.

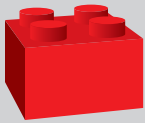
The relaxation of borrowing constraints on councils is not likely to be a viable solution, given that debt still needs to be paid back, and there are many other politically important priorities to fund on other fronts (not least a tidal wave of national direction to implement). Some may suggest we should look to the amalgamation of councils, or at least seek efficiencies from using shared services.²²⁶

Separate reform conversations about three waters are well underway, with the preferred option seeming to be institutional change (creating a series of arm's length, inter-regional water service delivery bodies) alongside funding changes (more user-charging and cross-subsidisation) to achieve equity and efficiency. From a marine perspective, it may be significant whether this includes stormwater functions as well as wastewater, given that the diffuse nature of stormwater runoff makes it both a more insidious threat and one that may, overall, prove even more expensive to address than wastewater upgrades.²²⁷ If billions are needed to give effect to the NPS for Freshwater Management, how much may be required to give effect to a revised NZCPS that took a much harder line approach to contaminants in stormwater? That is anyone's guess.

Greater standardisation of regional coastal plans would be possible and could save money – whether mandatory (through a template under national planning standards and reflecting a revamped NZCPS) or as an “off the shelf” model councils could choose to adopt if they wished. Alternatively, there could be more structure and principles around when councils should delegate functions, such as habitat mapping or enforcement, to others (eg the EPA or an Oceans Agency). Local Government New Zealand has floated the idea of allowing struggling councils to escalate tricky issues to a Crown agency if they choose to.²²⁸ That may well cover many marine issues, which are particularly complex.

However, a future system could instead tackle funding tools themselves. General rates are not likely to cut it when it comes to the kinds of marine biodiversity functions one might expect of regional councils in the wake of *Motiti*, given that these are highly dependent on the political will to raise funds. Opposition can be strong especially if the benefits are not apparent to existing residents who are paying for it (and who are the ones voting in local elections). If rates have not proved adequate to maintain core infrastructure in the face of community pressure to keep rates low, what chance does marine protection have? Will ecological infrastructure be subject to a “run to fail” approach like water pipes? If so, we might be faced with more than just the financial consequences of fixing it. Once ecosystems reach tipping points, it is hard to get them back.

Currently, targeted rates can be used by councils to reflect that some groups benefit or cost more than others when receiving services, but not to reflect the environmental outcomes they cause.²²⁹ This could be changed to enable greater use of targeted rates on land uses having an adverse impact on the moana. Value uplift capture could also be used in novel ways. This would involve public investments in the ecological health of the marine environment, such as MPAs with scenic and recreational values, being partially recouped by taxing any related bump in coastal property values upon sale.²³⁰



Targeted rates could be broadened to enable their use to charge land uses causing adverse impacts on the moana.

Greater central government funding in the form of grants could be forthcoming, although that may need some long-term predictability around where and whether (and for what purpose) funds will be paid. A project by project funding partnership may not address more systemic elements (eg ecosystem assessment, habitat mapping and the spread of complex or novel marine activities). The Biodiversity Collaborative Group has recommended that central government provide assistance for the proactive mapping of significant indigenous vegetation required by a proposed NPS for Indigenous Biodiversity, and the same may be required at sea.²³¹

Some things might be taken out of council hands entirely, and funded centrally. For example, new MPA legislation could place MPA identification and marine habitat mapping firmly with the Department of Conservation. Indeed, central government already has a relatively substantial degree of control over marine planning (eg in the Minister’s approval power over regional coastal plans), so it may be fair for funding responsibilities to follow suit.²³²

Central government is not as constrained as local government in its sources of funding. But fluctuating political priorities and pork-barrel politics mean that flows of money through the budget process can still be haphazard. The influence of this on the Department of Conservation’s advocacy role is a case in point, which is particularly noticeable for marine matters. It is to be hoped that a living standards budget may provide greater stability,²³³ and environmentally focused principles in the Public Finance Act itself might help. However, there is a risk that funding prioritises land-based protections over marine ones. A future system will need to consider tools that can defend the moana specifically, especially where funding is needed for things that have no direct short-term economic benefit.



Central government could provide greater funding assistance to regional councils to support marine management, and take over specific functions such as marine mapping.



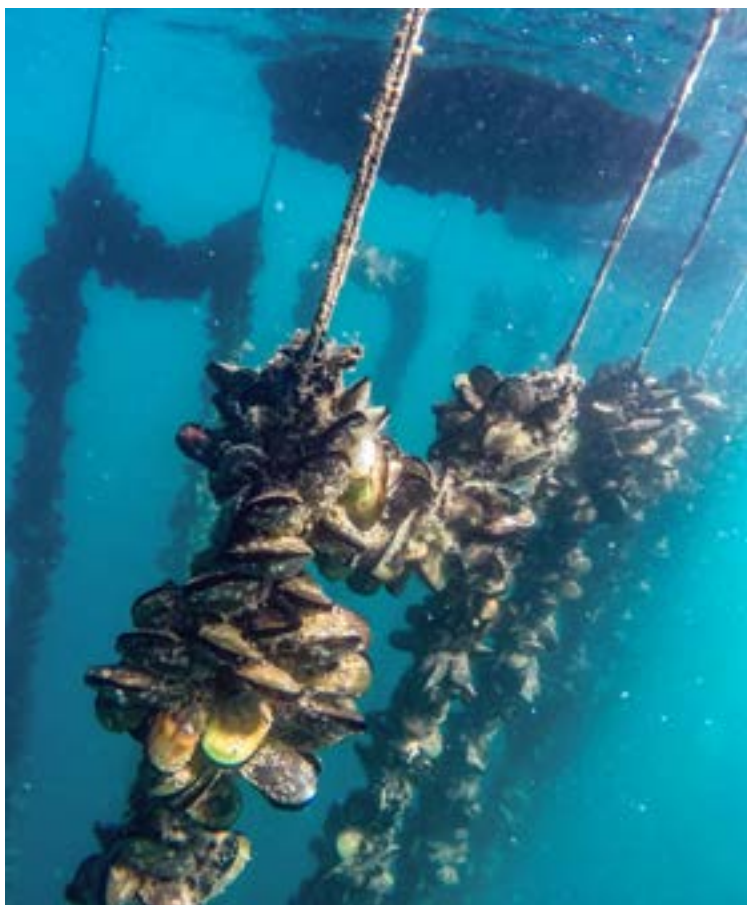
Waka ama, Orakei Basin, Auckland

Raewyn Peart

One potential funding tool available under the RMA (and presumably the NBA) would be resource rentals or charges. This is not about the allocation of rights, but rather about the allocation or distribution of value that is created *by* those rights, recognising that a non-private resource should see some of its value returned to the public and iwi/hapū. It raises money that can be earmarked for agencies to spend on environmental improvements or other actions that may be less politically expedient.

As explained below, charges are possible at the moment under the RMA, but are not uniform or consistent. One option would be to make charges compulsory through amendment to the Act, or to provide greater policy guidance through the use of national direction on the subject. A sub-option might be to charge for some types of use (eg where occupation of coastal space is exclusive) but not others. Essentially, that would create a subsidy for “positive” activities (eg potentially some forms of shellfish or seaweed aquaculture), creating not just a funding mechanism, but also an economic incentive for environmental enhancement.

Raewyn Peart



Mussel line, Kaiaraara Bay, Aotea

A spotlight on charging for use of the marine environment

Prior to the RMA, charges for the use of the coastal environment were levied under the Harbours Act 1950 by a variety of regulators, including Harbour Boards and the Marine Division of the Ministry of Transport.²³⁴ Other charges were levied under special Acts of Parliament relating to the coastal marine area.²³⁵ The result was an ad hoc charging regime which was applied inconsistently. During the mid-1990s, a report was commissioned to review charging regimes and it concluded that coastal charges could valuably employ the market mechanism of supply and demand, both to allocate coastal space in a manner that acknowledged its scarcity and to promote allocation of the resource to its best use. The report concluded that the existing system should be discontinued and replaced by some form of user or occupation charge, applying as an adjunct to the coastal planning regime and appropriately tied to local circumstances.²³⁶

The Resource Management Amendment Act 1997 provided regional councils with the express powers to charge for occupation of coastal space. Regional councils must now include statements in their regional coastal plans as to whether charging will be employed, and consider the balance of public and private benefits when determining whether or not to employ a regime.²³⁷

However, few councils have gone ahead with charging.²³⁸ Explanations put forward for not introducing such regimes include that, while it is generally considered appropriate to charge, the risks at this point in time are too high due to lack of clarity in the legislation, a number of barriers to implementation, and issues regarding equitable implementation.²³⁹ Others include (rather unbelievably) uncertainty around what coastal occupation charges are; the low level of coastal occupation in a district; uncertainty over future ownership and management of the foreshore and seabed; and the likelihood of a lengthy plan change process holding up other priorities.²⁴⁰

Occupation charges could be imposed whenever a coastal permit is granted, but another mechanism might be to raise money by auctioning or tendering rights via a competitive process (if that were a means through which rights were allocated in the future). However, care would need to be taken here; allocative choices may not just be about raising

the most funds, but rather seeking to prefer activities that would best achieve wider objectives (eg environmental enhancement, food security, social development etc).

Resource rentals could also be reintroduced for commercial fishing. That could, conceivably, even be extended to recreational fishing if there were a requirement for fishers to be licensed and report their catch. Recreational charges would be hotly contested on the grounds that fish for personal use should be free as long as sustainably caught. However, commercial resource rentals are by no means a new concept.

A spotlight on charging for fisheries

When the QMS was first introduced, quota owners were required to pay resource rentals to the government for utilising the fisheries resource. The purpose of the rentals was to enable government to recoup some of its fisheries management costs, as well as to disincentivise speculation in the quota market.²⁴¹ Resource rentals were, however, later replaced with cost recovery levies. These levies can cover a wide range of fisheries management costs, including management activities, enforcement, research, stock assessment and addressing the adverse effects of fishing on the marine environment and protected species. The legislation sets out a range of principles that apply to determining who should pay what costs, based on the user-pays or beneficiary-pays principle.

However, some have pointed out that this can incentivise opposition to research or management actions where they are not in the financial interest of quota holders. This is particularly the case when research is targeted at the stocks most at risk and therefore at those “for which TAC reductions are most likely”.²⁴²

We need to consider what the purpose of charging should be in a future system. Is it to enable a more efficient pooling of resources to improve stock management (utilisation) for those holding ITQ? If so, one might dispense with charging altogether and support quota holding groups to fund their own research directly. Or is it to fund research to support minimising or eliminating environmental impacts? If so, there might need to be clearer principles around how this is funded vis-a-vis research on stocks themselves. Or is it to assist the industry as a whole to innovate and transition towards the use of more sustainable gear and

methods? That would essentially be a cross-subsidy for those unable to do so. Or is it a mechanism to provide a return to the community by extracting some of the value conferred on users from granting exclusive access to a shared resource? There is a similar debate happening in the context of whether commercial interests should pay to bottle and export drinking water.²⁴³ At present, it can be unclear who the “customer” of levies is.

A future system could conceivably reimagine the idea of a resource rental as a funding tool to be used for the benefit of the marine environment as a whole, not just as a cost recovery mechanism or a stock-specific research tool. It could cover not just stock-related matters, but also fundamental biological studies relating to non-fished and protected species, and ecosystems-based research and habitat mapping. It could extend to tangible restoration initiatives to support not just fisheries but also broader ecosystem services. A rental could be calculated according to the value of quota,²⁴⁴ and be payable either annually or when quota is sold (although that may have challenges).²⁴⁵

Either way, funds could be distributed across various bodies responsible for pursuing positive marine outcomes, including mana whenua, government agencies and regional councils, or managed by an independent fund manager operating under a series of clear statutory principles. It could even be hypothecated to fund the activities of independent bodies like an Oceans Commission (see Chapter 12) which would require a secure funding streams.

At a bigger picture level, a more strategic and principled approach to charging for the commercial use of the marine environment might mean that multiple independent funding streams could be coordinated in the service of a wider vision. For example, charges could be imposed on land-based activities using the moana as a receiving environment for waste, simultaneously raising money, providing an incentive for reducing impacts, and more equitably sharing costs between activities in the marine environment (eg fishers) and those affecting them (eg farmers, foresters and urban developers). A nationally consistent approach to financial contributions under the RMA might be transformed to perform part of

this role. From an ecocentric perspective, all of these funding mechanisms might be recast not just as a resource rental (implying ownership), but rather as a payment or koha to nature held on trust by a co-governed entity charged with speaking for its interests (see Chapter 12). However, across all of the above, great care will need to be taken when it comes to relationships under te Tiriti o Waitangi. Charging for the use of resources (and where any funds are directed) can raise significant issues around tino rangatiratanga (eg the prospect of iwi paying to use resources where ownership is contested, where rights have previously been taken, or where settlements over the use of resources have been made),²⁴⁶ and detailed design would need to involve mana whenua.

A future system will need to carefully consider how it funds the activities needed to achieve its objectives. Charging for the use of marine resources would need to involve mana whenua in detailed design to address any issues around te Tiriti o Waitangi, settlements and where revenue is directed.



Greater use could be made of resource rentals and charges which could be imposed more consistently across all marine users.

How the system funds the activities of public authorities can have powerful side-effects on the behaviour of private persons. However, a future system could explore the more intentional use of economic incentives to drive positive outcomes. These may also offer alternative (and arguably more effective) ways to reduce harm, and can unleash the creativity of the private sector to find innovative solutions that command and control regulation may not. However, a regulatory “safety net” will still be required to establish firm environmental limits.

We explored economic incentives and other non-regulatory tools in our work on resource management reform, and summarise some possibilities in Figure 8.3 in the context of the moana. These go well beyond what is typically thought of as the oceans management system.

Craig Potton

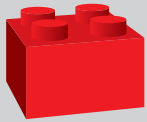


Sand mining, Pakiri Beach

Charges and taxes	These are tools that impose a cost on activities, inputs or outputs. They can be described as fees, levies, taxes, prices, etc but can be regarded as a tax if they are not connected to the provision of a service.²⁴⁷ In a future system, there could be a green tax targeting environmentally-damaging activities. Some have proposed imposing taxes on inputs that can lead to pollution (eg fertilisers) if outputs cannot adequately be measured. Tax or rates rebates could be offered for activities that achieve positive outcomes like environmental restoration. Resource rentals, targeted rates and value uplift capture mechanisms have been mentioned above. A distortionary tax system is not necessarily a bad thing. Embedding incentives within the tax system could encourage behaviour to become durable societal norms, not just a regulatory obligation. But although the RMA has encouraged the use of economic instruments,²⁴⁸ the specific tools (or legal ability) to implement them have not been forthcoming and amendment would be needed for something like a green tax. Yet “regional councils in particular could be well placed to make use of environmental taxes to help carry out their environmental stewardship responsibilities”,²⁴⁹ which would be an interesting line of inquiry to pursue given their new-found jurisdiction over fishing activity under the RMA. There are also questions of equity to consider. If we were to impose taxes on, for example, commercial fishing for benthic impacts, it may be unjust not to impose comparable taxes on land-based activities or aquaculture. And the regressive impacts of green taxes need to be considered too; some fishers, for example, may not be in a position to alter their behaviour in response and require assistance to (for example) purchase new gear.
Subsidies	Subsidies provide financial rewards for particular activities or outcomes, and can be used to drive positive behaviour. In a future system, subsidies could be deployed in a more structured and strategic way (akin to the Land Transport Fund) to support private activities envisaged in a marine spatial plan (eg regenerative aquaculture) or to transition to more sustainable methods of use (eg to subsidise new fishing gear or commercial fishing boats). Subsidies could come from hypothecated funding raised by resource rentals or green taxes.
Deposit refund schemes	These involve the payment of a deposit when a product is purchased, which is repaid when the product is returned after use. This system can provide a strong financial incentive for returning products to a centralised facility to better ensure product reuse, safe disposal or recycling, including plastic products that might otherwise end up in the marine environment. However, it would have limitations when it came to things like microplastics.
Feebates	Feebates involve people being charged a fee (charge) if their behaviour or performance falls below a set level, or being provided a reward (subsidy) if they exceed it. This could, for example, potentially be used to provide incentives to invest in new fishing gear or mitigation measures for aquaculture.
Bonds	Bonds are provided by a person conducting an activity as a surety against potential harm and non-compliance with regulatory measures. The prospect of having to offer funds for remediation in advance incentivises preventative measures.
Ecosystem valuation	Putting a price on ecosystem services and natural capital could ensure that environmental harm is appropriately valued. Much depends on methods of valuation, and in an ecocentric view it may be inherently objectionable to commoditise the natural world in this way. It may be very hard, however, to measure ecosystem services in the marine environment, and even more broadly “ecosystem services analysis is regarded as not having progressed to a stage where there is an accepted framework for analysis”.²⁵⁰

Behavioural nudging	Nudging works by providing subtle interventions based on key elements of human psychology. For instance, people take actions when it is relatively easy to do so, when they get enjoyment or benefit from it, when they are reminded about it in a timely way, where consequences are direct and observable, and when they are in competition with others in their peer group. So while there is considerable peer pressure in Aotearoa New Zealand not to litter on beaches and ease of access to rubbish bins, there is much less social stigma attached, and few alternatives to, washing one's car where cleaning chemicals enter stormwater drains flowing to the sea. A future system could be more geared towards psychological nudging by establishing a "nudge unit" within government and by requiring consideration be given to such methods. It might be particularly valuable in the context of encouraging compliance.
Public service messaging	Simply making people aware, through education campaigns, can sometimes change their behaviour. This could focus more specifically on the moana in a similar way that the Energy Efficiency and Conservation Authority engages in education on climate change and energy.
Green certification schemes	A company or a product can be independently verified as meeting robust environmental standards. The overall idea is that businesses will choose to be bound by higher standards and pursue them rigorously, because consumers are becoming more environmentally discerning. A lot of private certification already goes on in the marine environment, under a number of brands.²⁵¹ This concept could be expanded to include more activities or sectors impacting on the marine environment (eg for wastewater and tourism). It could also see an independent entity like the Parliamentary Commissioner for the Environment or an Oceans Commission be involved in certification (or at least accrediting those doing the certification), including linking corporate performance to statutory targets. Certification would often require close oversight of a complex supply chain, and an onus of proof on those seeking certification that standards have been met all the way along it. It could generate a domino effect of environmental improvement through corporate peer pressure. Large, publicly facing companies, like those directly selling products and services, would have an incentive to put pressure on their less publicly accountable suppliers, like manufacturers or primary producers, so that they could obtain certification. This could incentivise a move towards higher value use of marine resources. For example, bulk harvesting methods can have greater impacts on the environment, potentially reducing the market appeal of the products in high value markets.²⁵² Smaller vessels, or artisanal fishing, could arguably get "more bang for one's buck".
The school curriculum	A review of the school curriculum could include considering the mandatory teaching of civics, environmental sustainability, and climate change, with a particular focus on the moana and its connection to land. There could also be a review of tertiary programmes, including the core requirements of vocational qualifications leading to careers in jobs central to oceans management, such as fisheries management and marine spatial planning.
Strengthening directors' duties	Directors' duties under the Companies Act could be expanded to focus not just on the interests of shareholders, but also on the interests of future generations, the public interest and the moana itself. This would reflect that (especially large) corporations can often be better placed than individuals to make a real difference to environmental outcomes. ²⁵³ The United Kingdom's Companies Act expressly provides that directors must take into account a company's impacts on the environment.
Expanding corporate financial disclosure requirements	Disclosing environmental risks forces firms to assess and understand their risks, and allows investors and shareholders to make decisions based on them. Recent years have seen significant international developments on this front. ²⁵⁴ While most of this is about climate risks, that could be expanded to include the disclosure of risks from environmental performance in the marine environment.

Figure 8.3: Summary of economic incentives and other non-regulatory tools



A number of non-regulatory tools could be explored to provide economic and behavioural incentives in a future system, including charges and taxes, a more systematic use of subsidies, feebates, bonds, nudging, reform of the school curriculum and professional training programmes, directors' duties and corporate disclosure requirements.

8.9 Concluding comments

In this chapter we have explored options for reforming the toolkit in a future system. This is by no means intended to be comprehensive. Other tools will be possible, and there will be a vast number of detailed policy questions to consider in how both existing and new tools are designed. All options will have pros and cons, and the best way forward will depend not just on how effective different tools are, but also on what objectives we want them to achieve.

Tools are also a product of the worldviews underpinning the system. Do we trust market mechanisms to deliver on public good outcomes? Should we, for example, establish stronger tradeable property rights in aquaculture as we have done for wild fisheries? Is the taxation of

environmentally harmful activities the most efficient or ethical way to reduce harm? Or do we turn to regulatory tools – to set limits, to allocate (or reallocate) resources to the most publicly desirable use, and to support industries that the government of the day thinks are the way of the future?

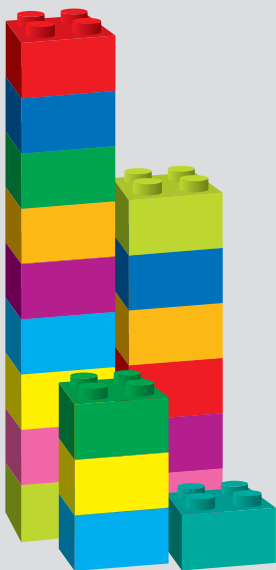
Some of this might depend on the role we expect these tools to perform. It is reasonably clear that the setting of environmental limits cannot be done through non-regulatory means. On the other hand, it is equally obvious that non-regulatory tools have not been used to their full potential in, for example, enhancing the environment. Tools can overlap, too, and there may be a need to clarify when this is and is not appropriate. For example, there can be opposing views as to what fisheries regulations and RMA plans should cover, the extent to which market forces under the QMS render the need for some regulatory measures redundant, and whether we should rely on the ETS or broader regulations to reduce emissions of greenhouse gases.

It is also useful to think about not just individual tools, but also types of tools. That includes broad categories like regulatory and non-regulatory interventions, but also tools that operate in certain common ways or exist in close-knit relationships. These include things like property rights, planning frameworks and economic instruments. Another particularly important kind of tool is a MPA, which we turn to now in Chapter 9.



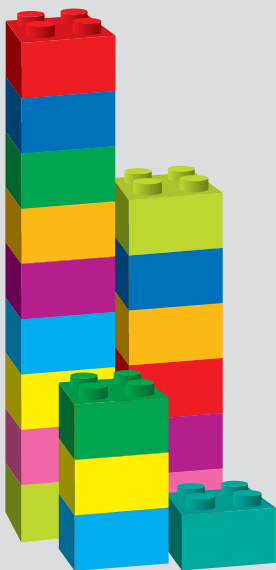
Muriwai beach

Raewyn Peart



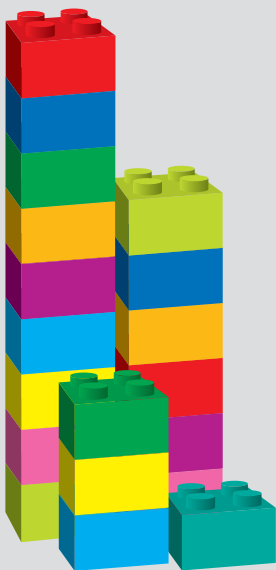
Summary of options for reform: Reconsidering the toolkit

- A National Planning Framework envisaged under the NBA provides an opportunity for marine matters to be more thoroughly integrated into other parts of national direction. New marine-related national direction could be created and existing documents reviewed through a marine lens. The NZCPS itself could be strengthened.
- The NZCPS could be paired with new national level regulations (an NES) to give effect to its objectives and policies, and/or it could be strengthened to provide for more extensive “implementation” provisions.
- An EEZ policy statement could provide a much stronger framework for consenting in the EEZ, increasing certainty for applicants and the environment.
- Combined plans under the NBA should provide more effective tools for marine management. Conservation planning could be strengthened to have a focus on marine bio-regional areas. A future system could also see the creation of a more developed planning framework for fishing.
- Fishing permits could be brought under a more environmentally policy-driven framework, which could operate alongside the QMS.
- Consenting could be applied more broadly to waste minimisation frameworks.
- Environmental limits contemplated by the NBA could be more targeted to the marine context, including by being more specific about what things limits must be created for. To be useful, a provision classed as a limit would need to have clear consequences different to other provisions.
- There are many regulatory tools available under the Fisheries Act that have been underutilised. A future system could provide more structure and direction around how (and why) they are to be deployed, and could characterise some of them as environmental limits.
- The Harvest Strategy Standard, which provides a more nuanced approach to setting TACs, could be formally incorporated into legislation.
- A hard “cap” could be placed on recreational take (a “total allowable recreational take”) as well as a commercial TACC. However, that could have challenges.
- Greater spatial separation could be created between recreational and commercial fishing activities by creating dedicated recreational fishing areas.
- A future system could provide more framing around how to set localised catch limits within QMAs, requiring boundaries to be redrawn based on ecological factors, or providing a more agile process (and trigger points) by which QMAs are (or must be) revised.



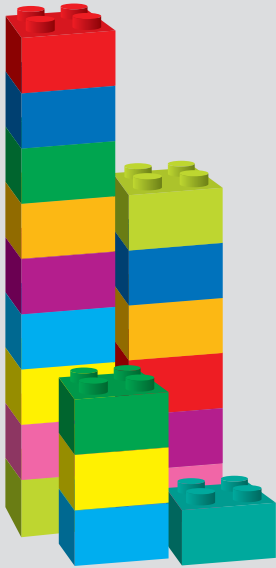
Summary of options for reform: Reconsidering the toolkit (continued)

- Tools under conservation legislation could be strengthened so that they provide for more powerful species-based environmental limits. In particular, the process for creating population management plans could be made simpler and/or focused only on the biological needs of protected species (rather than the impact on other users of the sea)
- The system could provide that a breach of environmental limits has clear and immediate consequences as a matter of law, including (to the extent necessary) overriding existing land use rights.
- The QMS could be expanded to include commercial operators of recreational fishing activities (eg charter boats), by requiring such operators to cover their catch by purchasing ACE.
- A parallel system of quota could be established for all recreational fishers (replacing tools like bag limits).
- Recreational fishing could be included in the same market as commercial quota, so (at least in theory) fisheries would go to their highest value use.
- Coastal permits under the RMA and EEZ Act could be made more akin to property rights by allowing greater tradability and longer duration, especially when it comes to aquaculture and other activities requiring a long-term presence (eg wind turbines affixed to the seabed or desalination facilities).
- Property rights in aquaculture could be established that are not linked to particular places or the need for coastal occupation (eg for mobile aquaculture operations based on a particular biomass rather than the area of operation).
- Aquaculture rights could be made more fungible with quota rights, meaning that trading of rights could occur across sectors.
- Cap and trade markets for some forms of diffuse pollution (eg nutrients) could be rolled out more proactively across relevant catchments, and include estuaries. Depending on the ability to measure or estimate runoff from individual properties, that could include sediment.
- Property rights could be eschewed in a future system by removing “ownership” over some things (eg buyback of private title and a different status for Crown owned minerals), and by declining to use market based tools for others (eg greenhouse gas emissions and occupation rights for aquaculture).
- Some have suggested altering or even replacing the property rights based QMS system. It could be undone through buyback of quota and implementing a permitting system. Alternatively, more targeted changes (eg more aggregation controls, creation of a public quota holder, and earmarking some quota for particular types of commercial fishers) could be made to soften the social impacts of market forces and incentivise environmental improvements.



Summary of options for reform: Reconsidering the toolkit (continued)

- Existing use rights for land could be overridden where environmental limits were threatened (eg to avoid significant impacts on protected areas in or near estuaries).
- National guidance could state what durations for resource rights are appropriate for different activities, in order to provide adequate commercial certainty, while also avoiding locking in sub-optimal uses.
- One option for allocating marine resources would be to use a first in time permitting system, whereby the first user to apply receives rights as long as the environmental impacts of an activity are acceptable. However, that can have a number of issues.
- More proactive, structured and competitive allocative mechanisms could be used (or made mandatory) in a future system, such as auctioning or attribute weighted tendering.
- A more proactive allocation of rights in particular spaces or zones could be achieved through marine spatial planning. This could distribute rights between different uses based on public interest principles, and potentially stakeholder consensus, although not necessarily different users.
- A formal forum could be established whereby new entrants or sectors wishing to use the marine space in a way that conflicts with existing uses could have some legal pathway to negotiate access rather than being excluded.
- The Public Works Act or minerals-type access arrangements could be used to accommodate publicly important uses of the marine environment.
- Rights in a future system could be made more spatially agile, especially when it comes to fixed occupation rights. That is particularly relevant to aquaculture operations, which may need to shift or become more operationally mobile, but it could also apply in the future to other activities as environmental conditions change (eg floating wind farms or tidal energy facilities).
- A future system could enshrine human rights to a healthy marine environment (eg in the Bill of Rights Act). However, that would have challenges in practice, and may not be a silver bullet solution to addressing environmental issues.
- A future system could recognise that the moana itself has legally enforceable rights. The normative basis of recognising personhood for nature will be important, but potentially difficult to establish given the different worldviews of te ao Māori and te ao Pākehā.
- The scale at which the moana is granted rights is important. At one end of the scale, the ocean as a whole could be recognised as a person with rights. At the other end, particular places, species or features could be given personhood. Layers of personhood could even be established, just as we have layers of personhood for companies.



Summary of options for reform: Reconsidering the toolkit (continued)

- If the moana was granted legal rights, the system would need to be clear as to what those rights involve. They could be much wider than current environmental protections in the RMA or Fisheries Act. They could be as broad as the rights enjoyed by humans.
- Emergency orders could be utilised in a future system where environmental limits were imperilled. That could be one basis on which legally binding rāhui could be deployed.
- The Waste Minimisation Act could contain a duty for ministers to progress regulatory tools like prohibitions and product stewardship schemes to meet mandatory targets for the reduction or elimination of plastic dangerous to marine life.
- The Building Code, wastewater and stormwater infrastructure standards, and vehicle emissions standards could be strengthened to reduce the impacts of buildings and runoff on the marine environment.
- Regulatory tools could be used to drive positive outcomes in the marine environment. Duties on public authorities to pursue restoration and enhancement could be strengthened; sectoral accords could be reached which industries to provide improvements; a marine biobanking framework could be deployed; and fast track processes (or less stringent activity status) could provide an incentive for projects having public interest/environmental benefits.
- Targeted rates could be broadened to enable their use to charge land uses causing adverse impacts on the moana.
- Central government could provide greater funding assistance to regional councils to support marine management, and take over specific functions such as marine mapping.
- Greater use could be made of resource rentals and charges which could be imposed more consistently across all marine users.
- A number of non-regulatory tools could be explored to provide economic and behavioural incentives in a future system, including charges and taxes, a more systematic use of subsidies, feebates, bonds, nudging, reform of the school curriculum and professional training programmes, directors' duties and corporate disclosure requirements.

Endnotes

- 1 Depending on how granularly one looks at tools, they could include very specific things like consent conditions or controlled activity standards.
- 2 Which may not even look the same if our legislation is designed or split differently in a future system: see Chapter 11 on legislative design.
- 3 Geoffrey Palmer “Law-Making in New Zealand: Is There A Better Way? [Harkness Henry Lecture]” (2014) 22 Wai L Rev 1.
- 4 They contain objectives and policies, but the main method by which those are achieved are through rules and consenting mechanisms.
- 5 For example, detailed drafting about how consenting frameworks work (eg changes to notification requirements, timeframes for decisions, rules around provision of additional information and so forth).
- 6 Some elements may not be regulatory, eg other methods to achieve objectives. However, these other methods are often referred to in plans but use other tools outside the plan itself to achieve (eg public projects funded by councils).
- 7 This direction would be consistent with the NZCPS, policy 22(3): “Control the impacts of vegetation removal on sedimentation including the impacts of harvesting plantation forestry.” While the NES for Plantation Forestry allows councils to impose more stringent restrictions to give effect to policy 22(3), the NES itself does not require it (and, if anything, makes it more difficult to do by requiring strong justification through section 32 reports).
- 8 As policy 22(3) is relatively weakly worded compared to others. On the need to include estuaries in the NPS for Freshwater Management, see Parliamentary Commissioner for the Environment *Managing our estuaries* (Office of the Parliamentary Commissioner for the Environment, Wellington, August 2020).
- 9 For example, onsite stormwater disposal and how urban form interacts with the coast (eg providing a buffer to avoid hardening of coastlines).
- 10 Recognising that not all sedimentation is about estuaries.
- 11 *Attorney-General v The Trustees of the Motiti Rohe Moana Trust* [2019] NZCA 532.
- 12 Furthermore, the NES for Marine Aquaculture does not give effect to the broader policies in the NZCPS.
- 13 For example, Environment Canterbury is only just now embarking on a review of its regional coastal plan which was notified in 1994 and made operative in 2005.
- 14 *Draft National Policy Statement for Indigenous Biodiversity* (Ministry for the Environment, 2019) at 6.
- 15 At 18, [3.4].
- 16 Much effort has focused on what the general precautionary principle and purpose of the Act say.
- 17 For example, it could provide for regulations to be promulgated classifying areas for a range of reasons, including because they are especially vulnerable due to their biophysical characteristics.
- 18 See Deidre Koolen-Bourke and Raewyn Peart *Conserving Nature* (Environmental Defence Society, Auckland, 2021), ch 4.
- 19 Influential in the sense that “avoid” can essentially impose bottom lines.
- 20 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 144.
- 21 The first place-based fisheries plan is set to be created in the Hauraki Gulf as a result of the Seachange Tai Timu Tai Pari process. Contrast the more general Fisheries New Zealand *National Inshore Finfish Fisheries Plan* (Discussion Paper 2019/18, November 2019), which is still in a draft phase.
- 22 The Prime Minister’s Chief Science Advisor has recommended to “improve the processes for input and engagement in fisheries management, particularly in regards to undertaking effective iwi and stakeholder engagement, public involvement, and adequate checks and balances.” Office of the Prime Minister’s Chief Science Advisor *The Future of Commercial Fishing in Aotearoa New Zealand* (February 2021) at 22.
- 23 Fishing permits can be revoked for non-compliance with things like an obligation to pay deemed values for fish that cannot be covered by ACE.
- 24 In the same way that an “objective” is a specific type of provision with a particular legal status under the RMA, but is also a general name for something we are trying to achieve.
- 25 On an Oceans Act, see the discussion in Chapter 11.
- 26 See the spotlight in Chapter 3.
- 27 See Kiely McFarlane and others *Environmental limits - A Proposed Framework for Aotearoa New Zealand* (Cawthron Institute, Report No 3530, June 2020) at 155; see also Parliamentary Commissioner for the Environment *Managing our estuaries* (Office of the Parliamentary Commissioner for the Environment, Wellington, August 2020).
- 28 Ministry for the Environment *Natural and Built Environments Bill* (Exposure Draft, 2021), cl 7(4) (c) and (d). And for all domains limits include the minimum biophysical state of the natural environment or of a specified part of that environment: see cl (3)(a).
- 29 The direct amendments to Southland’s Regional Coastal Plan through special legislation for Fiordland illustrates how the RMA could be used more proactively to achieve marine protected areas that are more nuanced than marine reserves. See the concept of “China Shops” in Fiordland (Te Moana o Atawhenua) Marine Management Act 2005, sch 12, Policy 4.1.4.
- 30 On biobanking, see later in this chapter and Marie A Brown *Banking on Biodiversity: the Feasibility of Biodiversity Banking in New Zealand* (Environmental Defence Society, Auckland, October 2017)
- 31 Moreover, some marine systems can change gradually rather than having a single “threshold” at which it tips into a new, less healthy state, so limit setting is not just about identifying ecological tipping points.
- 32 For example, based on a definition of ecological integrity or some other measure.
- 33 Kiely McFarlane and others *Environmental limits - A Proposed Framework for Aotearoa New Zealand* (Cawthron Institute, Report No 3530, June 2020).
- 34 Kiely McFarlane and others *Environmental limits - A Proposed Framework for Aotearoa New Zealand* (Cawthron Institute, Report No 3530, June 2020) at 155. at 155.
- 35 See the later spotlight on the Harvest Strategy Standard on arguments about the relevance of social and economic factors in determining limits under the Act.
- 36 Fisheries New Zealand “Central area fishing rules” Ministry for Primary Industries (2 July 2021) <www.mpi.govt.nz>; see also Fisheries New Zealand “Fishing rules” Ministry of Primary Industries (1 December 2021) <www.mpi.govt.nz>
- 37 Stephen Eayrs, Tony Craig and Katherine Short *Mitigation Techniques to Reduce Benthic Impacts of Trawling: MIT2019-02 A review for the Department of Conservation by Terra Moana Limited* (Terra Moana, April 2020). Technology is currently available which could be rapidly deployed to significantly reduce impacts: see Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 87.
- 38 *LegaSea Rescue Fish Ika Rauora: A pathway to fish abundance and marine ecosystem recovery* (May 2020).
- 39 Compare Seachange Stakeholder Working Group Sea Change Tai Timu Tai Pari *Hauraki Gulf Marine Spatial Plan* (Hauraki Gulf Forum in partnership with others, April 2017).
- 40 Department of Conservation and Fisheries New Zealand *National Plan Of Action – Seabirds 2020* (Fisheries New Zealand, November 2019) at 13.
- 41 Source: Stephen Eayrs, Tony Craig and Katherine Short *Mitigation Techniques to Reduce Benthic Impacts of Trawling: MIT2019-02 A review for the Department of Conservation by Terra Moana Limited* (Terra Moana, April 2020), adapted from Robert McConnaughey and others “Choosing best practices for managing impacts of trawl fishing on seabed habitats and biota” (2020) 21 Fish and Fisheries 319.
- 42 That said, some tools are not the most effective; Cabinet papers have sought to address issues with them. See Minister for Ocean and Fisheries “Fisheries Amendment Bill: Strengthening fishing rules and policies: landings and discards” (2 July 2021); Minister for Oceans and Fisheries “Fisheries Amendment Bill: Strengthening fishing rules and policies: offences and penalties and agile decision-making” (2 July 2021); and Minister for Oceans and Fisheries “On-board cameras across the inshore fishing fleet” (2 July 2021) (eg landing and discard rules, cost recovery for cameras on boats)
- 43 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 67.
- 44 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 68.
- 45 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018), chapter 4.
- 46 In the form of the Magnuson-Stevens Fishery Conservation and Management Act 16 USC §5 1801-1891d (US).
- 47 See Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018).
- 48 As opposed to sustainability measures, which can be set under the Fisheries Act but also where the RMA has jurisdiction.
- 49 *Royal Forest and Bird Protection Society v Minister of Fisheries* [2021] NZHC 1427.
- 50 At [47].

- At [92].
- At [199] and [200].
- At [108].
- At [117].
- At [144].
- At [166].
- Terrance Quinn "Ruminations on the Development and Future of Population Dynamics Models in Fisheries" (2003) 16(4) *Natural Resource Modelling* 341.
- See Fisheries New Zealand *Fisheries Assessment Plenary: Stock Assessments and Stock Status* (Ministry for Primary Industries, Volume 1, May 2021), preface.
- For example, Fisheries New Zealand *Aquatic Environment and Biodiversity Annual Review 2019-20: A summary of environmental interactions between the seafood sector and the aquatic environment* (Fisheries Science Team, Ministry for Primary Industries, Wellington, June 2020) at 427 and following, at 427 and following.
- Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019).
- John McKoy "Fisheries resource knowledge, management, and opportunities: Has the Emperor got no clothes?" (paper presented to Royal Society of New Zealand Conference, Wellington, November 2006).
- Parliamentary Commissioner for the Environment *A review of the funding and prioritisation of environmental research in New Zealand* (Office of the Parliamentary Commissioner for the Environment, Wellington, December 2020).
- Parliamentary Commissioner for the Environment *A review of the funding and prioritisation of environmental research in New Zealand* (Office of the Parliamentary Commissioner for the Environment, Wellington, December 2020).
- Experience being, for example, the presence of a super abundance of marine life for diving, tourism and so forth.
- This would not be without its challenges, including the inevitable tension between sectors when establishing where such areas should be located.
- For example, in the Hauraki Gulf none of the QMAs coincided with known biological fish stocks. See Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 14.
- See Chapter 2.
- See Fisheries Act 1996, ss 24-26.
- Fisheries New Zealand *Fisheries Assessment Plenary: Stock Assessments and Stock Status* (Ministry for Primary Industries, Introductory Section, November 2021) at 257.
- Fisheries New Zealand *Fisheries Assessment Plenary: Stock Assessments and Stock Status* (Ministry for Primary Industries, Introductory Section, November 2021) at 283.
- David E Pollock *Evolution of Life-History Patterns in Three Genera of Spiny Lobsters* (1995) 57 *Bulletin of Marine Science* 516.
- John D Booth and Bruce F Phillips "Early Life History of Spiny Lobster" (1994) 66 *Crustaceana* 271.
- Ivan A Hinojosa and others "Differing environmental drivers of settlement across the range of southern rock lobster (*Jasus edwardsii*) suggest resilience of the fishery to climate change" (2016) 26 *Fisheries Oceanography* 49.
- Stephen M Chiswell and John D Booth "Sources and sinks of larval settlement in *Jasus edwardsii* around New Zealand: Where do larvae come from and where do they go?" (2008) 354 *Marine Ecology Progress Series* 201.
- Stephen M Chiswell and John D Booth "Sources and sinks of larval settlement in *Jasus edwardsii* around New Zealand: Where do larvae come from and where do they go?" (2008) 354 *Marine Ecology Progress Series* 201.
- Stephen M Chiswell and John D Booth "Sources and sinks of larval settlement in *Jasus edwardsii* around New Zealand: Where do larvae come from and where do they go?" (2008) 354 *Marine Ecology Progress Series* 201.
- Fisheries New Zealand *Fisheries Assessment Plenary: Stock Assessments and Stock Status* (Ministry for Primary Industries, Introductory Section, November 2021) at 291-311.
- Fisheries New Zealand *Proposal to review the recreational rules for the CRA 2 rock lobster fishery* (Ministry for Primary Industries, Discussion Paper No 2018/17, November 2018) at 5.
- Fisheries New Zealand *Fisheries Assessment Plenary: Stock Assessments and Stock Status* (Ministry for Primary Industries, Introductory Section, November 2021) at 297-311.
- See Deidre Koolen-Bourke and Raewyn Peart *Conserving Nature: Conservation System Reform Issues Paper* (Environmental Defence Society, Auckland, 2021).
- Fisheries Act 1996, s 15(1).
- Marine Mammals Protection Act 1978, s3H(1)(n).
- Kate Mulcahy and Raewyn Peart *Wonders of the Sea: The protection on New Zealand's marine mammals* (Environmental Defence Society, Auckland, 2012) at 80-82.
- The distinction between the two can sometimes become blurred, as we explored in Chapter 6 (whether the QMS is still regarded as a "tool" or just as part of the fabric of a capitalist society).
- Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Synthesis Report* (Environmental Defence Society, Auckland, December 2018) at 230; Garrett Hardin "The tragedy of the commons" (1968) 162 *Science* 1243.
- Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) chs 2-3, chs 2-3. See Chapter 7 on the "economic" worldview that might give rise to such thinking. That is not to say such a view is *bad* – it is simply one way among many that things can be valued.
- Although that is not always necessarily the case, as it assumes the operation of an effective market.
- More accurately, some people's later interpretation of Ronald Coase's work. See RH Coase "The problem of social cost" (1960) 3 *Journal of Law and Economics* 1.
- Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 19-21, 57.
- Despite a 2005 strategy for managing the environmental effects of fishing
- Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018), ch 5.
- See, for example, Eric Crampton *Refreshing Water: Valuing the Priceless* (New Zealand Initiative, 2019).
- Although there has more recently been a recognition that other tools (including regulatory ones) may be needed to achieve climate objectives (especially where whole sectors. Such as agriculture, are not included in the emission trading scheme).
- Largely based on financial return to public coffers through royalty payments, although that conversation about how to manage minerals in the public interest – including for climate change reasons – is a live one.
- Small areas of foreshore and seabed became privately owned as a result of their historical incorporation into land or seabed titles or as a result of the sea eroding away land.
- Marine and Coastal Area (Takutai Moana) Act 2011.
- See David Grinlinton "The nature of property rights in resource consents" (2007) 7 *BRMB* 37; Laura Fraser "Property rights in environmental management" (2008) 12 *NZJEL* 145; Barry Barton "The nature of resource consents" in *Environmental law: National issues intensive* (New Zealand Law Society, 2009); Resource Management Act 1991, s 122.
- Ownership is not possible in the EEZ anyway under international law.
- Although a true cap for commercial fishing is not technically there either, given the ability to pay deemed values where ACE cannot be obtained. That said, deemed values are designed to ramp up as more fish are landed without ACE cover, effectively limiting catch.
- See Raewyn Peart *Farming the Sea: Marine aquaculture within resource management reform* (Environmental Defence Society, Auckland, 2019) at 104.
- See the project's working paper at 94-96.
- On including estuaries within the management units in the NPS FM more broadly, see Parliamentary Commissioner for the Environment *Managing our estuaries* (Office of the Parliamentary Commissioner for the Environment, Wellington, August 2020).
- A lack of "ownership" might require a shift in language from "royalties" to "resource rentals" or "koha" – see later in this chapter.
- In the impacts of bottom trawling, see Karen McVeigh "Bottom trawling releases as much carbon as air travel, landmark study finds" *The Guardian* (online ed, 17 March 2021).
- On marine spatial planning, see Chapter 10.
- Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 26 and following.
- Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 101.
- Quota themselves are not linked to or conditional upon measures to protect the environment in the same way that permit issued under the RMA is.
- Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 17.
- Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 28 and following.

- 111 LegaSea *Rescue Fish Ika Rauora: A pathway to fish abundance and marine ecosystem recovery* (May 2020).
- 112 LegaSea *Rescue Fish Ika Rauora: A pathway to fish abundance and marine ecosystem recovery* (May 2020) at 3.
- 113 Michael Neilson "Land Back: Green Party calls for revamp of Treaty settlements" NZ Herald (online ed, 7 February 2022) <www.nzherald.co.nz>
- 114 Retirement of quota would, to have any effect, rely on a proportional reduction in the TAC and, ultimately, a higher abundance target than MSY.
- 115 Fisheries New Zealand "Crown fishing quota and ACE tenders" (10 February 2022) Ministry of Primary Industries <www.mpi.govt.nz>
- 116 Offshore would be more difficult as there is less potential for an independent fleet so far from shore.
- 117 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 96.
- 118 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 105.
- 119 Indeed, conferring rights by regulatory means can have advantages over relying on the trading of private property, since the former can be more closely linked to regulatory obligations.
- 120 Resource Management Review Panel *New Directions for Resource Management in New Zealand* (June 2020) at 200, 267, 268, and 281.
- 121 Resource Management Review Panel *New Directions for Resource Management in New Zealand* (June 2020) at 200, 267, 268, and 281.
- 122 Resource Management Act 1991, ss 123-123A.
- 123 Derek Nolan "Coastal" in Derek Nolan (ed) *Environmental and Resource Management Law* (6th ed, LexisNexis, Auckland, 2018)
- 124 *Golden Bay Marine Farmers v Tasman District Council* EnvC Wellington W42/2001, 27 April 2001.
- 125 Waitangi Tribunal *Ahu Moana: The Aquaculture and Marine Farming Report* (Wai 953, 2002) at 4-5.
- 126 Resource Management (Aquaculture Moratorium) Amendment Act 2002.
- 127 Aquaculture Reform (Repeals and Transitional Provisions) Act 2004.
- 128 Māori Commercial Aquaculture Claims Settlement Act 2004, ss 7-18 and 19-31.
- 129 Aquaculture Reform (Repeals and Transitional Provisions) Amendment Act 2011.
- 130 See generally Te Ohu Kaimoana *Annual Report 2017* (Te Ohu Kaimoana, September 2018) at 15.
- 131 Resource Management Act 1991, s 165G.
- 132 Ministry for the Environment *Water Programme of Action: Water Allocation and Use – Technical Working Paper* (ME 561, December 2004) at 22.
- 133 Waikato Regional Council "RFT SAS2016-2017-1800 – for release of fish farming space Coromandel marine farming zone" (26 January 2017) New Zealand Government Electronic Tenders Service (GETS) <www.gets.govt.nz/>
- 134 Resource Management (National Environmental Standards for Marine Aquaculture) Regulations 2020.
- 135 *Central Plain Water Trust v Ngai Tahu Properties* [2008] NZRMA 200 (CA) at [92]. See generally Philip Milne "Allocation of public resources under the RMA: Implications of *Aoraki Water Trust v Meridian*" [2005] RM Theory & Practice 146 at 156; Peter Skelton "Project Aqua reveals flaws in regional planning" *The National Business Review* (online ed, Auckland, 19 September 2003).
- 136 Although other mechanisms are possible (eg where there is no competition).
- 137 And the courts have made some creative interpretations of the RMA to suggest that the merits of different applications close in time can be compared: see *Central Plains Water Trust v Synlait Ltd* [2009] NZCA 609, [2010] 2 NZLR 363 (CA) at [89]; *Synlait Ltd v Central Plains Water Trust* [2010] NZSC 32, [2010] NZRMA 257; *Ngai Tahu Property Ltd v Central Plains Water Trust* [2009] NZSC 24
- 138 For example, this could be similar to the idea of monitoring demand through housing and business land assessments under the NPS on Urban Development, where the appearance of pressures trigger the use of tools.
- 139 *Ngāi Tai ki Tāmaki Tribal Trust v Minister of Conservation* [2018] NZSC 122.
- 140 It is interesting to ponder whether allocative choices could in the future effectively be made by holders of customary marine title, given that they will have powers to refuse permission for resource consents and concessions within their areas of customary marine title.
- 141 That said, such problems are equally likely to appear under a first-in-time consenting approach.
- 142 And for space to be auctioned, tendered or consented on a first in time basis within those areas.
- 143 See generally Raewyn Peart *Farming the Sea: Marine aquaculture within resource management reform* (Environmental Defence Society, Auckland, 2019).
- 144 See *Seachange Stakeholder Working Group Sea Change Tai Timu Tai Pari Hauraki Gulf Marine Spatial Plan* (Hauraki Gulf Forum in partnership with others, April 2017) at 4, 42 and 81.
- 145 Desalination would have a small marine footprint, but its impacts (eg discharge of highly saline water) could be spatially wider and have impacts on other activities such as aquaculture (see Chapter 2).
- 146 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Synthesis Report* (Environmental Defence Society, Auckland, December 2018) at 233.
- 147 Although not problem free; there can still be interventions relating to aggregation and competition, and issues of equity and efficiency where a market is not operating as intended.
- 148 That is not necessarily the case; much quota is held as investments by large holders.
- 149 See *Aoraki Water Trust v Meridian Energy Ltd* [2005] 2 NZLR 268 (HC) at [36]; *Dart River Safaris Ltd v Kemp* [2000] NZRMA 440 (HC) at [27]; *Southern Alps Air v Queenstown Lakes District Council* [2008] NZRMA 47 (HC).
- 150 *Queenstown Lakes District Council v Hawthorn Estate Ltd* [2006] NZRMA 424 (CA).
- 151 Renewable electricity generators have highlighted the need for very long period of certainty, including in the marine space.
- 152 On compensation declarations and aquaculture agreements, see Fisheries Act 1996, ss 186ZD-186ZR.
- 153 On the idea of unitisation agreements, where arrangements can be beneficial for both parties, see Barry Barton, Kimberley Jane Jordan and Greg Severinsen *Carbon capture and storage: Designing the legal and regulatory framework for New Zealand* (Centre for Environmental, Energy and Resources Law, University of Waikato, 2013).
- 154 On access arrangements, see Crown Minerals Act 1991, ss 47-80.
- 155 On issues with the extinguishment of consented rights for non-compliance, see Marie A Brown *Last Line of Defence: Compliance, monitoring and enforcement of New Zealand's environmental law* (Environmental Defence Society, Auckland, 2017); Marie A Brown *Last Line of Defence: Compliance, monitoring and enforcement of New Zealand's environmental law* (Environmental Defence Society, Auckland, 2017).
- 156 See for example Skara Bohny "Aquaculture experts drawn to Nelson for open ocean expo" *Stuffmedia* (online ed, 6 August 2019).
- 157 Raewyn Peart *Farming the Sea: Marine aquaculture within resource management reform* (Environmental Defence Society, Auckland, 2019) at 104.
- 158 See generally Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018). Although that may still raise similar issues of social equity similar to what we have seen with the fisheries QMS.
- 159 Paola Rosa-Aquino "Floating wind turbines could open up vast ocean tracts for renewable power" *The Guardian* (online ed, 29 August 2021).
- 160 Article 1 describes a right that humans have to adequate conditions of life in an environment of a quality that permits a life of dignity and wellbeing.
- 161 See: United Nations "Access to a healthy environment, declared a human right by UN rights council" (media release, 8 October 2021); The human right to a clean, healthy and sustainable environment HRC Res 48/13.
- 162 Klaus Bosselmann *The Principle of Sustainability: Transforming Law and Governance* (Ashgate Publishing, Surrey, 2008) at 6, 111, 114.
- 163 Klaus Bosselmann *The Principle of Sustainability: Transforming Law and Governance* (Ashgate Publishing, Surrey, 2008) at 6, 111, 127.
- 164 Tim Hazledine "Economics and the resource management system" in Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Working Paper 3* (Environmental Defence Society, Auckland, September 2018) at 154.
- 165 Greg Severinsen, Raewyn Peart and Bella Rollinson *The Breaking Wave: A conversation about reforming the oceans management system in Aotearoa New Zealand* (Environmental Defence Society, August 2021) at 84-86 for a more in-depth explanation.
- 166 See Greg Severinsen, Raewyn Peart and Bella Rollinson *The Breaking Wave: A conversation about reforming the oceans management system in Aotearoa New Zealand* (Environmental Defence Society, August 2021) at 86.
- 167 See the English translation at Georgetown University "Constitution of the Republic of Ecuador" Political Database of the Americas (31 January 2011) <www.pdba.georgetown.edu>
- 168 See the English Translation at World Future Fund "Law of Mother Earth: The Rights of Our Planet - A Vision from Bolivia" (7 December 2010)<<https://www.worldfuturefund.org/>

Projects/Indicators/motherearthbolivia.html>

- 169 See arts 2(1), 3, and 6.
- 170 This could even be reflected in a regional agreement in the South Pacific, building upon shared understandings of the moana as an ancestor.
- 171 Lin Edwards “Scientists say dolphins should be treated as non-human persons” *Phys.org* (6 January 2010); Thomas White *In Defense of Dolphins: The New Moral Frontier* (Blackwell Publishing, Malden, 2007); “BBC Dolphins deserve same rights as humans, say scientists” *BBC News* (21 February 2012).
- 172 Marine Mammals Protection Act 1978.
- 173 Animal Welfare Act 1999.
- 174 Rata Pryor Rodgers “The Connection of Māori to Whales” (Gateway Antarctica Supervised Project Report, University of Canterbury, 2017). This source also mentions whakatauki that attest to taonga status such as “He taonga no Tangaroa, I waihotia mo tātou, Ko te tohorā ki uta”: “This whale cast on the beach is the treasure left to all of us by the great god of Tangaroa”.
- 175 See for example Ngāi Tahu Claims Settlement Act 1998.
- 176 See Katharina Kropshofer “Whales and dolphins lead ‘human-like lives’ thanks to big brains, says study” *The Guardian* (online ed, 16 October 2017). See also the discussion in Greg Severinsen, Raewyn Peart and Bella Rollinson *The Breaking Wave: A conversation about reforming the oceans management system in Aotearoa New Zealand* (Environmental Defence Society, August 2021) at 84. at 84.
- 177 For example, Tangaroa embodies the spiritual significance of the marine environment, not just living things.
- 178 Earth Law Centre “Earth Law Framework for Marine Protected Areas” (12 September 2017) <<https://www.earthlawcenter.org/oceanrights>>
- 179 For example, if there were benefits to be had in shifting one MPA to create a better connection to another.
- 180 Tineke Lambooy, Jan van de Venis and Christiaan Stokkermans “A case for granting legal personality to the Dutch part of the Wadden Sea” (2019) 44(6-7) *Water International* 786.
- 181 Section 12.
- 182 Section 13.
- 183 Section 4.
- 184 Section 5.
- 185 See Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation Synthesis Report* (Environmental Defence Society, Auckland, December 2018) at 109.
- 186 They were first brought into the statutory framework in 1981 through an amendment to the Water and Soil Conservation Act 1967 and were carried through into Part 9 of the RMA in 1991.
- 187 Resource Management Act 1991, s 201.
- 188 Section 199.
- 189 Resource Management Review Panel *New Directions for Resource Management in New Zealand* (June 2020) at 204.
- 190 Some fishers have noted the lack of support for those suffering from fisheries closures when support is forthcoming for farmers on land.
- 191 Biosecurity Act 1993, s 144; Fisheries Act 1996, s 16. Biosecurity New Zealand *The New Zealand Government Biosecurity Response Guide* (Ministry for Primary Industries, October 2018).
- 192 For example, temporary closures under ss 186A and 186B of the Fisheries Act 1996 are referred to as rāhui (although they differ slightly from traditional rāhui). See Kimberley Maxwell and Wally Penetito “How the use of rāhui for protecting taonga has evolved over time” (2007) 2 MAI Review.
- 193 Te Ahukaramū Charles Royal “Kaitiakitanga – guardianship and conservation - Understanding kaitiakitanga” (24 September 2007) Te Ara - the Encyclopedia of New Zealand <<http://www.TeAra.govt.nz/en/artwork/11563/haia>>
- 194 Nicola Wheen & Jacinta Ruru “Providing for Rāhui in the Law of Aotearoa New Zealand” (2011) 120(2) *Journal of the Polynesian Society* 169. This conservation rāhui is distinct from, for example, a tapu related rāhui that is placed following a death in the area; Kimberley Maxwell and Wally Penetito “How the use of rāhui for protecting taonga has evolved over time” (2007) 2 MAI Review.
- 195 Nicola Wheen and Jacinta Ruru “Providing for Rāhui in the Law of Aotearoa New Zealand” (2011) 120(2) *Journal of the Polynesian Society* 169
- 196 Kimberley Maxwell and Wally Penetito “How the use of rāhui for protecting taonga has evolved over time” (2007) 2 MAI Review.
- 197 Hirini Moko Mead *Tikanga Māori: living by Māori values* (revised edition, Huia Publishers, Wellington, 2016) at 197.
- 198 Nicola Wheen and Jacinta Ruru “Providing for Rāhui in the Law of Aotearoa New Zealand” (2011) 120(2) *Journal of the Polynesian Society* 169.
- 199 For example, the Tītī hunting season ends with a rāhui until the next season: Peter Garven, Marty Nepia and Harold Ashwell *Te whakatau kaupapa o Murihiku: Ngai Tahu resource management strategy of the Southland region* (Aoraki Press, Wellington 1997) at 29.
- 200 For example, a Ngai Tahu rāhui on shellfish required the approval of the marae group and certain practices to be followed in order to collect shellfish in the area: Peter Garven, Marty Nepia and Harold Ashwell *Te whakatau kaupapa o Murihiku: Ngai Tahu resource management strategy of the Southland region* (Aoraki Press, Wellington 1997) at 29.
- 201 At 122.
- 202 Maxine Jacobs “Legal interpretations of wāhi tapu and rāhui sought in landmark rights hearing” *Stuff* (online ed, 17 February 2022) <<https://www.stuff.co.nz/pou-tiaki/300519200/legal-interpretations-of-whi-tapu-and-rhui-sought-in-landmark-rights-hearing>>
- 203 For example, gear requirements under the Fisheries Act.
- 204 See Waste Minimisation Act 2008 and Litter Act 1979.
- 205 Local Government Act 2002 and Building Regulations 1992.
- 206 Territorial Sea, Contiguous Zone and Continental Shelf Act 1977, ss 8 and 27.
- 207 See Maritime Transport Act 1994, s 200.
- 208 As under the current Local Government Act 2002.
- 209 Required under the NPS on Urban Development.
- 210 The size and resourcing of councils is particularly significant; and there are variations in collection techniques and data quality. See Marie A Brown *Last Line of Defence: Compliance, monitoring and enforcement of New Zealand's environmental law* (Environmental Defence Society, Auckland, 2017) at 193-194; Beca Carter Hollings and Ferner Ltd *Stock Take of RMA Monitoring Across Selected Agencies* (BECA, Final Report, May 2012).
- 211 Beca Carter Hollings and Ferner Ltd *Stock Take of RMA Monitoring Across Selected Agencies* (BECA, Final Report, May 2012).
- 212 Marie A Brown *Last Line of Defence: Compliance, monitoring and enforcement of New Zealand's environmental law* (Environmental Defence Society, Auckland, 2017). at 194.
- 213 Environmental Reporting Act 2015.
- 214 Marie A Brown *Last Line of Defence: Compliance, monitoring and enforcement of New Zealand's environmental law* (Environmental Defence Society, Auckland, 2017).
- 215 For example, this is recognised in the mandatory approval by the Environment Court of mediated consent orders under the RMA. Parties to litigation are not free to come to any agreement they wish; any agreement must comply with the purpose of the Act.
- 216 *JF Investments v Queenstown Lakes District Council* EnvC Christchurch C48/2006, 27 April 2006 at [42]; *Royal Forest and Bird Protection Society of New Zealand Inc v Buller District Council* (No 2) [2013] NZHC 1346, [2013] NZRMA 293 at [54], [72], [122].
- 217 See *Report of the Biodiversity Collaborative Group* (Biodiversity (Land and Freshwater) Stakeholder Trust, October 2018); see also Fleur Maseyk and others *Biodiversity Offsetting under the Resource Management Act* (Local Government New Zealand, September 2018)
- 218 See Tim Denne “Resource management law reform and economics” in Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Working Paper 3* (Environmental Defence Society, Auckland, September 2018). See also Joan Martínez-Alier and Ronald Muradian “Taking stock: the keystones of ecological economics” in Joan Martínez-Alier and Ronald Muradian (eds) *Handbook of Ecological Economics* (Edward Elgar, Cheltenham, 2015) 1
- 219 James Good and others “Emerging Details Of The New Biodiversity Net Gain Regime” (24 January 2022) Mondaq <www.mondaq.com/uk/>
- 220 See Marie A Brown *Last Line of Defence: Compliance, monitoring and enforcement of New Zealand's environmental law* (Environmental Defence Society, Auckland, 2017).
- 221 See Chapter 7.
- 222 For example, that of sparsely populated Northland is larger than that of Auckland.
- 223 Such as the forestry pressures faced in the Marlborough Sounds alongside aquaculture development.
- 224 R Ward “Three waters review” (Paper presented to the New Zealand Building Nations Conference, 2018). See also New Zealand Council for Infrastructure Development (now Infrastructure New Zealand) *Integrated governance, planning and delivery: A proposal for local government and planning law reform in New Zealand* (Productivity Commission, August 2015) at 9.

- 225 Greg Severinsen *Reform of the Resource Management System: The urban context* (Environmental Defence Society, Auckland, August 2020).
- 226 In 2013, provisions were added to the Local Government Act to encourage councils to use shared services models. A review of local government settings more broadly is being undertaken, which might see boundaries changed in the future.
- 227 For example, buying urban tracts of land to install wetlands, diverting some stormwater flows to wastewater, imposing restrictions on what can go into stormwater drains.
- 228 Martin Jenkins A “blue skies” discussion about New Zealand’s resource management system (Local Government New Zealand, December 2015) at 39.
- 229 See New Zealand Society of Local Government Managers *Rating knowhow: A guide to the Local Government (Rating) Act 2002* (Local Government New Zealand, November 2013).
- 230 This tool recognises that people benefit in that the value of their land goes up from proximity to publicly funded goods. For example, rapid transit has increased property prices near central Auckland stations by up to 20 percent (Infrastructure New Zealand correspondence, 25 October 2018). See generally Transport for London *Land value capture* (February 2017). While it is usually looked at as an infrastructure funding tool, environmental amenity can also improve property prices and the tool could be used in novel ways to protect or enhance those ecosystems.
- 231 *Report of the Biodiversity Collaborative Group* (Biodiversity (Land and Freshwater) Stakeholder Trust, October 2018) at 92.
- 232 That is part of a broader conversation about the extent to which increased obligations on councils should be met with central government funding to discharge them.
- 233 Treasury “Our Living Standards Framework” (12 April 2022) <www.treasury.govt.nz/>
- 234 Derek Nolan “Coastal” in Derek Nolan (ed) *Environmental and Resource Management Law* (6th ed, LexisNexis, Auckland, 2018) at [5.62].
- 235 At [5.62].
- 236 At [5.62].
- 237 Resource Management Act 1991, s 64A(1).
- 238 At present, Environment Southland is the only regional council to have introduced coastal occupation charges, although Marlborough District Council has recently included a coastal occupation charge in its proposed Marlborough Environment Plan which will be levied through the annual plan process.
- 239 See, for example, Tasman District Council *Draft Plan Change 56: Coastal Occupation Charges - Report on Assessment of Alternatives under Section 32 of the Resource Management Act* (February 2015).
- 240 See, for example, Gisborne District Council *Proposed Variation 15 to the Proposed Regional Coastal Environment Plan* (date unknown) at 9.
- 241 Raewyn Peart *Voices from the Sea: Managing New Zealand’s Fisheries* (Environmental Defence Society, Auckland, 2018) at 18.
- 242 Raewyn Peart *Voices from the Sea: Managing New Zealand’s Fisheries* (Environmental Defence Society, Auckland, 2018) at 18.
- 243 See Jörn Scherzer and Jim Sinner “Resource Rent: Have you paid any lately?” (Ecologic, Research Report No 8, December 2006) at 9; Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation Synthesis Report* (Environmental Defence Society, Auckland, December 2018) at 260.
- 244 Fishers used to be charged a proportion of the overall research budget based on the value of their quota holdings, until 2001 saw a change of focus to charge only those directly benefiting or causing the need for expenditure.
- 245 It relies on actual trading within market to raise funds.
- 246 Including fisheries and aquaculture, but also extending to things like the recognition of customary marine title and protected customary rights under the MACA Act.
- 247 Tim Denne “Resource management law reform and economics” in Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Working Paper 3* (Environmental Defence Society, Auckland, September 2018).
- 248 The RMA even lists the investigation of such things as a specific function of the Minister for the Environment, without providing any framework or authorisation for creating them.
- 249 New Zealand Productivity Commission *Local government funding and financing* (2019) at 160.
- 250 Tim Denne “Resource management law reform and economics” in Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Working Paper 3* (Environmental Defence Society, Auckland, September 2018). See also Lin Roberts and others *The nature of wellbeing: How nature’s ecosystem services contribute to the wellbeing of New Zealand and New Zealanders* (Department of Conservation, March 2015).
- 251 For example, the Marine Stewardship Council’s Fisheries Standard.
- 252 Raewyn Peart *Voices from the Sea: Managing New Zealand’s Fisheries* (Environmental Defence Society, Auckland, 2018) at 107.
- 253 Jared Diamond *Collapse: How Societies Choose to Fail or Survive* (2nd Edition, Penguin Books, London, 2011).
- 254 Daniel Kalderimis and Nicola Swan *Sustainable Finance Forum Legal Opinion 2019* (Chapman Tripp, October 2019).

9 Spatial protections in the toolkit



Bay of Islands

9.1 Introduction

Not all tools are inherently spatial. For example, many measures applied to achieve sustainable fisheries (eg gear restrictions) and prevent marine pollution (eg wastewater standards and ship design requirements) have no real spatial element. However, many regulatory tools can apply (or apply differently) to a particular geographical area within Aotearoa New Zealand's waters. For example, regional coastal plans apply to a whole region, but can impose different rules, policies and objectives in particular zones. The same can be achieved under the EEZ Act (although it has not really been done in practice). Conservation frameworks provide for the establishment of wildlife and marine mammal sanctuaries, fisheries are generally managed according to spatially defined fisheries management areas, and spatial protections are provided for submarine cables and pipelines. Shipping lanes and AMAs are also geographically fixed.

However, the concept of a “marine protected area” is more than just a blanket term for any tool having a spatial dimension. It is focused on the benefits that restrictions or controls in a particular place could have for marine *biodiversity*. Conversations about MPAs generally focus on the extent to which we have the right tools in the regulatory toolbox to achieve that.

The system can have many tools that are expressed spatially. However, it is possible to think about a distinct category of spatial tool called an MPA, which focuses on protecting an area in ways that will improve marine biodiversity.

Defining an MPA

It is not always clear what an MPA is. Its definition is broad and vague¹ and the term is used to mean different things across the world.² Some consider MPAs to be any area that has a degree of protection higher than its surroundings.³ By that logic we already have many tools capable of creating protected areas. A more ambitious definition of an MPA from the IUCN is “a clearly defined geographical space, recognised, dedicated and managed, through legal or other effective means, to achieving the long-term conservation of nature with associated ecosystem services and cultural values”⁴ and “only those sites where the main goal or outcome is conserving nature should be considered MPAs”.⁵ That is quite a different approach.

Aotearoa New Zealand's non-statutory MPA Policy (2005) defines an MPA as “an area of the marine environment especially dedicated to, or achieving, through adequate protection, the maintenance and/or recovery of biological diversity at the habitat and ecosystem level in a healthy functioning state”.⁶ This definition echoes the conception of MPAs found in

the Convention on Biological Diversity. The Aotearoa New Zealand policy definition does not require the main *goal* of a tool to be conservation so long as the *effects* meet the New Zealand MPA Protection Standard.⁷

To meet this standard, a management tool must enable a healthy functioning state to be achieved through maintenance or recovery.⁸ Maintenance and recovery include “where feasible, the avoidance of change from human induced pollution, sedimentation, fishing, tourism or visitor-based disturbance, undersea or seafloor commercial activities, or scientific/research activities”.⁹ The tool must provide for this at the levels of:¹⁰

- a) physical features and biogenic structures that support biodiversity;
- b) ecological systems, natural species composition (including all life-history stages), and trophic linkages; and
- c) potential for biodiversity to adapt and recover in response to perturbation.

Significantly, MPAs may be subject to some level of extractive use so long as the standard is otherwise met.¹¹ The MPA Policy also recognises two categories: type 1 and type 2 MPAs. Permanent no-take marine reserves, where no disturbance is allowed, are established under the Marine Reserves Act and are classified as type 1 MPAs. Type 2 MPAs are other protected areas established outside of the Marine Reserves Act, which provide enough protection from the adverse effects of fishing to meet the MPA Protection Standard.¹² Type 2 MPAs can include mātaimai reserves and undersea cable and pipeline protection areas, if – and only if – site assessment shows that such areas meet the Standard. Many management tools have the effect (either intended or incidental) of protecting marine habitats and ecosystems.¹³ The division between type 1 and type 2 MPAs bears no relationship to any internationally recognised MPA classification system.¹⁴ The tools set out in Figure 9.1 arguably amount to MPAs in Aotearoa New Zealand.¹⁵



Motu Manawa-Pollen Island marine reserve

Raewyn Peart

Management tool	Instrument	Restrictions	Application	Responsible Minister/ Department
Marine reserves	Marine Reserves Act 1971	Generally no-take areas – all extractive activities are prohibited. A broad range of activities can be managed, controlled or excluded	Territorial sea	Department of Conservation; concurrence of Minister of Fisheries required
Marine mammal sanctuaries	Marine Mammals Protection Act 1978	A range of restrictions necessary to protect marine mammals, depending on the marine mammal sanctuary	Territorial sea and EEZ	Department of Conservation
Marine parks/ specially protected areas	Fisheries (Amateur Fishing) Regulations 2013; Special legislation – eg Hauraki Gulf Marine Park Act 2000 and 2001 amendment; Sugar Loaf Islands Marine Protected Area Act 1991	A range of restrictions depending on the marine park	Territorial sea	Department of Conservation; bespoke management bodies
Submarine cables and pipeline protection zones	Submarine Cables & Pipelines Protection Act 1996; Submarine Cables & Pipelines Protection Order 2009	No fishing or anchoring except for ships being used for research by or for the Ministry for Primary Industries, as long as the research is undertaken without directly or indirectly attaching a vessel to the seabed	Territorial sea and EEZ	Ministry of Transport
Mātaihai reserves	Fisheries Act 1996, s 186; Declaration of Mātaihai Reserve and Appointment of Tāngata Kaitiaki/tiaki Notice	In general, commercial fishing is prohibited, and recreational/customary fishing is regulated by the tāngata kaitiaki/tiaki	Territorial sea	Minister of Fisheries; local iwi/hapū
Taiāpure	Fisheries Act 1996, Part 9; Fisheries Order	Special management region for areas which have special customary significance to iwi or hapū as a source of food	Estuarine or coastal areas	Minister of Fisheries; management committee nominated by local Māori community can make recommendations
Section 186 temporary closures	Fisheries Act 1996 (Temporary Closure) Notice	A range of restrictions depending on the particular area, such as the closure of fishing areas, restrictions on fishing methods, or the closure of fisheries for up to two years. Designed for customary use. Must be supported by tangata whenua	Territorial sea	Minister of Fisheries
Benthic protection areas	Fisheries Act 1996; Fisheries Act (Benthic Protection Areas Regulations 2007)	Prohibits the use of dredge and restricts the use of trawl nets within 100 metres of the seabed	EEZ	Minister of Fisheries

Management tool	Instrument	Restrictions	Application	Responsible Minister/ Department
Seamount closures	Fisheries Act 1996; Commercial Fishing Regulations	Prohibits seabed and midwater trawling on some specified seamounts	EEZ	Minister of Fisheries
Areas closed to specific fishing activity	Fisheries regulations	For example, numerous spatial trawl, set net, and dredging closures. Spatial seasonal closures to protect various nursery and spawning grounds	Territorial sea and EEZ	Minister of Fisheries
Regional coastal plans	RMA	May be used to prohibit certain activities in identified areas	Territorial sea	Regional councils
New Zealand Coastal Policy Statement	RMA	May be used to prohibit or control certain activities in a dedicated area	Territorial sea	Minister of Conservation
Regulations	EEZ Act	May be used to prohibit certain activities in a dedicated area	EEZ	Minister for the Environment

Figure 9.1: Tools which arguably amount to MPAs in Aotearoa New Zealand

MPAs can be defined in quite different ways. Spatial protection tools in the current system are, under existing policy, regarded as MPAs if they meet a protection standard. Their primary purpose does not have to be the maintenance or restoration of biodiversity.

But why does it matter if something is called an “MPA” if it has no legislative meaning in the domestic context? Some may question whether the blanket term itself is useful, and whether focusing on the design of more specific tools is a better approach. Indeed, it can be something of a red herring if MPAs are treated as the only tools to be deployed.

However, defining an MPA can be a useful way to gauge the effectiveness of marine biodiversity protection, given the widespread acceptance that spatial protections of some kind are needed to achieve meaningful progress. In other words, if we define it well, we can set targets and measure progress more easily.

Equally, if MPAs are defined too loosely, then progress can become misleading. A wide definition allows countries to meet international targets by counting protection measures that are less restrictive; they can rush to meet quantitative targets, neglecting the quality of marine protection in

favour of spatial coverage. Having a common understanding of what an MPA does and does not include can also be helpful in determining what is missing from the current toolkit and what might be needed to fill gaps.

The extent to which the current toolkit for MPAs is adequate depends on many things: what we are wanting to protect, for what purpose, how and by whom they are established, by when they must be achieved, and where they are located. Many have argued convincingly that Aotearoa New Zealand’s current toolbox for spatially protecting the marine area falls short in a number of ways. For example:

- There is a lack of a legal mechanism to create marine reserves or other highly protected areas beyond the boundaries of the coastal marine area.¹⁶
- The purpose of establishing a marine reserve is very limited, focused on scientific research and not biological diversity or species conservation.
- Type 1 MPAs are limited to marine reserves, but these do not allow any take for cultural purposes and can be insensitive to the expectations of mana whenua.



Cape Rodney-Okakari Point marine reserve

- The locations of benthic protection areas, established under the Fisheries Act, are arguably not in the best locations to achieve ecosystem protections (given that they only protect against the impacts of trawling and dredging fishing methods, and most are too deep to trawl/dredge).¹⁷
- Many sensitive ecosystems remain largely unprotected, such as seamounts and reefs.

Perhaps most fundamentally, overall coverage of MPAs is limited, and protected areas that are in place are not representative or connected. At the seventh meeting of the Parties to the Convention on Biological Diversity in 2004, targets were set for “comprehensive, effectively managed, and ecologically representative national and regional systems of protected area”. This means that MPAs should cover a representative sample of the ecosystems and species present in a country’s oceans. In terms of international expectations, we are well short of that.

A spotlight on MPA coverage

Aotearoa New Zealand’s marine environment is vast, stretching from the subtropical waters off Rangitāhua/Kermadec Islands to the subantarctic waters surrounding Motu Ihupuku/Campbell Island. Yet only 0.4 percent of this is protected in no-take marine reserves.¹⁸ As a whole, 9.8 percent of the territorial sea is protected in no-take reserves, but zero percent of our vast EEZ has this level of protection.¹⁹

About 28 percent of the EEZ and territorial sea combined is protected under a “variety of other protective measures”.²⁰ Protections from fishing impacts on the benthic marine environment cover 27.4 percent of the area, and seamount protection from trawl impacts covers 2.6 percent. Marine mammal sanctuaries – spatial conservation measures to manage risks to marine mammals – cover 0.7 percent. Fishing management tools, classified domestically as meeting a protection standard for an MPA, require at minimum a prohibition on bottom trawling, seining and dredging. So although a total of over 30 percent of New Zealand’s ocean was under some form of protection, as of April 2018, most of these areas did not meet MPA policy protection standards.²¹

The distribution of marine protection is also uneven across Aotearoa New Zealand’s 14 coastal marine biogeographic regions, with 96.5 percent of marine reserve coverage located around offshore islands in the northern and southern extremes of the territorial sea (the Rangitāhua/Kermadec Islands and Subantarctic Islands).²² The remaining 3.5 percent of marine reserves and other “marine protection measures” in the mainland territorial sea are poorly spread across biogeographic regions. Consequently, our current coastal marine protection network does not protect a fully representative range of habitats, let alone in a connected way.²³

Some are of the view that single-species protections and bans on damaging fishing gear, as is possible under current tools, “will not lead to the long-term conservation of the whole ecosystem and therefore do not qualify as MPAs”.²⁴ It is hard to find marine zoning tools in regional coastal plans that achieve this, and the EEZ Act does not really engage in spatial management at all (yet).

However, it may be possible to make better use of the existing toolkit. Just because tools have not been used in the past does not necessarily mean we have exhausted their potential. It may be necessary to make some tweaks to the legal framework, but it is possible to argue that overhaul is not required.

There are some reasonably obvious deficiencies in the current toolkit when it comes to MPAs. However, existing tools could be used more effectively than they have been in the past.

9.2 MPAs under the RMA

The RMA (or future NBA) offers one potential pathway for creating MPAs. Indeed, creating MPAs through a regional coastal plan, as was achieved around Motiti Island, has been described as innovative.²⁵

A spotlight on Motiti Island and MPAs

After the oil spill disaster caused by the grounding of the MV Rena, local hapū found that the temporary exclusion zone (based on navigational safety under the Maritime Transport Act and a Bay of Plenty Regional Council bylaw) had contributed significantly to biodiversity recovery.²⁶ The Motiti Rohe Moana Trust sought protection of the area on a longer-term basis, but its application for a closure under the Fisheries Act was unsuccessful.²⁷

After several years of legal action, the Trust was successful in its appeal of the Bay of Plenty Regional Coastal Environment Plan.²⁸ The Court directed the Council to “implement new rules within its Regional Coastal Environment Plan to protect three reef systems near Motiti and complete scientific monitoring, in collaboration with tangata whenua and multiple agencies, to inform future integrated marine management solutions for the wider Motiti Natural Environment Management Area”.²⁹ The new rules prohibit all forms of fishing from several areas, creating a tool that was previously only the preserve of marine reserves and fisheries controls.

This *Motiti* decision signals a “new phase” in the relationship between councils and central government agencies in the marine space.³⁰ It is not inconceivable that regional coastal plans may organically become a more widely used mechanism for establishing MPAs in the future, now that their potential to do so has been highlighted. The question is whether

councils will take up that opportunity. To some extent this will depend on capacity issues, as well as political will. To increase take-up, councils could be supported by central government to identify and protect biodiverse areas of their marine environment in the future. Another question is how jurisprudence will evolve; as explored in Chapter 11, there is still a great deal of uncertainty about the extent to which a network of MPAs (or other habitat protections) could be achieved under the RMA.

However, the case of Motiti highlights another potential mechanism by which *central* government could drive the use of the RMA for spatial protections. Aotearoa New Zealand’s non-statutory MPA policy recognises that a revised NZCPS could provide, not only more specific policy guidance on managing the effects of sedimentation, discharging, and dumping *on* an MPA network, but also recognise “the types of values at the national, regional and local level that would merit some form of marine protection”.³¹ And, given the broad powers of Ministers to create policy and regulation under the RMA (which may be even broader under the NBA), it is not inconceivable that regional or even place-specific national direction could be developed for particular networks or even individual MPAs.

Case law has confirmed that a regional policy statement can contain “rule-like” policies and include specific lines on maps, and there is no reason to think a regionally specific NPS would be any different. While planning mechanisms and jurisprudence is underdeveloped compared to the RMA, the EEZ Act could conceptually be used in the same kind of way to establish spatial protections that restrict fishing, mining and other activities in vulnerable places like seamounts.

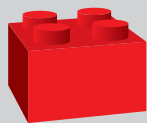
Establishing MPAs under these existing frameworks would have the benefit of maintaining a single planning process through which other “zones” (eg for ports and aquaculture) could be established alongside protected areas and be reflected in a single regional coastal plan. It could also be used to integrate consideration of land uses³² and catchments with the establishment and protection of MPAs (eg to ensure upstream activities do not harm biodiversity within protected areas). As broad, effects-based frameworks, these acts have the advantage of being able to apply to fishing, mining, pollution and other seabed disturbances.

However, the RMA and EEZ Act provide a relatively fragile legislative environment in which to create MPAs. For one, the purposes of the statutes are broad when compared to the kind of “conservation” focused purposes needed for area-based protections. Regulatory restrictions are not permanent, and can be undone through politically driven changes in national direction or changes to regional coastal plans. On land, it is

telling that conservation areas are established and managed by dedicated conservation legislation; we certainly do not rely on the RMA to create and manage national parks and reserves.³³

That said, the proposed NBA provides an opportunity to strengthen the legislative foundations for MPAs. For example, the Act could conceivably have a new part, with a dedicated protective purpose (about intrinsic value and preservation of biodiversity), for the establishment of protected areas in the coastal marine area. That would not be dissimilar to the “system within a system” for water conservation orders, which are not subject to the purpose of the RMA.³⁴

Alternatively, the NBA could include the concept of an MPA as one expression of an “environmental limit” necessary to defend minimum levels of marine biodiversity (see Chapter 8). Those limits are already proposed to have a separate, more protective, purpose than other regulatory restrictions under the Act. The same approach could be reflected in amendments to the EEZ Act, which could have a third purpose added to its existing ones.



The RMA (and NBA) as well as the EEZ Act provide opportunities for the more proactive deployment of MPAs in the future at both national and regional levels.



Te Matuku marine reserve

9.3 MPAs under the Fisheries Act

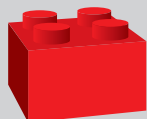
The existing Fisheries Act could also be used to proactively create more protected areas. That is possible already, but might be better supported if its purpose and principles were expanded. An amended purpose might, for example, see the location of benthic protected areas revisited, trawl corridors identified, an ongoing plan to spatially protect seamounts, and the establishment of recreational fishing parks (to the extent those could be regarded as MPAs). However, it would still lack the RMA/NBA's ability to control a wide range of activities (eg mining) and therefore represents a less coordinated approach.

That said, it might be possible to create protected areas by using existing Fisheries Act *tools* in novel ways through separate, bespoke legislation. The Kermadec islands is a good example, which could be replicated in other geographically specific legislation (which may, for example, come out of the South East Marine Protection Forum and implementation of the Seachange Tai Timu Tai Pari initiative in the Hauraki Gulf).³⁵ As described in Chapter 7, the Kermadec proposal involved setting the TAC to zero for an entire fisheries management area. This effectively removed the ability to exercise commercial fishing rights in the future, but did not formally extinguish the quota itself. That has all sorts of te Tiriti and natural justice issues associated with it (see Chapter 6 on existing rights and Chapter 7 on procedural justice). But for present purposes it begs a different question: could a future system use fisheries tools like the TAC to effectively create no-take MPAs? It may be possible if a TAC could be set for a more specific area *within* a QMA, or if a new, smaller scale QMA were to be created within a larger one. That would be more complicated than in the Kermadec situation, where the prospective boundaries of the sanctuary coincided with an entire existing QMA.

The use of the TAC in the service of an MPA raises other, more nuanced, possibilities. For example, should the law allow a lower (but above zero) commercial catch limit to be set for reasons broader than managing the fish stock itself (ie biodiversity protection)? Here, rights would not be extinguished, only reduced. Another possibility might be that regional councils under the RMA – if they were equipped to do so – could be empowered to deploy their own catch limits in the service of broader biodiversity outcomes. However, this would likely create tricky issues of overlap with the Fisheries Act catch limit setting functions.

The process of applying the TAC as a tool to create an MPA, in cases such as the Kermadec Islands, has arguably emerged because we lack a more comprehensive framework for establishing protected areas in the EEZ. It is

easier for bespoke legislation to piggyback on tools under other legislation, for which compensation would not be payable, rather than create a totally new regime. It raises the issue of whether it is appropriate for a tool developed for one purpose (to maximise the sustainable yield of fish) to be used for another (to create a protected area for broader, value-based reasons including intrinsic values and meeting international commitments). A better approach may be to create more modern MPA legislation directly.



The Fisheries Act could be used to deploy MPAs more systematically in the future, which might be supported by strengthening or clarifying its purpose and sustainability principles. As well as using spatial fisheries closures, the TAC itself could be adjusted to provide spatial biodiversity protection.

9.4 Dedicated MPA legislation

The Kermadec Ocean Sanctuary proposal provides an example of a bespoke, area-based statute for the creation of MPA, partly reflecting the inadequacies of the Marine Reserves Act. Similarly bespoke legislation has been floated for the Hauraki Gulf to implement more nuanced MPAs coming out of the Seachange Tai Timu Tai Pari spatial planning process, and to implement MPAs off the South East Coast of the South Island/te Wai Pounamu. Future reforms could focus on making the tools in the Marine Reserves Act more fit for purpose, rather than bypassing it in favour of bespoke legislation every time a new MPA needs to be created.



Whanganui A Hei (Cathedral Cove) marine reserve

A spotlight on marine reserves

The Marine Reserves Act 1971 emerged off the back of lobbying by the scientific community after a University of Auckland professor asked the government to establish a marine reserve adjacent to the Leigh campus and was told there was no legal mechanism to establish it.³⁶ The Act's purpose indicates this history, with the purpose and mechanisms for marine reserves reflecting scientific and research interests. The Act provides for "the setting up and management of areas of the sea and foreshore as marine reserves for the purpose of preserving them in their natural state as the habitat of marine life for scientific study."

The first marine reserve was established in 1975.³⁷ At the time, the legislation was pioneering, and the marine reserve was potentially a world first.³⁸ But that was close to 50 years ago, and the legislation is now outdated. The protection of biodiversity is not mentioned in the legislation, nor are cultural and recreational uses. As a result, it has "not been the regulatory tool of choice" for creating MPAs.³⁹ That is putting it politely. Even when it comes to creating new *marine reserves* – the tool at the heart of the Act – special legislation has often been preferred.⁴⁰

Under the Act, a marine reserve can only be established if the marine area contains underwater scenery, natural features or marine life of such distinctive quality, or so typical, or beautiful, or unique that its continued preservation is in the national interest. The decision-making criteria are limited to whether establishment is in the "best interests of scientific research" and "for the benefit of the public".⁴¹ This is clearly deficient as a normative foundation for MPAs. It protects small areas that are beautiful and healthy if they are considered to have scientific value, but it does nothing to stop the broader decline of biodiversity in the marine environment.

This framework does not reflect contemporary values or problems, including the desire to protect the representativeness of marine ecosystems around the country.⁴² There are no environmental criteria in the Act. Somewhat astonishingly, the protocol to guide decision-making between Fisheries New Zealand and the Department of Conservation states: "[w]hether or not the marine area that is the subject of an application is under threat or at risk of

destruction is not a consideration relevant to s 3 of the Act”.⁴³

The framework of the legislation prioritises existing rights and uses, as marine reserves cannot “interfere unduly” with a range of interests including commercial fishing and recreational uses. The tension between the purpose, culture, and values of fisheries and conservation bodies has contributed to stalled progress on marine reserves and generated pushback on marine protection generally.

The Marine Reserves Act could be amended to address the specific issues highlighted in the spotlight above. Alternatively, it could be fundamentally reimagined with a new name, purpose, and toolkit. Indeed, there have been attempts to do so. The latest was in 2012, when the National-led government undertook a review of MPA legislation. It decided to withdraw the stalled Marine Reserves Bill and introduce a new one which would align with the recently enacted MACA Act and the (then) proposed EEZ Act.⁴⁴ The government consulted on a proposed Marine Protected Areas Act in 2016, but a Bill was never introduced – presumably stalled by the same government’s Kermadec Islands controversy.

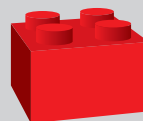
The consultation document indicated that the proposed Act would have four protection categories: marine reserves, species-specific sanctuaries, seabed reserves and recreational fishing parks.⁴⁵ Unlike an earlier proposal, the Act would apply only to the territorial sea.⁴⁶ It also introduced a new decision-making framework. Proposals would only be considered by the leading Minister if they contained adequate information about the environment, the benefits of protection, and the economic impacts on current and future uses.⁴⁷ From there, the Minister(s) would decide whether to proceed with a collaborative community and stakeholder or Board of Inquiry approach.⁴⁸ Both approaches were to require full public consultation and an independent economic assessment.⁴⁹

The proposed legislation also envisioned that other tools, such as customary or recreational fishing tools under the Fisheries Act, might form part of a proposal as a “suite of protection and management measures in an area”.⁵⁰ Regional coastal plans under the RMA would have to recognise MPAs and take them into account when determining consents.⁵¹ There would be a mechanism to review marine protected areas, which in exceptional circumstances could lead to them being revoked.⁵²

While such an approach could be resurrected in a future system, further changes could be made. The most obvious is that a new suite of MPAs could be made available in the EEZ, where no meaningful framework for

cross-sector spatial protections is available, let alone proactively used. The bar for provisionally accepting a proposal for some MPAs could be set lower (eg where environmental indicators showed decline), at which point the onus could be placed on anyone opposing the measure to demonstrate why it is *inappropriate*.⁵³

The legal “reach” of MPAs could also be extended by strictly requiring that activities operating beyond its boundaries not have an appreciable impact on the values being protected within it. That could, for example, create a strong legal obligation for councils to prevent activities on land or catchments (eg clear fell harvesting) that generate sedimentation impacts, in a similar way that the Department of Conservation can prohibit the flying of drones over national parks from private land.⁵⁴ That might lead to existing use rights on land being extinguished or the review of consent conditions. In fact, an MPA Act could even create *commercial* fishing protection zones, in which undue adverse effects of land-based activities on habitats of value to fisheries could be controlled.



The Marine Reserves Act could be reimagined in a future system as an MPA Act, which could go further than previous proposals (including by applying MPAs to the EEZ, broadening its purpose, and triggering land use change under the RMA).



Long Bay-Okura marine reserve

Raewyn Peart

9.5 Design features of MPAs

Choices will need to be made as to exactly what activities are to be restricted in MPAs, and what things are to be protected. Most of the focus of debate tends to be on extractive activities (ie fishing and mining), and “no-take” MPAs. The IUCN also states that industrial activities and infrastructural developments are not compatible with MPAs and should be excluded. But the values to be protected may be broader than just those threatened by these things.

For example, the 2005 MPA Policy does not directly address protection of marine historic or cultural heritage, or protection for non-extractive use (eg diving) or values, tourism or recreational opportunities. Such issues, it was envisioned, would be considered “following the development of the Oceans Policy”.⁵⁵ That never eventuated.

Do we need bespoke MPAs for the active protection of marine heritage, wāhi tapu and culturally significant sites? What should we be protecting those things *from*? And will customary marine title areas already provide strong enough links to the RMA (ie through veto rights on coastal permits) to prevent inappropriate activities on such things in practice? Arguably not, if recreational fishing and gathering shellfish remain permitted activities and beyond the scope of the RMA. A wider range of MPAs may be necessary.

And do we require MPAs for green infrastructure like offshore wind farms or tidal generation, similar to the protection of submarine cables? Or, indeed, for ecological infrastructure provided by, for example, commercial seaweed or shellfish farming operations? These might be characterised as exclusion zones, but still contemplated by an MPA Act given their potential benefits for biodiversity.

A wide range of protected area types could be provided for. For example, legislation in California envisages marine reserves,⁵⁶ marine conservation areas,⁵⁷ marine parks⁵⁸ marine recreational management areas⁵⁹ and special closure areas⁶⁰ with each having a different focus and restrictions.⁶¹ In Canada, “marine refuges” have recently been provided for in federal fisheries legislation. These are long-term fisheries closures to protect sensitive ecosystems and fish populations.⁶²

The process by which MPAs are developed is also important to consider. Previously, EDS listed the pros and cons of government versus stakeholder-led MPA development, and we made a number of recommendations for MPA reform, many of which still have relevance

today.⁶³ On the one hand, a collaborative process is important in order to build consensus and move beyond adversarial positions on marine management. Where consensus cannot be reached an independent board of inquiry could evaluate competing interests against policy goals in order to make a recommendation to the ultimate decision-maker (eg the responsible Minister). Limiting a board of inquiry’s terms of reference strengthens collaborative processes and incentivises participants to achieve agreement. Such terms of reference should, however, be focused on achieving the biodiversity goals of MPAs so that the process does not simply become one of negotiation between the interests that happen to be present at the table (or who are selected to be there). The role of iwi/hapū would need to be carefully considered in any process (see below).

Alternatively, proposals could be considered by the Environment Court as an independent entity, which would make recommendations to the Minister based on legally rigorous assessment of criteria in the legislation. That approach could be similar to the current process for considering water conservation order applications, and could guard against the process being politicised.

Because all of this takes time, consideration could be given to conferring interim protection to sites identified as “potential” protected areas. Recent changes to Canada’s Oceans Act have allowed for this to happen via ministerial order, and for the footprint of current human activities in such areas to be frozen. Proposed MPAs have to be fully designated within five years (or else repealed).⁶⁴ This “two stage” process is based on a strengthened precautionary principle in the Act, in that a lack of scientific certainty is not good reason to delay or fail to designate an MPA.

It will be important to consider various design features of MPAs. Those include the things being protected and the process by which they are created and managed.



A more comprehensive set of MPAs could include spatial protections for heritage, wāhi tapu areas, recreational sites and green infrastructure. The process for creation could be made more collaborative and/or independent, with interim protection conferred.

9.6 MPAs and mana whenua

A new approach to MPAs needs to be cognisant of the Tiriti obligations, and the need to reconcile two core purposes of area-based protections that are usually talked about: protection of biodiversity and indigenous interests. These things are generally compatible, but much depends on specific design features.

A Western approach to spatial protections – strict separation of “conservation” areas from “exploitable” areas – has been one reason for the failure of MPAs. There can be tensions between highly protective MPAs (such as no-take reserves) and the exercise of customary rights, which is how Māori maintain connections with te moana.

Part of this comes down to the process by which MPAs have been created in the past. Involvement of Māori has been inconsistent at best. The Kermadecs controversy demonstrates that legally, albeit not politically, MPAs may be developed with arguably insufficient consultation with all relevant interests. That said, some MPAs have been created with strong support of mana whenua (eg the Whanganui-A-Hei (Cathedral Cove) Marine Reserve).

The government’s non-statutory MPA Policy was also developed on the heels of the Foreshore and Seabed Act 2004 (now repealed). Perhaps reflecting this, a principle of the MPA Policy was that planning should reflect “the need to *take into account* obligations that arise from Treaty of Waitangi commitments to tangata whenua that are included in marine management legislation and Treaty settlement legislation” – rather than *give effect to* (or “comply” with) them.⁶⁵ Tangata whenua were to be “involved early” and “effective participation” would be provided but they would have no right of veto over MPA proposals and no formal role in their management.⁶⁶

More to the point, none of this has statutory force. There is no legal process – and therefore no visible or predictable process – by which MPAs are created other than the deficient and outdated process for creating marine reserves. Ad hoc and quite different mechanisms have been used in the Hauraki Gulf, the South East Marine Protection Forum and Rangitāhua/Kermadec Islands.⁶⁷ Other ones might be used in the future, raising the prospect of the same kinds of legal battles that have defined the Rangitāhua experience.

While mana whenua have, to some extent, been marginalised in the MPA *development* and *design* process over the years, they may have

considerable powers to prevent their deployment in coming years. If customary marine title is established under the MACA Act, the title group may give or decline permission and this determines whether an application for a marine reserve can proceed.⁶⁸ However it remains that, when setting sustainability measures under the Fisheries Act, the Minister must only “have regard” to any planning document that may have been lodged by the marine title group. The upshot is that customary marine title allows marine reserves to be stopped as of right, but it does not allow fishing activity to be halted. This may uphold settlement rights in quota, but it does not necessarily reflect the broader interests of Māori in protecting customary marine title areas *from* the impacts of fishing.

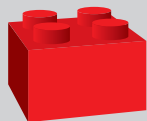
So what does that mean for the design of MPAs in a future system? Because many applications for customary marine title are yet to be decided, it is too early to say what implications may evolve in practice. Only three customary marine title applications have been heard at the time of writing,⁶⁹ although those awaiting recognition have the potential to cover large swathes of Aotearoa New Zealand’s coastal marine area.⁷⁰

On the one hand, this might facilitate iwi/hapū-created and managed MPAs if permission (ie veto) rights under the RMA are coordinated with efforts to establish local mātaihai reserves or temporary closures. To support that outcome, one option could be that customary marine title holders are given comparable influence over fisheries sustainability measures and planning as they have over plans and consents under the RMA. At its most extreme, title could even allow holders to require permission to engage in commercial fishing in their areas (including conditions relating to gear and methods). Some MPAs authorised by title holders might even be temporary (eg with a moratorium on activities), better reflecting the idea of a rāhui.

Alternatively, title holders could be given *less* influence (ie no veto right) over decisions to create MPAs that are less restrictive than marine reserves, as long as mana whenua are involved in the design of these more nuanced areas through a collaborative or co-development process, and have a hand in their management. Such areas could allow some recreational take, reflecting the fact that many Māori rely on recreational catch to provide food for the marae. The purpose of those areas might even be quite different from that of the Fisheries Act, and result in different restrictions being placed through (for example) bag limits, temporary closures (by rāhui) or gear requirements. The Seachange Tai Timu Tai Pari process in the Hauraki Gulf has suggested the creation of new “Ahu Moana” areas that could be developed to restrict activities without severing ancestral connections to the sea and its resources.⁷¹

Finally, we note that if the RMA were to be used as a mechanism for controlling the impacts of fishing (as highlighted in *Motiti*) there may be a “back door” way for the MACA Act to be used by customary marine title holders to do the same. For example, if a resource consent was required to undertake commercial fishing (eg in reef environments where fishing were a discretionary rather than a prohibited activity), then a title holder could exercise a permission right to refuse it. Management plans could also be used to influence the fishing-related provisions of a regional coastal plan, which in turn could be used to establish no-take protected areas. While indirect, this could be a way to progress more nuanced and culturally sensitive MPAs in the coastal marine area. However, that would not be possible in the EEZ or in parts of the coastal marine area which are not subject to customary marine title.

MPAs will need to be developed, designed and deployed in a way that is sensitive to the interests of mana whenua and obligations under te Tiriti o Waitangi.



The recognition of customary marine title might provide a mechanism by which title holders could themselves deploy MPAs, and that roundabout mechanism could be strengthened.



Wainui, Gisborne

A spotlight on Australian indigenous protected areas

Currently, marine reserves established under the Marine Reserves Act are no-take zones which do not allow for customary fishing. While marine protection options that do allow for customary fishing can be facilitated through fisheries management tools, it is worth considering whether more can be done outside of fisheries management to facilitate marine protection that is sensitive to, or is dictated by, the cultural needs of iwi. This is especially so given government’s acknowledgement that mātaihai and taiāpure cannot be established for the purpose of biodiversity protection – this outcome is merely an ancillary effect of sustainable utilisation.⁷² An option for the future is to create more nuanced MPAs that weave indigenous connections together with the preservation of biodiversity.

Australia has advanced the concept of “indigenous protected areas” (IPAs). IPAs are “areas of land and sea managed by Indigenous groups as protected areas for biodiversity conservation through voluntary agreements with the Australian Government”.⁷³ As of 2019, there were nine IPAs with marine components, and 25 with saltwater elements such as beaches, estuaries and islands.⁷⁴ Seventeen projects are in the consultation stage, but when complete they will cover over 2.6 million hectares of sea country.⁷⁵

IPAs are non-legislated, policy-based protected areas.⁷⁶ To create an IPA, an indigenous community must declare this status and commit to managing their land to maintain biological diversity along IUCN guidelines.⁷⁷ The concept is based on indigenous understandings of territory rather than “ownership” per se (not conceptually too far from the idea of customary marine title in Aotearoa New Zealand).⁷⁸ Funding discussions and management planning follow the declaration of an IPA.⁷⁹ IPAs may be funded by government or nongovernmental entities such as environmental or philanthropic organisations.⁸⁰ The IPA is formally recognised by government after the completion of the management plan.⁸¹ The plan sets out “the Traditional Owners’ priorities for their Country, the values, goals, and objectives of the IPA, and identifies relevant partners, as well as their shared commitment to achieving those goals and objectives”.⁸²

Implementation for Sea Country IPAs typically occurs in partnership with government fisheries and other relevant agencies, research institutions, nongovernment conservation organisations, commercial and recreational fisheries representatives, tourism operators and others.⁸³ Most IPAs operate with an associated indigenous rangers programme to facilitate on-the-ground management by Indigenous people.⁸⁴ These management activities may include research and management of marine species, removal of abandoned fishing gear and other marine debris, constructing visitor infrastructure, supporting Traditional Owners to visit sea country, habitat mapping, cultural site maintenance, fisheries compliance patrols, biosecurity checks and other activities in collaboration with partner agencies.⁸⁵

IPAs in Australia represent a move away from a “wilderness” model of conservation to one that recognises the enduring cultural, social and economic links indigenous people have with the land and waters.⁸⁶ IPA management is also shifting away from a co-management or joint management approach and towards an indigenous-governed approach.⁸⁷ Furthermore, it signals the

growing acceptance of plural approaches – when an IPA overlaps with an existing MPA, “the IPA becomes a complementary, collaborative governance mechanism that does not threaten the [MPA’s] management”.⁸⁸

There are several lessons for MPA management in Aotearoa New Zealand. IPAs appear to be similar to iwi management plans under the RMA and potentially planning documents under the MACA Act.⁸⁹ The limitation of the latter is that it can be difficult for groups to establish marine title. It may be worth following Australia’s approach of moving away from the requirement of formal marine title (not relying on the MACA Act) to allowing a more autonomous definition according to tikanga that can be implemented through a new category of MPA.

IPAs also appear to be more integrative than iwi management plans in that they apply broadly to activities in an area, including fishing. Lastly, it is an example of how two separate but complementary systems – indigenous and state – can operate to achieve marine protection. This conceptualisation could be helpful for thinking about the differences between involving Māori in traditional government-led MPA processes versus such MPA processes operating alongside autonomous Māori-led MPA processes.



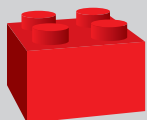
Bruny Island, Tasmania

Raewyn Peart

9.10 Concluding comments

Overall, a more nuanced set of MPAs may be needed in a future system to encompass more things than they do at present. A key question is the extent to which strict no-take reserves are still necessary at the core of such networks to achieve biodiversity protection goals, or whether the mass deployment of more culturally sensitive spatial protections could do as good a job. In California, for example, legislation requires that no-take reserves are implemented within a broader network of MPAs.⁹⁰

Yet while there are a number of options when it comes to MPAs in a future system, some of which we have canvassed above, there is also an opportunity to fundamentally reimagine what an MPA is. Instead of being just one tool in the toolkit to protect biodiversity through a set of area-specific regulations, an ecocentric approach may treat such areas as “persons” in their own right, or at least as the domain or property of a person. Such persons could be given rights to defend (just as a human has rights when it comes to his or her bodily autonomy) rather than being protected by rules. We discussed this concept of legal personhood earlier (giving rights to nature), and note that particular areas could be given that status via MPA legislation.



MPAs could be made their own legal persons, reflecting a rights for nature approach.

MPAs are generally thought to be a fixed marine use, like coastal aquaculture or ports, as opposed to a mobile one, like shipping, tourism and recreation. However, there is an emerging idea of mobile and dynamic MPAs which may shift, for example, as the effects of climate change are realised.⁹¹ It is interesting to consider whether a marine reserve – which we are used to thinking of as protecting a particular place – could instead abandon that place in favour of another where greater overall biodiversity protections could be achieved. What are we wanting to protect? It may be a mix of place-based values in some situations and overall contribution to biodiversity values in others.⁹²

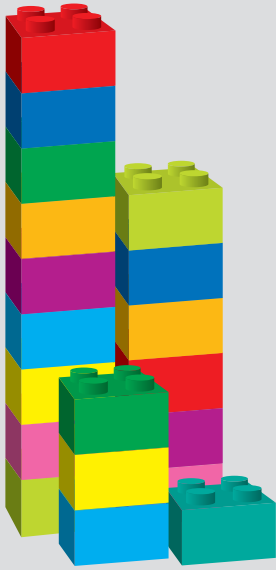


In the future, a process for shifting some MPAs from one place to another (based on the values being protected rather than the space) could be provided for, recognising that climate and environmental change may demand greater agility.

This highlights the importance of situating MPAs within a long-term and multi-sector plan for the marine space, and thinking about them both strategically and in an integrated way. That is not just the case with MPAs; indeed, as we explore in the following chapter, the toolkit as a whole may benefit from greater strategy and integration. The location and design of MPAs need to be thought about alongside many other activities and protections.



Scuba divers, Cape Rodney-Okakari Point marine reserve



Summary of options for reform: Spatial protections in the toolkit

- The RMA (and NBA) as well as the EEZ Act provide opportunities for the more proactive deployment of MPAs in the future at both central and regional levels.
- The Fisheries Act could be used to deploy MPAs more systematically in the future, which might be supported by strengthening or clarifying its purpose and sustainability principles. As well as using spatial fisheries closures, the TAC itself could be adjusted to provide spatial biodiversity protection.
- The Marine Reserves Act could be reimagined in a future system as an MPA Act, which could go further than previous proposals (including by applying MPAs to the EEZ, broadening its purpose, and triggering land use change under the RMA).
- A more comprehensive set of MPAs could include spatial protections for heritage, wāhi tapu areas, recreational sites and green infrastructure. The process for creation could be made more collaborative and/or independent, with interim protection conferred.
- The recognition of customary marine title might provide a mechanism by which title holders could themselves deploy MPAs, and that roundabout mechanism could be strengthened.
- MPAs could be made their own legal persons, reflecting a rights for nature approach.
- In the future, a process for shifting some MPAs from one place to another (based on the values being protected rather than the space) could be provided for, recognising that climate and environmental change may demand greater agility.



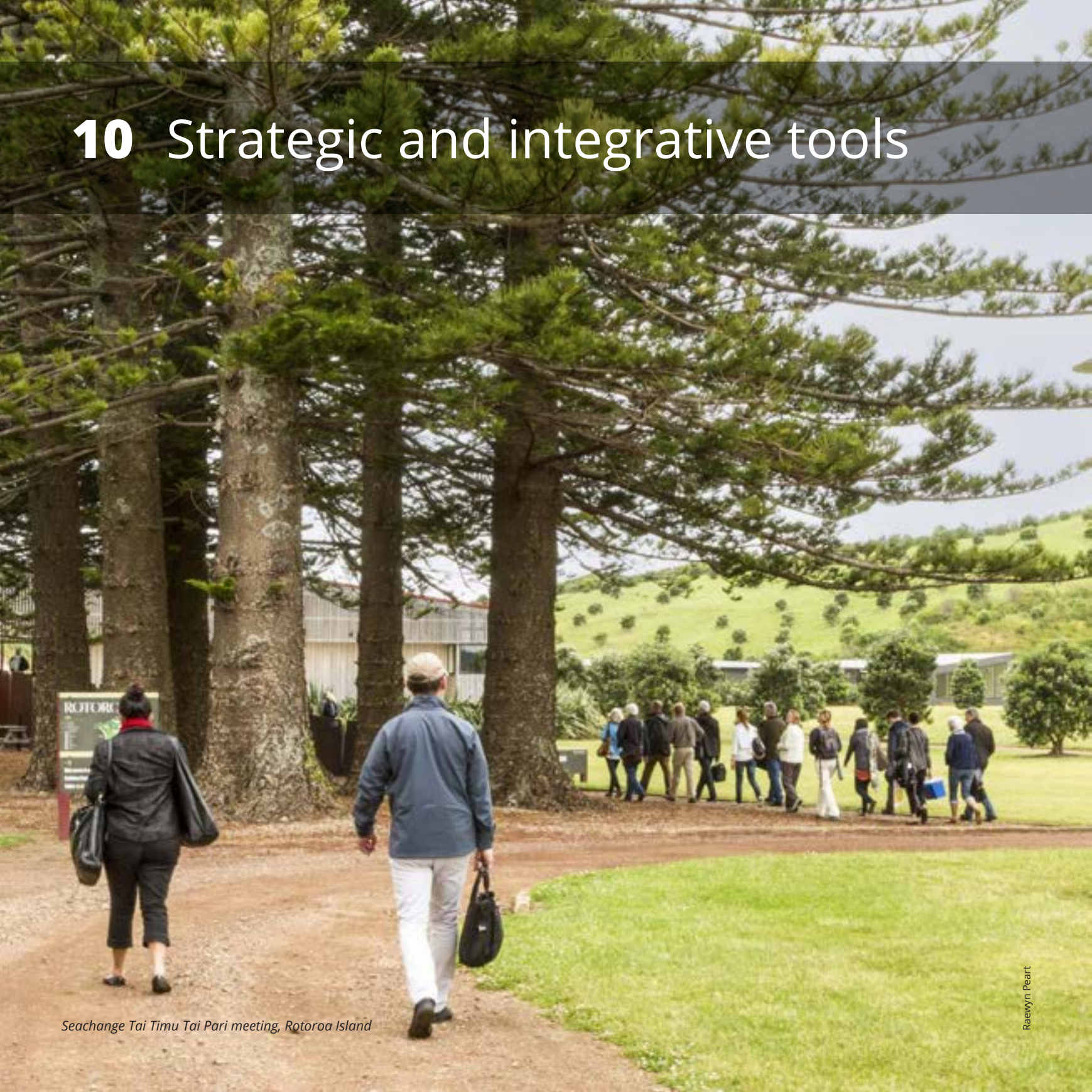
Tawharanui marine reserve

Endnotes

- 1 Bill Ballantine "Fifty years on: lessons from marine reserves in New Zealand and principles for a worldwide network" (2014) 176 Biological Conservation 297 at 298.
- 2 Bill Ballantine "Fifty years on: lessons from marine reserves in New Zealand and principles for a worldwide network" (2014) 176 Biological Conservation 297 at 298.
- 3 Under the framework of the Convention on Biological Diversity, the term "marine and coastal protected area" is used to refer to: Any defined area within, or adjacent, to the marine environment, together with its overlying water and associated flora, fauna, historical and cultural features, which has been reserved by legislation or other effective means, including custom, with the effect that its marine and/or coastal biodiversity enjoys a higher level of protection than its surroundings. See *Report of the Ad Hoc Technical Expert Group on Marine and Coastal Protected Areas* (UN Doc UNEP/CBD/SBSTTA/8/INF/7) (13 February 2003) at [30].
- 4 Jon Day and others *Guidelines for applying the IUCN protected area management categories to marine protected areas* (IUCN, 2019) at 2. This is very similar to the MPA definition provided in the international guidance on marine spatial planning provided in the report: Intergovernmental Oceanographic Commission and Directorate General for Fisheries and Maritime Affairs *MSPglobal International Guide on Marine/Maritime Spatial Planning* (IOC-UNESCO, IOC/2021/MG/89, 2021).
- 5 Jon Day and others *Guidelines for applying the IUCN protected area management categories to marine protected areas* (IUCN, 2019) at 8.
- 6 Department of Conservation and Ministry of Fisheries *Marine Protected Areas Policy and Implementation Plan* (Department of Conservation and Ministry of Fisheries, Wellington, 2005) at [14], [MPA Policy 2005].
- 7 See Department of Conservation "Type 2 Marine Protected Areas" <www.doc.govt.nz>; Department of Conservation and Ministry of Fisheries *Marine Protected Areas: Classification, Protection Standard and Implementation Guidelines* (Department of Conservation and Ministry of Fisheries, Wellington, 2008).
- 8 MPA Policy 2005 at [83].
- 9 At [84].
- 10 At [83].
- 11 At [85].
- 12 See Department of Conservation "Type 2 Marine Protected Areas" <www.doc.govt.nz>
- 13 See MPA Policy 2005 at [26] - [61]. The IUCN protected area framework assigns protected areas to one of seven categories depending on the levels of protection from extractive activities that the area offers. While the IUCN categories are useful for comparing protected areas between countries, its protection standards are often applied inconsistently and minimum internationally-agreed standards of protection are lacking. While Aotearoa New Zealand is a member of the IUCN, and the Department of Conservation acts as the State party representative and has developed a national version of the IUCN Threatened Species Classification Scheme, the IUCN categories are not reflected in legal measures for national MPA policy.
- 14 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening marine protection in New Zealand* (EDS, Auckland, 2012) at 122.
- 15 Adapted from Ministry for the Environment Ministry for the Environment "Marine Areas with Legal Protection: Environment Report Card" (December 2008) <www.environment.govt.nz> <https://environment.govt.nz/assets/Publications/Files/Environmental-Report-Card-Marine-Areas-with-Legal-protection_0.pdf>; Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening marine protection in New Zealand* (EDS, Auckland, 2012); and Office of the Prime Minister's Chief Science Advisor *The Future of Commercial Fishing in Aotearoa New Zealand* (February 2021) at 113.
- 16 Marine Reserves Act 1971, s 2: the Act only applies to the territorial and internal waters of Aotearoa New Zealand (not the EEZ).
- 17 Office of the Prime Minister's Chief Science Advisor *The future of commercial fishing in Aotearoa New Zealand* (February 2021) at 91-92.
- 18 Department of Conservation *New Zealand's latest National Report under the Convention on Biological Diversity: Reporting period 2014-2018* (Department of Conservation, Wellington, 2019) at 117. These no-take marine reserves meet the strictest definition of IUCN categories, see further below.
- 19 At 117.
- 20 At 117.
- 21 Alberto Rovellini and Megan Shaffer *Quantitative targets for marine protection: a review of the scientific basis and applications* (Department of Conservation, Project 4792, June 2020) at 42.
- 22 Department of Conservation *New Zealand's latest National Report under the Convention on Biological Diversity: Reporting period 2014-2018* (Department of Conservation, Wellington, 2019) at 117.
- 23 At 117.
- 24 At 8.
- 25 Office of the Prime Minister's Chief Science Advisor *The Future of Commercial Fishing in Aotearoa New Zealand* (February 2021) at 134.
- 26 At 130.
- 27 At 131.
- 28 At 131.
- 29 At 132.
- 30 At 132.
- 31 See MPA Policy 2005 at 6 and 9.
- 32 Land uses are regulated primarily through district plans, but regional councils have some jurisdiction over land use for environmental purposes and can include broader land use policies (some of which can resemble "rules") within regional policy statements.
- 33 We have a dedicated National Parks Act and Reserves Act.
- 34 Resource Management Act 1991, pt 12.
- 35 See Department of Conservation "Consultation on south-eastern South Island marine protected areas" <<https://www.doc.govt.nz/get-involved/have-your-say/all-consultations/2020-consultations/consultation-on-south-eastern-south-island-marine-protected-areas/>>
- 36 Bill Ballantine "Fifty years on: lessons from marine reserves in New Zealand and principles for a worldwide network" (2014) 176 Biological Conservation 297 at 298; and Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening marine protection in New Zealand* (EDS, Auckland, 2012) at 92.
- 37 Bill Ballantine "Fifty years on: lessons from marine reserves in New Zealand and principles for a worldwide network" (2014) 176 Biological Conservation 297.
- 38 At 298.
- 39 Office of the Prime Minister's Chief Science Advisor *The Future of Commercial Fishing in Aotearoa New Zealand* (February 2021) at 114.
- 40 For example, in Kaikōura and Fiordland.
- 41 Marine Reserves Act 1971, s 5(9).
- 42 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening marine protection in New Zealand* (EDS, Auckland, 2012) at 98.
- 43 Hugh Logan and Warwick Tuck *Marine Reserves Protocol* (Department of Conservation and Ministry of Fisheries, August 2003)
- 44 Marine Reserves Bill 2012 (224-1) (select committee report).
- 45 Ministry for the Environment *A new Marine Protected Areas Act: Consultation Document* (ME 1224, January 2016)
- 46 At 16.
- 47 At 22.
- 48 At 22-23.
- 49 At 22-23.
- 50 At 24.
- 51 At 24.
- 52 At 25.
- 53 The tool could even provide for a level of compensation for lost "rights" (or a significant reduction in value), such as where a no-take MPA effectively extinguished or appropriated fishing quota across a significant portion of a QMA.
- 54 We tend not to have that same problem for the conservation estate on land, as most of this is higher up catchments. Most lowland parts of Aotearoa New Zealand have been used for agriculture, and this does not impact on eg national parks (exception of pests like possums).
- 55 See MPA Policy 2005 at [22]. Development of the Oceans Policy was discontinued.
- 56 Where it is unlawful to injure, damage, take, or possess any living geological, or cultural marine resource, except under a permit or specific authorisation from the managing agency for research, restoration, or monitoring purposes. The purpose is the protection of all marine resources.
- 57 Where it is it is unlawful to injure, damage, take, or possess any living, geological, or cultural marine resource for commercial or recreational purposes, or a combination of commercial

- and recreational purposes, that the designating entity or managing agency determines would compromise protection of the species of interest, natural community, habitat, or geological features.
- 58 Where it is it is unlawful to injure, damage, take, or possess any living or non-living marine resource for commercial exploitation purposes.
- 59 Where it is unlawful to perform any activity that, as determined by the designating entity or managing agency, would compromise the recreational values for which the area may be designated.
- 60 An area designated by the Fish and Game Commission that prohibits access or restricts boating activities in waters adjacent to seabird rookeries or marine mammal haul-out sites
- 61 See California Department of Fish and Wildlife "California Marine Protected Areas (MPAs)" <<https://wildlife.ca.gov/Conservation/Marine/MPAs>>
- 62 See Stephanie Hewson "Ocean Law Developments in Canada 2015-2019" (29 August 2019) West Coast Environmental Law <https://www.wcel.org/sites/default/files/publications/ocean_law_roundup_final5.pdf>
- 63 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening marine protection in New Zealand* (EDS, Auckland, 2012) at 125 – 130.
- 64 See Department of Fisheries, Oceans, and the Canadian Coast Guard "Changes to the Oceans Act" (1 August 2019) Government of Canada <<https://www.dfo-mpo.gc.ca/oceans/act-loi/index-eng.html>>
- 65 MPA Policy 2005 at [86]–[87].
- 66 MPA Policy 2005 at [116].
- 67 See Department of Conservation "Consultation on south-eastern South Island marine protected areas" <<https://www.doc.govt.nz>>; and Ministry for the Environment "Kermadec Ocean Sanctuary Bill" (4 April 2021) <www.environment.govt.nz>
- 68 Marine and Coastal Area (Takutai Moana) Act 2011, s 71.
- 69 Chapman Tripp "Tone set for whānau customary marine title claims" (13 October 2021) <<https://chapmantripp.com/trends-insights/tone-set-for-whanau-customary-marine-title-claims/>>
- 70 Under the MACA Act, applications for a High Court order for recognition of protected customary rights and customary marine title had to be made by 3 April 2017. There are 205 active marine and coastal recognition applications spanning an expansive cumulative area. See Courts of New Zealand "Marine and Coastal Area (Takutai Moana) Act 2011 applications for recognition orders" <<https://www.courtsofnz.govt.nz/the-courts/high-court/high-court-lists/marine-and-coastal-area-takutai-moana-act-2011-applications-for-recognition-orders/>>
- 71 Seachange Stakeholder Working Group *Sea Change Tai Timu Tai Pari Hauraki Gulf Marine Spatial Plan* (Hauraki Gulf Forum in partnership with others, April 2017) at 4.
- 72 MPA Policy 2005 at [50].
- 73 Department of Agriculture, Water and the Environment "Indigenous Protected Areas" (19 October 2021) <<https://www.awe.gov.au/agriculture-land/land/indigenous-protected-areas>>
- 74 Phil Rist and others "Indigenous protected areas in Sea Country: Indigenous-driven collaborative marine protected areas in Australia" (2019) 29(S2) Aquatic Conserv: Mar Freshw Ecosyst 138 at 141-142.
- 75 Department of Agriculture, Water and the Environment "Indigenous Protected Areas" (19 October 2021) <<https://www.awe.gov.au/agriculture-land/land/indigenous-protected-areas>>
- 76 Phil Rist and others "Indigenous protected areas in Sea Country: Indigenous-driven collaborative marine protected areas in Australia" (2019) 29(S2) Aquatic Conserv: Mar Freshw Ecosyst 138 at 138.
- 77 Lee Godden and Stuart Cowell "Conservation planning and Indigenous governance in Australia's Indigenous Protected Areas" (2016) 24(5) Restoration Ecology 692 at 695.
- 78 Phil Rist and others "Indigenous protected areas in Sea Country: Indigenous-driven collaborative marine protected areas in Australia" (2019) 29(S2) Aquatic Conserv: Mar Freshw Ecosyst 138 at 147.
- 79 Lee Godden and Stuart Cowell "Conservation planning and Indigenous governance in Australia's Indigenous Protected Areas" (2016) 24(5) Restoration Ecology 692 at 695.
- 80 Phil Rist and others "Indigenous protected areas in Sea Country: Indigenous-driven collaborative marine protected areas in Australia" (2019) 29(S2) Aquatic Conserv: Mar Freshw Ecosyst 138 at 142.
- 81 At 142.
- 82 At 142.
- 83 At 142.
- 84 National Indigenous Australians Agency "Indigenous Protected Areas (IPAs)" <<https://www.niaa.gov.au/indigenous-affairs/environment/indigenous-protected-areas-ipas>>.
- 85 Phil Rist and others "Indigenous protected areas in Sea Country: Indigenous-driven collaborative marine protected areas in Australia" (2019) 29(S2) Aquatic Conserv: Mar Freshw Ecosyst 138 at 145.
- 86 Lee Godden and Stuart Cowell "Conservation planning and Indigenous governance in Australia's Indigenous Protected Areas" (2016) 24(5) Restoration Ecology 692 at 692.
- 87 At 693.
- 88 Phil Rist and others "Indigenous protected areas in Sea Country: Indigenous-driven collaborative marine protected areas in Australia" (2019) 29(S2) Aquatic Conserv: Mar Freshw Ecosyst 138 at 143.
- 89 Marine and Coastal Area (Takutai Moana) Act 2011, s 85.
- 90 Review adapted from Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening marine protection in New Zealand* (EDS, Auckland, 2012) at 182.
- 91 See Intergovernmental Oceanographic Commission and Directorate General for Fisheries and Maritime Affairs *MSPglobal International Guide on Marine/Maritime Spatial Planning* (IOC-UNESCO, IOC/2021/MG/89, 2021).
- 92 Conceptually this is not so different from the idea of tenure review on land, where some pastoral lease land was put into private ownership and (in theory) land with higher conservation value shifts into the public conservation estate.

10 Strategic and integrative tools



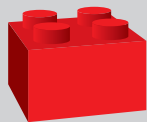
Seachange Tai Timu Tai Pari meeting, Rotorua Island

10.1 Introduction

In Chapters 8 and 9 we looked at various tools that could be used differently in a future system and new ones that could be added. But a future system will need to be concerned not just with making the right tools available, but also with how (and whether) they are actually deployed. Two related questions arise here, which we explore in this chapter. First, how might we make the toolkit more *strategic*, so it is focused on driving change for the future rather than just managing the present? And, secondly, how might we ensure that multiple tools are used in an *integrated* or *coordinated* way to ensure they work well together?

10.2 Injecting strategy into the system

There are a number of ways in which the existing toolkit might be made more strategic or future focused, reflecting a new system's potential role of pursuing positive outcomes. As described in Chapter 7, the purpose of the legislation under which tools are deployed matters. For example, the intention under the proposed NBA is for its purpose to be much more focused on achieving positive outcomes and not just mitigating adverse effects, reflected in a long list of outcomes that authorities are directed to promote.¹ If these are expressly tied to the tools required to achieve them (eg green taxes, biobanking, public funding, subsidies, behavioural incentives and so forth), and not just a list of aspirations, then that could change the entire orientation of the legislation from passive management to pursuit of change. Indeed, it would be possible for quite specific objectives concerning the marine environment (eg MPA coverage) to be codified in the purpose of primary legislation (see Chapter 7). These could be complemented by specific duties placed on public authorities, more closely resembling the objective-driven tools under the Climate Change Response Act.



A future system could be made more strategic, by recasting the purposes and principles of legislation to ones that drive towards a different future, rather than maintaining or protecting things or seeking static outcomes (eg wellbeing or sustainability).

New tools could also make the system more strategic. For instance, accountability mechanisms could be deployed to encourage or require legislated outcomes to be achieved. The progress of authorities towards attaining marine objectives could, for example, be subject to a scorecard issued by an independent authority like an Oceans Commission (see

Chapter 12), to which authorities like central or local government would be required to respond.²

The use of statutory “targets” is another option. Instead of statutes themselves setting specific objectives, they can provide a framework under which mandatory targets must be set, measured and pursued by public authorities. Admittedly, some formal targets already exist in the current system. However, they are for the most part “static” ones. For example, maintaining a biomass of fish likely to achieve MSY seeks to maintain a relatively stable status quo. Where they *are* more oriented at change, targets have been largely left to political, rather than legal, accountability mechanisms. For example, the latest generation of national direction under the RMA (eg the NPS for Freshwater Management and proposed NPS for Indigenous Biodiversity) contain measurable targets for improvement, but these could always be undone by Ministers (with no legal consequence). Tellingly for the marine environment, the NZCPS is an older generation instrument that does not seek specific change, and its language is largely that of protection and avoiding effects. It could have a facelift to be more future focused like the NPS for Freshwater Management.

The system could also deploy the concept of targets in a much more structured and systemic way. Indeed, mandatory targets formed a core part of the Randerson Panel's recommendations for a new NBA (although they seem to have disappeared from the exposure draft subsequently produced by the government).³



Koheru, Mokohinau Islands

Tanya Peart

A spotlight on targets under the RMA and NBA

The Randerson Panel has suggested that the NBA contain mandatory targets for a number of things, including the restoration of ecosystems and “viable populations of indigenous species”, within the coastal marine environment and elsewhere. Such targets could be built upon further, as a tool to pursue quite specific marine objectives through the NBA, something which the RMA has not been used for in the past. That could include (for example) a percentage target for MPA coverage, population targets for threatened species,⁴ or even catch limits to achieve “viable” populations for “indigenous” fish above MSY (although much would depend on how the interface with the Fisheries Act was phrased and what legal influence targets had). Mandatory targets could be used to directly implement some of the specific future focused marine objectives of *Te mana o te taiao – the Aotearoa New Zealand biodiversity strategy 2020*.⁵ They could be timebound, and there could be an accountability framework around how they are set (eg review by an independent commission or standing board of inquiry) and what happens if they are not met (eg an automatic review or even transfer of some powers).

It may be particularly important for environmental limits (see Chapter 8) to be treated as binding targets if a limit has already been infringed. This transforms the nature of what a limit is – from a strict regulatory standard or prohibition (to prevent things getting worse) to a set of proactive tools (eg regulation, funding, active management and incentives) that will return an indicator to a safe space within a certain timeframe. That requires a plan of action. Restoring degraded habitats (eg kina barrens) and threatened species populations can be thought of in this way, but so too can collapsed fish stocks.

Instead of consulting on multiple options for rebuilding stocks, a future system could have a more structured, directive and precautionary approach to reopening a closed fishery involving pre-set targets. As mentioned in Chapter 4, the government has proposed to introduce features to the Fisheries Act allowing automatic changes to TACs and TACCs. This could be accompanied by strong independent assessments to identify where trigger points for more precautionary targets are set.

Targets could also apply to the rollout of various tools, not just the achievement of biophysical outcomes. For example, targets could be set for the deployment of marine spatial plans (discussed later in this chapter), for the transfer of powers to mana whenua (eg under section 33 of the RMA), or for stepping-stones towards the deployment of a network of MPAs. The latter could reflect international approaches to measuring progress as set out in *The MPA Guide* (2021), which establishes a process to systematically document the global state of MPAs according to their stage of establishment and level of protection.⁶ One problem with marine reserves is that they have simply not been deployed in a way or to an extent that would achieve systemic biodiversity benefits across te moana. Instead, they are a tool that sits waiting to be used when there is sufficient will to do so. In contrast, Canada’s Oceans Act 1996 provides that:⁷

the Minister shall lead and coordinate the development and implementation of a national network of marine protected areas on behalf of the Government of Canada.

In performing his or her duties and functions ... the Minister shall ensure that:

- (a) clearly identified objectives are set with regard to each marine protected area; and
- (b) the network of marine protected areas covers diverse habitat types, biogeographic regions and environmental conditions.

Statutory targets could even be made mandatory across all statutes that manage things impacting te moana. They could be inserted into the Waste Minimisation Act (stepping-stones towards phasing out all single use plastics that impact the marine environment), the Fisheries Act (targets for improvements to fishing gear or the rollout of cameras on boats), the Maritime Transport Act (targets for decarbonising shipping) and the Wildlife Act (targets for rebuilding indigenous species populations and the rollout of tools to achieve them). In Canada, for instance, the federal government established a timeline for listing marine species under the Species at Risk Act 2002 (for which orders are made to protect critical habitat).⁸ The Minister must do so within 36 months of being given a species status assessment conducted by the Committee on the Status of Endangered Wildlife in Canada. The idea is to achieve greater accountability for doing something that is now seen to be beyond the realm of just politics.

The extent to which targets should be *enforceable*, with legal rather than just political consequences flowing from a failure to meet them, is debatable.

A failure to meet targets by some institutions could conceivably result in some kind of penalty or action, such as a transfer of powers from regional councils to the EPA if biodiversity targets were not met, or a transfer of wastewater services from councils to an independent entity if infrastructure failures were not remedied in a timely fashion. An alternative would be for a mandatory review or audit of policy or regulations to be triggered.



Mandatory targets could be used more systemically across a future system to drive positive change. Accountability mechanisms could be established around them to measure progress. Binding targets could cover many things, but may be particularly useful in returning to a safe ecological space if environmental limits have already been infringed.

Existing tools could also be made more strategic by planning ahead for how they might change over time. They would be more agile and responsive. Change might be designed to happen as certain trigger points are reached. Spatial tools (eg MPAs and AMAs) might, for example, move to more appropriate pre-planned locations based on changing environmental impacts or climate change (and this could, for example, address issues that have been faced with inshore aquaculture operations in the Marlborough Sounds – see Chapter 3).⁹

It could also apply to the use of other kinds of tools. For example, instead of responding to a problem by instigating a lengthy regional plan change process or review, a different “version” of the plan could be pre-prepared, waiting in the wings. If a pre-defined trigger point is reached (whether it be water quality, species mortality, or some proxy for ecosystem health) the planning environment could switch over automatically to one that is more protective and oriented towards restoration. Everyone would know what the trigger point was and what would happen when it was reached.¹⁰ Conversely, when indicators improved, a framework could automatically become a more enabling one. Such temporal layers of planning could prevent the delays that come about from what is currently a highly reactive system that is slow to respond to environmental threats.

Triggers could also require immediate and corrective action under conservation legislation. At present, when a species is allocated to a higher threat category upon review under the (non-statutory) New Zealand Threat Classification System (administered by the Department of Conservation), there is no statutory requirement to take action (eg the development of

a population recovery plan). Where national environmental reporting datasets identify an emerging or growing risk to the environment, there is no statutory compulsion for regulatory agencies to respond or even indicate their intended response. A future system could have points at which thresholds (eg a change to threatened status of marine mammals or seabirds) automatically triggers a graduated set of corrective actions (eg the release of funding or regulatory restrictions). It could be an “alert level system” for the marine environment.

This is not a radical proposition. Pre-planning has, for example, been embraced in the urban context, where certainty around the rules that will apply to the future release of land allows developers to hit the ground running when rezoning occurs. Some have even suggested that new planning rules could enter into force automatically if objectively measured indicators were triggered (eg a price differential between urban and rural land).

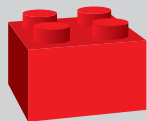


A future system could establish a more comprehensive range of trigger points that result in automatic or immediate management measures being taken. Here, the system would be more proactive in preparing for the future, by providing for greater agility when things change.

A future focused system also needs to be constantly scanning ahead to identify new challenges and opportunities that will require a response. Arguably the current system does not do this well. For example, there is a high degree of risk for an applicant wishing to undertake novel activities that the law is ill-equipped to regulate, such as deep sea mining (where there is next to no policy guidance under the EEZ Act), offshore fish farming (which lacks a meaningful policy framework even in the coastal marine area) or marine carbon capture and storage (where there are deeper questions about which legal frameworks even apply). As technology advances, the system tends to lag behind and react to innovations rather than pre-empt them. It means we may lose benefits that come from undertaking such activities (if they are declined consent) and incur unnecessary costs in determining whether proposals can or cannot proceed.

To address this, a “futures scanning” role could be given to an independent Oceans Commission (see Chapter 12) or added to the statutory functions of relevant government departments. In short, there needs to be stronger anticipatory governance – “a real, nationally-focused effort at looking ahead”.¹¹ This could resemble the Welsh approach, where the government is obliged to release a Future Trends report within one year of a general

election.¹² An exercise in looking ahead could be closely linked to monitoring and reporting, not just for environmental indicators but also identify how economic and social factors are changing in the marine space, and how past trends might play out in the future.¹³



Monitoring and reporting in a future system could be linked to obligations to conduct futures scanning exercises, to ensure that problems, opportunities and changes are pre-empted rather than leaving gaps in policy and regulatory frameworks to develop.

10.3 Towards a more integrated toolkit

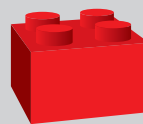
If tools are to be used in a strategic way to drive change, then they need to be well coordinated with each other. That is particularly the case if they are fragmented across different statutory frameworks or deployed by different institutions, as is the case now (and can be seen vividly in the case of the Hauraki Gulf). It is not enough to have a strategy for marine reserves if there is a separate and conflicting aquaculture or mining strategy; there is little point preventing fishing if a stock is going to collapse anyway due to land-based pressures like sediment; and even the best regulation in the world may fail to protect threatened species from extinction if there is no funding for people on the ground to restore habitats, eradicate pests, or enforce restrictions. This encourages us to think about a “toolkit” rather than just individual “tools”. An integrated toolkit could form the “glue” that holds the building blocks of the system together.

There are many potential options for how tools could be better integrated. One mechanism could be institutional. Here, a single institution could be charged with wielding multiple tools in the service of a clear and directive mandate. For example, one might assume that, if a more comprehensive suite of different MPAs was available to it, the Department of Conservation would be better placed to align these with other measures (eg coastal reserves, offshore islands, population management plans and funding for restoration). And if capacity and funding issues were addressed, regional councils might be able to integrate MPAs and fishing controls (and catchment activities) to achieve biodiversity goals through regional policy statements and plans. That said, putting all one’s management eggs in the same institutional basket can have risks, especially if the responsible entity is defunded and has to make hard choices, or its attention is diverted elsewhere. For instance, most regional councils put relatively little resource into marine management, and have, for the most part, not used habitat protection powers available to them.¹⁴

A spotlight on United States legislation

The United States Magnuson-Stevens Fishery Conservation and Management Act (1976)¹⁵ seeks to address impacts on fisheries by providing a formal linkage between fisheries management and other activities. Where another federal agency seeks to authorise, fund or carry out an action that might adversely impact on an essential fish habitat, it must consult with the National Marine Fisheries Service (the federal fisheries management agency). The agency is then required to provide formal recommendations, including on measures to avoid, minimise or offset the impact of the proposed activities. It is also proactive in restoring essential fish habitats.

This cross-cutting mechanism could be replicated in Aotearoa New Zealand, in that Fisheries New Zealand could be required to engage deeply in land use and coastal planning under the NBA. Indeed, the avalanche of plan reviews that will be necessary in the wake of the RMA being replaced by a new act provides an opportunity to do so. We note that the pendulum may already be swinging back towards more regional level engagement,¹⁶ in that Fisheries New Zealand has recently established a small coastal planning team to work with regional councils. That could be a good start towards a more formal or structured role in coastal planning.



Tools could be better coordinated in a future system by extending the responsibilities of institutions. If one institution has responsibilities for deploying (or engaging with) multiple tools, then they may be used in a more integrated way.

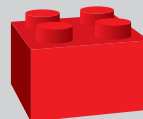
Another way forward could be stronger legislative cross-referencing. This could be used to make boundaries between statutes clearer; for example, the EEZ Act and Maritime Transport Act specifically explain how and why each statute deals with particular elements of marine pollution. The same could be done to clarify the respective jurisdictions of central government and regional councils for managing marine biodiversity under the RMA and Fisheries Act. At present, those boundaries remain uncertain (see Chapter 11).

Furthermore, cross-references could be made between MPA legislation and the Fisheries Act, specifying principles for when a reduction in value in fishing rights (quota) through protected areas is justified or warrants compensation. At the moment, such questions are left to political discretion. Cross-references could also be made between the Fisheries Act and the MACA Act, linking concepts like *taiāpure* and *mātaihai* to the exercise of protected customary rights and customary marine title. A clearer legal relationship could also be established between emissions reduction plans under the Climate Change Response Act and MPA legislation as well as sectoral acts for fishing and mining. That could require tools under the latter legislation to give effect to the former, addressing big picture issues like the emissions implications of bottom trawling, the protection of benthic habitats as carbon sinks, and the long-term impacts of oil and gas exploration. Better cross-referencing could also see the timing of different instruments aligned (eg the development and review of spatially focused fisheries plans¹⁷ at the same time as the marine and catchment components of regional plans, relevant parts of an EEZ policy statement, the development of iwi management plans, and processes for the establishment of MPAs).

Going even further, tools created under one framework could be used to connect to decision-making under others. For example, the relevant parts of the NZCPS could be deemed to be an EEZ policy statement, or (if expanded in scope) be required to be given effect to through fisheries plans or decisions on sustainability measures under the Fisheries Act. The NZCPS could even outline a national strategy for the deployment of MPAs, to be implemented through other legislation (eg an MPA Act). This kind of approach has been experimented with on land (albeit not without problems), where tools created under RMA national direction (future development strategies under the NPS on Urban Development) are intended to influence infrastructure funding decisions under others (eg the Local Government Act).¹⁸

Greater normative alignment could also potentially be achieved either by amending (or creating) purpose and principles clauses for legislation

(eg to insert common principles like ecosystem-based management and environmental limits), or by creating guidance as to how they are intended to be used in a synergistic way. The purpose of something like the Fisheries Act might even be revised to bring it in line with the sustainable management purpose of the RMA or the new purpose of *te ora* o *te taiao* in the NBA. That way, tools might not pull in different directions. This could be reinforced by clarifying the mandate of the institutions responsible for administering or making decisions under various statutes. The Prime Minister's Chief Science Advisor has recently recommended, for example, that a future system "define the relationships between the different legislative requirements and strategic visions across Ministries, Departments and Agencies to provide clarity to stakeholders".¹⁹



There are a number of ways that connections could be improved between legislative frameworks, such as through cross-referencing, alignment of processes and the insertion of common principles.

10.4 The use of statutory strategies

The above measures may be useful to knit the system's tools together spatially, temporally and normatively. But that approach still relies on stitching different pieces together, one by one, rather than on creating a more holistic envelope within which they can be placed. What we are really lacking in the current system are broader strategies through which the hundreds of available tools can be deployed in the service of a bigger vision. One can compare the current system to a builder who has access to a lot of equipment and materials yet lacks a blueprint for what she or he is constructing.

The lines along which the system "strategises" can therefore be significant. In other words, it is not clear which "topics" should have strategies and how focused or broad they should be. Various strategies have been produced over the years. Some, such as *Te mana o te taiao – the Aotearoa New Zealand biodiversity strategy 2020*, focus on broad domains with their scope driven by international legal obligations.²⁰ Others, such as the *Strategy for Managing the Environmental Effects of Fishing*, have been focused on a single sector and exist within the parameters of a single piece of legislation (the Fisheries Act). Some new and more focused strategies have also been recommended, such as an environmental research strategy.²¹ An MPA strategy, which seeks to coordinate and deploy multiple spatial protection tools, has a particularly long and chequered history.

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Recreational fishing on mussel farm, Coromandel

A spotlight on the strategic use of MPAs

In 2000, following the ratification of the Convention on Biological Diversity, the government released the *Biodiversity Strategy* (2000).²² Its overall aim regarding MPAs was to “protect a full range of natural marine habitats and ecosystems to effectively conserve marine biodiversity, using a range of appropriate mechanisms, including legal protection”.²³ It stressed the need for a representative network of MPAs, committed to the internationally set target of 10 percent of marine space being protected by 2010, and committed to reviewing the Marine Reserves Act.²⁴ This led to consultation on a new Marine Reserves Bill, which was introduced into Parliament in 2002, but which stalled on the back of the foreshore and seabed controversy. The Bill set out a more detailed regime for the establishment, management and enforcement of marine reserves. However, it markedly lacked consideration of Māori interests. There was no provision for tangata whenua input and customary fishing activities were to be prohibited in marine reserves in all cases.²⁵ In addition, the Bill could not really be described as strategic, as it did not provide for a plan to roll out marine reserves in a coherent network.

Although progress with the Bill was stymied, the Biodiversity Strategy prompted the development of the *Marine Protected Areas Policy and Implementation Plan* in 2005 (MPA Policy).²⁶ This reiterated the goal of having 10 percent of Aotearoa New Zealand's marine environment under “some form of protection” by 2010.²⁷ It noted the protections would contribute to maintaining genetic diversity to protect the health of the wider marine environment (providing for a wider purpose than that contained in the Marine Reserves Act).²⁸ The MPA Policy sought to coordinate the range of existing management tools by setting out a planning process whereby an “inventory” would be taken to determine where current protections were lacking. Based on the inventory, planning for offshore MPAs was to be undertaken on a national scale, and for nearshore MPAs on a regional level. MPA planning processes would “be underpinned by a commitment to minimise the adverse impacts of new MPAs on existing users of the marine environment and Treaty settlement obligations”.²⁹ Guidelines released in 2008 sought to advance the MPA Policy by establishing an implementation framework. This proposed establishing regional forums to provide reports on their respective areas, but this approach stalled after only two fora were established.³⁰

Te mana o te taiao – the Aotearoa New Zealand biodiversity strategy 2020 (the current non-statutory biodiversity strategy) has three sets of goals for MPAs. The first is that by 2025 “a protection standard for coastal and marine ecosystems will be established with implementation underway”. The second is that by 2030 “significant progress is made in establishing an effective network of marine protected areas and other protection tools”; and the third is that by 2035 (on the way to 2050 goals) “an effective network of marine protected areas and other tools, including marine and coastal ecosystems of high biodiversity value is established and is meeting the agreed protection standard”.³¹ How that is to be achieved remains unclear, and a more recent discussion paper prepared on MPA reform has yet to be publicly released.

The picture above is one in which there is little legal accountability for failing to progress MPAs at all, let alone in a coordinated way. What legal effect should be given to cross-cutting strategies (such as *Te mana o te taiao*) is therefore an important question for system design.

Very few strategies in the existing oceans management system have a statutory basis or legal influence. The link between such strategies, and the regulatory and funding tools needed to realise them, are also generally weak (with the possible exceptions of climate change and urban development).³² Strategies lack true integrative power because they do not legally influence implementation in much more detailed and contested areas of the law (including where strong rights exist, where legislative change is required or where regulations need to be altered).



Long Bay-Okura marine reserve

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This begs the question of whether such things need to be formalised and have clear relationships with the multiple tools needed to give them teeth. Indeed, there is some movement towards this kind of thinking in proposals for regional spatial strategies under a Strategic Planning Act. It remains unclear what legal weight (the extent to which they will be “binding”) these will have when it comes to regulatory and funding decision-making under the NBA and other acts. But there is a clear intention for them to have *some* legal import,³³ which recognises that tools are not just passive things to be used when the political will exists.

A similar approach could be adopted for fishing, where a strategy to manage environmental effects could, like the NZCPS, be made mandatory. It could be required to outline not just general objectives and policies concerning the environmental impacts of fishing, but also an *implementation* plan for how they will be addressed on the ground. In spirit, this might not look too different to the NPS for Freshwater Management, which has directive and legally binding provisions around implementation.

Another option would be to formalise a cross-cutting MPA strategy, linking it to statutory targets for deployment that are legally binding (or at least influential) in the same conceptual way that emissions reduction targets and budgets are binding under the Climate Change Response Act. Such a strategy could outline how MPAs are to be deployed under various acts (eg RMA/NBA, EEZ Act, Marine Reserves Act, Fisheries Act) and even include a programme for further legislative reforms (eg a new MPA Act). In this regard, Californian legislation provides an interesting example.

A spotlight on California's Marine Life Protection Act³⁴

California's approach to MPAs is an example of a “successful stakeholder-driven planning process which resulted in a network of marine protected areas spanning a large geographic area, designed in accordance with strong science, and managed as part of a state-wide system”.³⁵ It utilised a public-private partnership which brought together stakeholders, scientists and policy-makers.³⁶ The implementation of the legislation stalled twice before receiving significant funding and resourcing from this partnership.³⁷ Prior to the initiative, California's marine protection was in a similar state to Aotearoa New Zealand's: MPAs were established on a piecemeal basis, they had various levels of efficacy, and there was no coherent plan or purpose for their establishment and management.³⁸

California's Marine Life Protection Act 1999 establishes a process for the redesign of an MPA network and provides a set of goals to guide this. These include:³⁹

- the conservation of biological diversity and the health of marine ecosystems;
- recovery of wildlife populations;
- improvements to recreational and educational opportunities consistent with biodiversity conservation;
- protection of representative and unique habitats for their intrinsic value;
- ensuring that MPAs have defined objectives;
- effective management and enforcement;
- a design based on sound science; and
- ensuring that MPAs are managed, to the extent possible, as a network.

The network may include areas with various levels of protection, with some areas allowing specified forms of commercial or recreational fishing.⁴⁰ However, it must also contain no-take reserves.⁴¹

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Pateke, Aotearoa

A “master plan”, prepared by the Department of Fish and Game, was designed to guide implementation. As part of the public-private partnership, a Blue Ribbon Task Force was established to lead the development of MPA proposals.⁴² This Task Force included members “known for their integrity, intellect and experience in public policy and concern for common good”.⁴³ Development of the plan was assisted by a science advisory team and a regional stakeholder group. Stakeholder groups were responsible for developing alternative MPA proposals in each region. A science advisory sub-team worked directly with stakeholders and the Department to ensure proposals were scientifically sound, by advising stakeholders and evaluating their MPA proposals against science-based guidelines.⁴⁴

The plan was required to incorporate the best readily available scientific information, with the Department of Fish and Game obtaining the advice of specific parties, such as fisheries representatives, marine conservationists, marine scientists and other interested persons.⁴⁵ The Department was also required to confer with a host of government bodies and marine conservation personnel.⁴⁶ In addition, the legislation required the Department to consult and hold workshops alongside the development of the plan. The plan was to include a “preferred siting alternative”.

The Act’s guidelines provided that the preferred siting alternative must be designed in accordance with the following:⁴⁷

- 1) Each MPA shall have identified goals and objectives. Individual MPAs may serve varied primary purposes while collectively achieving the overall goals.
- 2) Marine reserves in each bioregion shall encompass a representative variety of marine habitat types and communities across a range of depths and environmental conditions.
- 3) Similar types of marine habitats and communities shall be replicated, to the extent possible, in more than one marine reserve in each biogeographical region.
- 4) Marine reserves shall be designed, to the extent practicable, to ensure activities that upset the natural ecological functions of the area are avoided.

- 5) The MPA network and individual MPAs shall be of adequate size, number, type of protection, and location to ensure that each protected area meets its objectives and that the network as a whole meets specified goals and guidelines.

A considerable area is now covered by some form of protected designation in Californian waters (over 16 percent),⁴⁸ and some commentators have noted signs that the initiative has been successful (eg in the form of more and larger fish and invertebrate life, especially in earlier MPAs).⁴⁹ But the extent to which it has achieved the aims of the Act is not yet evident. This may become clearer when the first “decadal review” of the MPA network is completed at the end of 2022.⁵⁰ However, it is notable that the state has taken a proactive, legally binding, approach in deploying a coherent network of MPAs according to clear objectives and principles. There are some similar design guidelines that appear in Aotearoa New Zealand’s MPA implementation plan, but these lack a statutory basis. In practice, despite some progress in recent years, MPA deployment has been slow and relatively ad hoc.

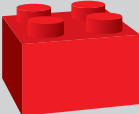
The California experience also shows the importance of complementary or supporting measures in an MPA strategy. It is not enough to establish “paper parks” on a map. Significant financial investment and active management, enforcement and education have been cited as factors that have set the Californian approach apart from others. For instance, recreational fishers catch and tag fish beyond (and, for some, within) MPAs to study the results of protection, mobile apps tell fishers where MPAs are located, and school groups survey coastal habitats.

Care needs to be taken to ensure that strategies are sufficiently focused to drive action; too broad, and there is the risk they will be worded in vague and indirect ways. But if they are too narrow, they might exclude considerations vital to their success.

Indeed, one of the reasons why MPA deployment has arguably failed so far is because spatial protections have not been considered in tandem with other measures, including those to protect, create or compensate for property rights and expectations (including in fisheries and aquaculture). The Biodiversity Strategy (2000), for example, noted that integration was needed between marine reserves and other marine protections under

the Fisheries Act, which has not been achieved. If proposals to establish MPAs fail to consider this broader picture, not only may opportunities for biodiversity enhancement be lost, but protected areas can have unintended impacts on other parts of the marine system (such as when fishing activity is displaced, increasing pressure elsewhere). A single-sector approach to spatial allocation, in the context of AMAs, ran into problems reinforcing the benefit of broader, higher-level spatial strategy.

The creation of cross-cutting strategies can be a mechanism through which different tools, potentially under multiple statutory frameworks, can be coordinated. However, most strategies in the current system lack legal influence or accountability around progress, thereby undermining their coordination function in practice.

 **Strategies in a future system could be made mandatory (guided by revised and carefully crafted statutory purposes) and have strong legal effect on the tools needed to realise their objectives.**

10.5 A strategic approach to marine information

Two recent reports have contained recommendations relevant to how Aotearoa New Zealand might better manage marine information. The first, released in December 2020 by the Parliamentary Commissioner for the Environment, Hon Simon Upton, undertakes a deep dive into the funding and prioritisation of environmental research.⁵¹ Although not directly focused on marine information per se (but on the more general category of “environmental” information), the report’s recommendations are still very relevant to the marine domain.

A key recommendation was that government (through the Ministry for the Environment) should prepare and regularly update an *environmental research strategy* to guide the funding of public good environmental research and inform the environmental reporting system. Such a research strategy could contain a specific section addressing *marine* research, which can be more technically difficult and expensive. Under the Parliamentary Commissioner’s proposals, public resources for environmental research would be ring-fenced and allocated to researchers in accordance with the research strategy. The funds would be allocated either by the Ministry of Business, Innovation and Employment (as is currently the case for much public good funding) or by a newly established *Environmental Research Council*. Such an

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Cape Rodney-Okakari Point marine reserve

Environmental Research Council would set criteria for funding allocation, develop and negotiate research platforms or programmes with long-term time horizons, and allocate a contestable funding pool for emerging issues and opportunities. In his report, the Commissioner commented that “I have reached the conclusion that environmental research is sufficiently distinct to merit its own funding allocation body with criteria that are tuned to the particular characteristics of research directed to environmental problem solving.”⁵² That could help provide a more balanced base of marine environmental research rather than the sectoral and fluctuating research that characterises the current system. How that is funded is important, and we looked at hypothecated funding tools in Chapter 8. A dedicated stream of work within a research strategy could be focused on mātauranga Māori, led by mana whenua and in which a Tikanga Commission (see Chapter 12) could have an oversight role.⁵³

A subsequent report released by the Prime Minister’s Chief Science Advisor, Professor Dame Juliet Gerrard, in February 2021 investigated the management of information related to commercial fishing specifically.⁵⁴ In the foreword to the report, Professor Gerrard described the current situation with fisheries data as follows:⁵⁵

We do have a lot of data about the ocean but in many ways, we also know frighteningly little. What we do know is often uncertain creating error bars in measurements which foster the differences in interpretations that fuel dissent. The data we do have is poorly integrated across different stakeholders. The mountain of electronic and other data collected for compliance purposes could be better mined for environmental, commercial, and social outcomes. New tools can support this if the data is shared. Aggregation of non-sensitive data from industry sources and integration with data from a wider range of scientists from different disciplines and regulators could radically change the amount of information available on which to based decisions, and the decision-making processes must be open to incorporate this data in a transparent way. Deep local knowledge and mātauranga Māori are also under-used and we could listen more to on-the-ground expertise.

Professor Gerrard goes on to highlight the lack of funding and connectivity within the fisheries information system:⁵⁶

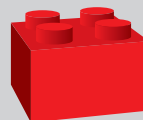
The industry levy funds vital data gathering and research for significant commercial species. It does not pay for basic public good research or research that would be valuable for other fished species. This creates a resourcing shortfall, unreasonable expectations on this funding, a lack

of trust and perverse incentives. There are many new high-tech tools and cool new ideas that could change the way we fish, but public good funded research is not always well connected to industry questions or environmental challenges ... Relationships between researchers looking at different aspects of the marine environment, housed in different institutions, mirror the poor relationships in the sector as a whole. A lot of energy is wasted trying to deconstruct an opposing narrative, which could be better spent coming to a shared understanding.

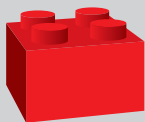
In her recommendations on ways to resolve these problems, Professor Gerrard echoes the Parliamentary Commissioner’s views by proposing a funding and research strategic action plan for fisheries. She also proposes that a data platform be established that “facilitates integration of data from a range of sources, compiles datasets in an accessible centralised platform, and turns them into information that can be readily applied in fisheries management and other areas of the marine domain, including state-of-the-art environmental reporting”.⁵⁷ In Canada, revamped fisheries legislation expressly recognises the importance of indigenous knowledge in decision-making and has established a public registry for fish habitat proposals and decisions. A cross-cutting and integrated approach to identifying research needs, funding it, using information across sectors in a strategic way, and sharing/presenting information in an efficient, user-friendly and transparent way are strong themes for policy makers to consider not just for commercial fishing but also for the marine environment as a whole.



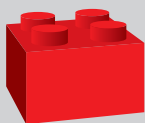
An environmental research strategy, containing a specific part on marine research and information, could be made mandatory and have legal influence over how integrated research is created, funded and deployed to achieve clearer cross-cutting objectives for the marine environment. The strategy could provide for large “one-off” exercises such as a national coastal habitat mapping project.



Marine information could be stored in a more consistent way in a national database, or using compatible databases across councils, national agencies and sectoral bodies, and be linked to clear reporting of both research and time-series monitoring data under the Environmental Reporting Act.



An Environmental Research Council or another independent agency such as an Oceans/Tikanga Commission could oversee the marine research and information system.



Funding for environmental research (or even marine components of it) could be ringfenced/ hypothecated using revenue from tools like resource rentals.

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Oyster processing, Clevedon

10.6 Marine spatial planning

Various strategies exist for particular sectors (eg fishing, aquaculture, mining) and domains (climate change, biodiversity, freshwater). Yet aside from in the Hauraki Gulf, a noticeably missing lens in our plethora of non-statutory strategies is one that looks at the *oceans* as a single space. Some have proposed marine spatial planning as an integrative tool to fill this gap.

If we were to develop a framework for marine spatial planning in Aotearoa New Zealand, we would need to have some clarity around what it means, and therefore what it could be expected to deliver. Marine spatial planning has become increasingly popular internationally, with UNESCO identifying initiatives (at various stages of progress) in around 70 countries.⁵⁸ All member states of the European Union were required to establish maritime plans by 2021, which has resulted in a plethora of marine planning activity in that region.⁵⁹

Various definitions have been proposed for marine spatial planning. This is partly a reflection of the evolving application of the tool over the last 30–40 years and the different drivers and outcomes sought in various contexts.⁶⁰ UNESCO's 2009 publication *Marine spatial planning: A step-by-step approach towards ecosystem-based management* provides the following definition:⁶¹

Marine spatial planning (MSP) is a public process of analyzing and allocating the spatial and temporal distribution of human activities in marine areas to achieve ecological, economic, and social objectives that are usually specified through a political process.

The European Union Directive on Maritime Spatial Planning 2014 defines maritime planning as “a process by which the relevant Member State’s authorities analyse and organise human activities in marine areas to achieve ecological, economic and social objectives”.⁶² In 2010 a group of 21 scientists offered a different definition which emphasises the delivery of ecosystem services:⁶³

Ecosystem-based MSP [marine spatial planning] is an integrated planning framework that informs the spatial distribution of activities in and on the ocean in order to support current and future uses of ocean ecosystems and maintain the delivery of valuable ecosystem services for future generations in a way that meets ecological, economic, and social objectives.

This definition expressly recognises the need to maintain ocean ecosystem services, and therefore brings a focus to the underlying health and productivity of marine ecosystems, consistent with ecosystem-based management. Drawing on this approach, we have adopted a working definition of marine spatial planning for the purposes of this analysis (although there are many different options and we would expect a more nuanced definition to be developed as a result of any formal policy development process):

Marine spatial planning in Aotearoa New Zealand is an integrated, strategic planning approach which guides the management of activities that cumulatively impact on the marine environment in order to maintain and restore healthy ecosystems while providing for kaitiakitanga and current and future uses.⁶⁴

Marine spatial planning in Aotearoa New Zealand

The key point is that spatial planning in the marine area can improve connections between parts of the management system that might have different regulatory processes with different purposes, different stakeholders, different timeframes, and different subject matter. It is something that focuses on place – through identifying and mapping specific values and actions – rather than on general principles and

outcomes. It is also important to note that it is as much about the process of *planning*; which brings together iwi/hapū and stakeholders to build relationships, and continues after a plan has been made; as it is about the resultant *plan* itself. This is closely linked to Treaty partnerships.

In Aotearoa New Zealand there is currently no legislative provision for integrated marine spatial planning. The RMA goes part of the way there, with the requirement to develop regional coastal plans (and the ability to link these to catchment plans), which include spatial measures. But these do not address matters outside the jurisdiction of regional councils (notably control of fishing for Fisheries Act purposes, and the creation of marine reserves).

Despite this legislative vacuum, we have seen the successful completion of a marine spatial plan for the Hauraki Gulf in late 2016 (see case study below). That project applied an ecosystems lens to the Hauraki Gulf's marine environment, identifying the strategic drivers of ecological decline and actions needed to address them. At the same time, it sought to make provision for cultural, social and economic uses of the Gulf, including through strengthening the role of mana whenua in marine management, supporting the fishing industry to move to higher value fisheries, and providing space for the expansion of aquaculture. It was about using our tools in a purposive and aligned way (and recommending new ones).



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Seachange Tai Timu Tai Pari field trip

A spotlight on Sea Change Tai Timu Tai Pari

The Sea Change Tai Timu Tai Pari project had its inception in the Hauraki Gulf Forum's 2011 *State of our Gulf Report*, which indicated that current management approaches were not sufficient to reverse the ongoing environmental decline of the marine system.⁶⁵ At the same time, there was growing awareness that marine spatial planning was becoming increasingly popular overseas. In order to understand what such an approach might contribute to the Hauraki Gulf, the Forum commissioned an international review of marine spatial planning. The resultant report, which was also released in 2011, concluded that "marine spatial planning is a well-accepted strategic planning process which could help achieve the purposes of the HGMPA [Hauraki Gulf Marine Park Act] including integrated management and the protection and enhancement of the life-supporting capacity of the Gulf."⁶⁶

The report generated considerable interest, and with the encouragement of the Hauraki Gulf Forum and EDS, Auckland Council and the Waikato Regional Council agreed to lead a marine spatial planning project in partnership with iwi, the Department of Conservation and the Ministry for Primary Industries.

A 16-member co-governance Project Steering Group was established to oversee the project, develop the terms of reference for the Stakeholder Working Group, and adopt the marine spatial plan once it had been written. Members consisted of eight representatives of the statutory bodies involved in managing the Gulf and an equal number of mana whenua representatives.

The Stakeholder Working Group undertook the actual work of developing the marine spatial plan. It consisted of representatives from commercial and recreational fishing, farming, aquaculture, infrastructure, community and environmental interests. Four positions on the group were made available to mana whenua. The group operated on a consensus basis which meant that "every member either supports or does not actively oppose (can live with) the decision".⁶⁷

The Stakeholder Working Group first convened in December 2013, and met approximately monthly up until late 2016 when

the plan was completed. An Independent Chair, appointed by the Project Steering Group, facilitated the group. During the early stages of the project six "Roundtables" were established to focus the plan development work on key elements of the overall picture as well as to involve a broader range of stakeholders. The topics for the Roundtables were fish stocks, water quality and catchments, aquaculture, biodiversity and biosecurity, accessible Gulf and Gulf infrastructure.

The resultant marine spatial plan was structured around four keke of knowledge: Kaitiakitanga and Guardianship; Mahinga Kai – Replenishing the Food Baskets; Ki Uta Ki Tai – Mountains to Sea; and Kotahitanga – Prosperous Communities.⁶⁸ The front end of the plan, which largely consists of objectives and actions, was supported by a summary of the scientific basis underpinning the plan in appendices.⁶⁹ There was no legislative straitjacket for the plan to work around, so it was able to be structured in a way that was sensitive to context and tikanga.

The plan identified 13 new aquaculture areas and 13 new protected areas as well as an extension in size of two existing marine reserves. In addition, an extensive area was identified as being unsuitable for aquaculture due to its proximity to the Auckland metropolitan area where there are many potentially conflicting uses of the water space. Fishing was also tackled, and the plan recommended that large benthic areas be protected through the retirement or mitigation of key stressors, such as bottom trawling, in order to allow natural regeneration. Smaller areas within these zones were to be the focus of more targeted passive restoration (through the establishment of marine reserves) and active restoration through the transplanting of species and/or establishment of new habitat patches.

Fishers were to be assisted to transition to methods such as long-lining, which produce higher quality fish, achieve a higher market price, and have less environmental impact. Without that broader framing – what should go where, and support for making changes – the deployment of protected areas and fisheries restrictions could struggle to get traction, being seen only as a removal of rights. Breaking down statutory silos creates opportunities for dialogue and synergies, rather than conflict, lobbying and ultimately stasis. While not free of problems,⁷⁰ the Sea Change

process offers a number of lessons for marine spatial planning in a future system. The government has recently announced a package of measures it intends to take to implement the non-statutory plan.⁷¹

There is currently no legislative framing for integrated marine spatial planning in Aotearoa New Zealand. However, a non-statutory marine spatial planning process was undertaken through the Sea Change Tai Timu Tai Pari initiative.

Benefits of marine spatial planning

Marine spatial planning provides a number of things that the current system does not.

- It is *strategic*, rather than reactive – it is a plan for what is going to happen in the future and when, and is focused on driving positive change rather than just mitigating harm. For example, it could contemplate how protected areas, aquaculture areas and other activities could move in response to a changing climate (warming seas), biosecurity risks or population change.
- It is *integrated* – it contemplates the deployment of multiple tools under different frameworks as a package. Currently there are a large number of marine-related statutes with no overarching mechanism to ensure they (or, rather the tools under them) are interacting coherently.
- Its *objectives are multi-faceted*, and it provides the opportunity to realise synergies (rather than just considering a single sector such as marine protection or aquaculture).

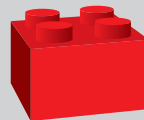
One commentator has said that marine spatial planning can overcome the “mismatch between the goals of EBM [ecosystem-based management] and the realities of their implementation across different spatial and temporal scales”.⁷² Because of its integrated nature, marine spatial planning provides an opportunity to fulfil the Crown's obligations under te Tiriti o Waitangi in the marine area, and a mechanism to integrate mātauranga Māori into marine management. Dealing with a particular place provides a much better opportunity to achieve this, by engaging with local knowledge

and tikanga. A future framework for marine spatial planning will need to draw on mātauranga Māori as a key source of knowledge and reflect the values of tangata whenua in the relevant marine area.

Marine spatial planning can be undertaken through a collaborative process, although not necessarily. The future may see a shift towards a more collaborative and negotiated style of environmental decision-making (noting that collaborative decision-making is a feature of tikanga Māori), although the potential for a consensus-based model to work may be reduced by increased diversity in the country's population, and tensions over resource use. Furthermore, there may be natural justice risks in using a collaborative or negotiated process (which cannot include everyone), particularly if it results directly in regulatory change under different legal regimes with their own statutory purposes. Moreover, international experience provides a warning:⁷³

where [ecosystem-based management] is equated with MSP [marine spatial planning], and in the presence of competing marine interests ... efforts have focused on the establishment of marine protected areas. This is often at the expense of environments outside of ‘pristine’ areas, and of local community or Indigenous rights which could be complementary to environmental objectives.

Integrated marine spatial planning has a number of potential benefits. It provides a forward-looking strategic lens and a pathway to achieve objectives, a mechanism to integrate different tools under separate statutory (and non-statutory) frameworks, and a forum for collaborative conversations to happen that are place-based and not just focused on one interest at a time.



A future system could provide for the mandatory creation of marine spatial plans to integrate or coordinate the use of other tools (regulatory and non-regulatory) in a particular place.

Content of marine spatial plans

There are different options for what a marine spatial plan could include. Some things would be common – plans should contain a robust description of the state of the marine area, its importance to those using it, and a broad and comprehensive description of pressures or issues faced.

Plans could include a vision statement, objectives and desired outcomes (including environmental bottom lines). They could also include the spatial identification of marine areas to be managed for specific purposes and areas suitable or not suitable for specific activities. Plans could be non-regulatory, in the sense that they would not have direct binding effect on individuals. However, they could be given force through links to regulatory plans, such as regional coastal plans under the RMA and fisheries plans under the Fisheries Act.

Alternatively, marine spatial plans might themselves contain regulatory provisions. These could, for example, create a layer of MPAs through the plan itself, without relying on clunky mechanisms under other legislation. However, that could cause more complexity when it comes to legislative design (see Chapter 11).

Raewyn Peart



The Noises

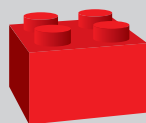
A spotlight on MPA planning within a spatial plan

Planning for MPAs alongside (or within) marine spatial planning enables all activities and environmental pressures to be considered simultaneously, providing more certainty for ocean users, communities and the environment. This can help enhance the performance of MPAs by zoning or providing incentives for synergistic marine-use areas (eg regenerative aquaculture or eco-tourism) close to MPAs, creating low impact buffer zones around vulnerable MPAs, and directing incompatible activities to where they would cause least harm. It can mitigate the effects of strict protection, which can concentrate fishing and other pressures elsewhere. This can provide “a broader perspective of how MPAs are nested within a marine spatial plan, and increase ecological representativeness through protection of important areas, including those not selected as sites for MPAs, where [other] conservation measures could be proposed and implemented”.⁷⁴

However, on the other hand, such an approach can result in diluted MPA proposals because the focus can move from biodiversity protection to other, user-related considerations. It risks becoming a negotiation, rather than being a science-based exercise, especially if a collaborative process is used to reach decisions. One option would be to take an ecologically focused approach to establishing MPAs (ie locating them where they would best protect biodiversity) and to only then consider how other activities can operate around them. This would be a two-stage exercise. However, separating MPA planning and marine spatial planning can increase costs, cause stakeholder confusion around the planning process, and risk the process stalling during the first stage which might be seen as just giving up rights without consideration of opportunities or compromises.⁷⁵

Spatial plans could also contain strategic measures. These might include an action plan of the tasks needed to be undertaken by agencies, iwi/hapū or stakeholders to achieve the desired outcomes. Provisions could set out matters such as marine research priorities, biosecurity measures, stricter catchment management, needed coastal infrastructure, and shellfish beds, wetlands and dunes to be restored. Funding arrangements to implement these provisions would need to be identified and could be encapsulated in “implementation agreements” as recommended by the Randerson Panel.⁷⁶

Finally, marine spatial plans could contain a “monitoring and evaluation” section. This could identify key performance indicators to allow progress towards the plan’s objectives and outcomes (including environmental limits) to be ascertained. Indicators could relate to matters such as nutrients, sediment, habitats, biodiversity values or fish stocks. Some outcomes could have aligned targets (interim steps on the way to achieving the long-term outcome) established. They could also include triggers whereby prescribed management action would be required if targets were not being met. This is not dissimilar to the structure of environmental limits, targets and outcomes set out in the proposed NBA.



Marine spatial plans could include environmental bottom lines, targets and outcomes for the marine area. They could be strategic only, and rely on implementation through other frameworks. Alternatively, they could include regulatory provisions themselves as a direct pathway for things like environmental limits and MPAs.

Triggers for marine spatial planning

Marine spatial planning can be applied “wall to wall” to a country’s entire marine space, or be deployed more specifically to some areas when there are pressures or conflicts to be addressed. Which option is preferable depends on how we frame the purpose of marine spatial planning. Is it a conflict resolution tool? Or something more? A middle road is to gradually roll out marine spatial planning to the entire marine area, over time, but to start in areas where there are currently significant issues.

The “wall to wall” approach has been favoured in other jurisdictions. The European Union Directive on Maritime Spatial Planning requires member states to “establish and implement” maritime plans for all their marine waters.⁷⁷ In response England, for example, is rolling out a series of 10 marine plans throughout its territorial sea and EEZ.

The Randerson Panel took a similar approach for the preparation of regional spatial strategies, which it recommended should include all the coastal marine area. After considering a more targeted approach, the Panel concluded that such spatial planning should be mandatory for all regions, but with provision for prioritisation and sequencing by the responsible Minister(s). This was because spatial planning was seen to have broad benefits which would be applicable around the entire country.⁷⁸

An important consideration, in deciding an approach to rolling out marine spatial planning in a targeted or broader manner, is the considerable resources required to mount an integrated planning exercise for the marine area. Fully integrated marine spatial planning can be resource intensive and it is therefore not something to be embarked on lightly; there should arguably be a pressing need, complex issues or conflicts to resolve, and an appetite for change (ie to implement it).⁷⁹



Marine spatial plans could be targeted to areas where there are particular issues or conflicts or they could cover all the country’s marine areas. There could be specific trigger points specified for when a planning process was deemed necessary.

Connection with terrestrial planning

It is important that there is a strong link between marine spatial planning and terrestrial spatial planning. This is because the latter is an important tool for addressing impacts *on* the marine environment, including on marine industries (eg through determining where agriculture and forestry can and cannot go and where cities are to expand). We have previously pointed out that in the Sea Change Tai Timu Tai Pari context:⁸⁰

The impact of poor water quality on the ecological health of the Hauraki Gulf was one of the greatest areas of concern, with the main stressor being sediment ... The approach set out in the plan is wide-ranging and includes measures to reduce soil erosion, to minimise sediment entering waterways and to stabilise sediment once it has reached the marine environment ... The plan recommends that a cap is placed on nitrogen discharge levels which are to be kept at or below current rates until sufficient scientific work had been completed to enable an appropriate nutrient load limit to be put in place.

This raises the question as to whether terrestrial and marine spatial planning are in fact so inter-linked that they should occur together, and whether the proposed framework for spatial planning under the Strategic Planning Act should include marine spatial planning (see Chapter 6 and Appendix 3). A recent Cabinet paper has confirmed that the Strategic Planning Act is intended to apply to the coastal marine area,⁸¹ and depending on how it is drafted, it may also be compatible with rollout in the EEZ. Alternatively, the provisions in marine spatial plans could serve to drive responses in terrestrial plans (a “sea to the mountains” approach), including through directly influencing or changing provisions in regional and district plans.

A spotlight on marine spatial planning in the United Kingdom

Marine spatial planning in the United Kingdom is undertaken within the auspices of the Marine and Coastal Access Act 2009. This establishes a regime for marine planning as well as broader regulation of the country's marine area. The Act was heralded as "a trailblazing piece of legislation" as "it marked the point at which marine spatial planning became an established part of the administration of marine activities in UK waters".⁸²

The Act applies to the entire British marine area including the territorial (or "inshore") seas, and the EEZ or "offshore" seas. It provides for a two-tiered approach to marine planning,⁸³ which includes the preparation of marine policy statements (which are discretionary), and marine plans, which are mandatory where there is a marine policy statement in place for the area, but which are otherwise discretionary.

Under the Act, a marine policy statement is to set out general policies for contributing to the achievement of sustainable development within the country's marine area.⁸⁴ Such a statement was adopted in 2011. It sets out a vision for the marine environment which is for "clean, healthy, safe, productive and biologically diverse oceans and seas".⁸⁵ It includes 21 high-level marine objectives grouped under the themes of achieving a sustainable marine economy; ensuring a strong, healthy and just society; living within environmental limits; promoting good governance; and using sound science responsibly.⁸⁶ It confirms that the statement and marine plans "form a new plan-led system for marine activities" providing for "greater coherence in policy and a forward-looking, proactive and spatial planning approach to the management of the marine area, its resources, and the activities and interactions that take place within it".⁸⁷

Because a marine policy statement exists for the whole of the country's marine area, marine plans are mandatory for the entire territorial sea and EEZ.⁸⁸ Under the Act, a marine plan is required to state policies for the sustainable development of the area, identify (by means of a map or otherwise) the marine plan area that it relates to, and conform with any relevant marine policy statement.⁸⁹ The marine policy statement further explains that "marine plans will provide a clear, spatial and locally-relevant expression of policy, implementation and delivery".⁹⁰ They are to be "based on an ecosystem approach" and be "participative and informed by data provided by consultees, stakeholders, regulators and relevant experts".⁹¹

The responsibility for preparing marine plans lies with the Secretary of State (for England), Welsh and Scottish Ministers, and Department of the Environment in Northern Ireland for their respective areas. These functions may be delegated to a public body, although the decision to adopt and publish a plan, once prepared, remains with the Ministers/Secretary of State/Department of the Environment.⁹²

In England, the Secretary of State's marine planning powers have been delegated to the Marine Management Organisation which is an executive, non-departmental public body with broad statutory marine functions (effectively acting as an integrated oceans agency – see Chapter 12). In Scotland, the development of regional marine plans has been devolved to marine planning partnerships which consist of "marine stakeholders who reflect marine interests in their region".⁹³ These include representatives from local councils, fishing and other industries, and environmental and recreational non-governmental organisations.⁹⁴

Before beginning work on developing the plan, the marine planning authority must give notice to councils, whose area of jurisdiction lies adjacent to the marine planning region.⁹⁵ It must also publish a statement of public participation.

The statement of public participation must identify the area for which a plan is being prepared⁹⁶ and invite representations on matters to be included in the proposed plan (ie before a consultation draft is prepared).⁹⁷ It must also include a proposed timetable for the preparation and publication of a consultation draft, the making of "representations" (ie submissions) on the draft, their consideration, and the adoption and publication of the plan.⁹⁸ Provision may be made for public meetings to be held. The marine plan authority must take "all reasonable steps" to comply with the statement and it must be kept under review.⁹⁹

Once proposals for the marine plan have been developed, a consultation draft must be publicly notified.¹⁰⁰ Prior to this, a sustainability appraisal must be carried out on the draft plan and the results of this published at the same time as the consultation draft.¹⁰¹ Submissions on the consultation draft can be made by any person, in accordance with the statement of public participation.¹⁰²

The marine plan authority must consider appointing an independent person to investigate the proposals contained in the consultation draft and to report on them, but is not required to do so.¹⁰³ If

appointed, the independent investigator makes recommendations on the plan and these must also be made public.¹⁰⁴

The marine plan is adopted once the marine plan authority has decided to publish the plan. This can only be done with the agreement of the Welsh and Scottish Ministers (for Wales and Scotland respectively), the Secretary of State (for England) and the Department of the Environment in Northern Ireland.¹⁰⁵ Thus, while a delegated agency may be responsible for preparing the marine plan, it operates within the framework of broader government policy.

Marine plans have legal status insofar as any public authority must take any authorisation or enforcement decision “in accordance with” them, and have regard to them when taking any other decisions affecting the marine area.¹⁰⁶ Having said that, enforcement and authorisation decisions do not have to comply with the marine plan if relevant considerations indicate that they should not. In that case, reasons for that decision must be stated.¹⁰⁷

The marine plan authority must keep the marine plan under review and report to Parliament every three years on the effect of the policies, their effectiveness in securing the objectives of the plan, progress made towards securing the objectives, and whether the objectives of the marine policy statement are being met.¹⁰⁸ After a report is published, the marine plan authority must decide whether or not to amend or replace the marine plan.

Ten years after the passage of the Marine and Coastal Access Act, two regional marine plans have been adopted in England (in 2014 and 2018) and an additional four are well advanced. Scotland adopted a national marine plan in 2015 and is now working on regional plans. Wales adopted a national marine plan in 2019.¹⁰⁹ A review of the effectiveness of the plans indicated that many of their policies were expressed in broad terms, and others were conflicting and/or ambiguous, meaning that they appeared to have little effect on licencing decisions. The reviewers recommended that policies needed to be strengthened and made more specific. They also noted that efforts were needed to change the culture of licensing officers who were reluctant to apply policies within marine plans to their decisions. In addition, they suggested that the plans should be subject to legal challenge, in a similar way to terrestrial plans, as this enables a pragmatic approach to be applied to their application.¹¹⁰

In Scotland, where regional planning has been devolved to stakeholder groupings, interviews with participants identified strong support for the approach, as opposed to plan making

being led by a central authority (eg the Marine Management Organisation in England). The devolved approach “supported learning regarding other perspectives and building of trust between organisations” and had assisted with “conflict avoidance by enabling developers to explore appropriate siting of activities”.¹¹¹ This was similar to the findings from a review of the Sea Change Tai Timu Tai Pari project where the collaborative process was seen as one of its biggest strengths.¹¹²

Lessons that can be drawn from the United Kingdom experience include the need to include clear and directive provisions in plans, to have a clear implementation pathway through linkages with consenting decisions, and to address planning and consenting culture in implementation. The Scottish experience also indicates the strength of adopting a devolved approach in bringing stakeholders together, building trust and helping to resolve conflicts.

International experience highlights the importance of marine spatial planning having clear and direct influence on decision-making, and a clear implementation pathway, as well as the value in collaboration between stakeholders.



Ilfracombe, North Devon, United kingdom

Raewyn Peart

10.7 From spatial planning to an oceans policy

Reforms to better integrate the ocean management system could go beyond the incorporation of a marine spatial planning framework, to think more broadly about an integrated oceans policy. As with marine spatial planning, that could mean different things.

At root, a national oceans policy is essentially a mechanism to provide a coherent approach for oceans management across a country's entire oceans realm. This is in the context of typically fragmented legislative and institutional arrangements. It is often seen as a way to address conflicts between different ocean uses, to support new uses, and to provide protection for ecosystems and species. It is not place-based like marine spatial planning, and instead sets out a vision for the oceans as a whole accompanied by a set of high-level principles.¹¹³

This could end up being little more than a politically driven agenda for reform – a manifesto for change – which is arguably what the government's recent establishment of a vision and objectives for the oceans is.¹¹⁴ However, an oceans policy could be conceived of as something more concrete – a structural and living feature of a future system (an actual instrument with ongoing influence). It could incorporate a strategy for *deploying* marine spatial plans.

Raewyn Peart



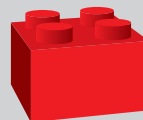
Tahunanui Beach

A spotlight on a Marine Spatial Planning Strategy

One way to achieve a coherent or principled roll out of marine spatial plans would be for the responsible Minister to develop a “Marine Spatial Planning Strategy”, which would set out a programme for developing marine spatial plans. This would be similar to the marine policy statement provided for under the United Kingdom Marine and Coastal Access Act, which creates the framework for the preparation of marine plans.

The Strategy could be reviewed at least every 10 years, and identify priority areas for marine spatial planning over the next 10-year planning period, based on a set of statutory criteria. Preparation of the Strategy would necessitate a strategic look at which parts of the coast and marine environment should be targeted, and in what order, allowing prioritisation of effort. High pressure and contested areas, or those that are particularly vulnerable, might be prioritised. The Strategy could also set out high-level objectives for marine spatial plans in the identified areas, akin to a terms of reference. This would enable, for example, an objective to be included on increasing MPA coverage in a proposed planning area, making greater provision for aquaculture or renewable energy or providing for public infrastructure.

A Marine Spatial Planning Strategy could also allow public funds to be set aside for plan development and implementation, and encourage agencies to align other planning and funding processes with the marine spatial planning process (eg regional councils could decide to defer changing regional coastal plans until the process was complete).



A Marine Spatial Planning Strategy, developed by the Minister of Oceans, could provide a policy framework for marine spatial planning and set out a programme for developing marine spatial plans in targeted areas.

However, the concept of an oceans policy goes beyond just marine spatial planning. For example, it could outline strategic actions to reduce pressures on the marine area that are not “spatial” or place-based, such as measures to reduce production of plastics or their release to the environment,

funding for the improvement of stormwater and wastewater infrastructure, and building controls to mitigate impacts on the marine environment.¹¹⁵

Public presentations given by the former Oceans Secretariat in the 2000s, before that oceans policy process was halted, indicated that the proposals developed at that time included a statement of a *vision* for the oceans; a set of overarching *objectives* to govern management of the oceans; and the development of a *National Oceans Plan* to set national priorities, baseline ecological standards, and processes and tools for reconciling competing uses. In the report of the Ministerial Advisory Committee, established as part of the oceans policy process, the Chair highlighted that the consultation process had identified a high degree of interest in the idea of preparing a “comprehensive national policy” and agreement that it was timely to try to “define a vision and develop an overarching framework to guide and rationalise the management of our interaction with the seas”.¹¹⁶

Of course, this was before the advent of the EEZ Act, when there was a glaring hole in the oceans management system for that space. However, commentators like Karen Scott still see a justification for an oceans policy as an integrative device, despite some gaps having been filled.¹¹⁷ She recently suggested that an oceans policy should include an articulation of values, goals and principles and development of processes to support bioregional and/or spatial planning.¹¹⁸ It could promote the implementation of ecosystem-based management across Aotearoa New Zealand’s marine area, facilitate integrated management, provide an ecologically coherent framework for area-based protection measures, provide a framework for co-governance of ocean resources with Māori, raise the profile of oceans as an economic and environmental priority, and help to implement Aotearoa New Zealand’s international commitments and demonstrate international leadership.¹¹⁹ An oceans policy could also provide a framework for futures scanning – to ensure that it remained a living document that was sensitive to changes in the marine environment, such as those driven by climate, technology and societal changes.

Scott suggests that a co-governance Oceans Council be established to develop, report on, and provide advice in relation to, a “New Zealand Oceans Policy”. A new “Ministry for Ocean Affairs” would then be given the mandate for the operational implementation of the policy along with other oceans matters.¹²⁰ We discuss institutional settings further in Chapter 12.

The Prime Minister’s Chief Science Advisor recently also recommended to government that it “develop a bold *Oceans Strategic Action Plan* for 2040 to protect and manage Aotearoa New Zealand’s territorial sea and EEZ, with a clear integrative framework to prioritise, coordinate, implement and measure outcomes to achieve 100% sustainably managed oceans.”¹²¹

In terms of the fisheries component of such a plan, she suggests that the Action Plan could include a definition and role for an ecosystems approach to be applied to fisheries; provide a clear framework for reporting, decision-making, planning and responsibilities; provide a framework for the development of fisheries-related plans; include actions to support a move from volume to value in commercial fisheries; and prioritise actions across a multi-year programme. The Oceans and Fisheries Minister was identified as the appropriate Minister to lead the development of the Action Plan.¹²²

EDS’s previous analysis of the development of national oceans policies in the USA, Australia, Canada and Aotearoa New Zealand identified five key elements of such policies. These still resonate today and provide a broad architecture to encompass many of the suggestions above.

- *A common purpose* for the management of oceans, which can be articulated in the form of a vision for the oceans and high-level management objectives and principles which are to apply to oceans management.
- *Integrative mechanisms* to jointly harness the actions of different oceans managers to achieve the common purpose. This may be through joint committees, working bodies, advisory bodies and/or the establishment of specialist oceans institutions.
- *Strategic actions* which are required to implement the policy. These often focus on addressing gaps and conflicts in the current oceans management system. This can include such matters as reforming outdated legislation (such as the Marine Reserves Act) and reviewing the management of specific sectors such as aquaculture or fisheries.
- *A framework for area-based planning* (marine spatial planning) which applies the principles of the policy to a specific marine ecosystem (place), as described above.
- *Performance management systems* which include the ongoing monitoring of the health of the oceans and regular review of the effectiveness of management actions. This enables progressive “learning by doing” and the adaptation of management approaches and the oceans system more broadly, in light of new information or changes in technologies, markets and community values. Oversight mechanisms could include the Parliamentary Commissioner for the Environment being tasked with undertaking regular independent reviews of the implementation of oceans policy.

Such an oceans strategy or policy could form the normative glue that holds disparate parts of the system together. Others have said a similar thing, describing it as a normative “anchor” for the system. It would be similar to the *Te mana o te taiao – the Aotearoa New Zealand biodiversity strategy*, which:¹²³

provides a holistic, integrated, intergenerational approach to protecting and preserving biodiversity based on *Te Ao Māori* (Māori worldview)... [and] recognises that humans are a part of nature, and have kinship relationships with living natural ecosystems, and seeks to set fundamental objectives and values applying across the regulatory regime.

That said, careful thought would need to be given to how effective this normative glue or anchor could be in practice when faced with prescribed statutory principles that might differ and take primacy when decisions were being made under other legislation. Nevertheless, in terms of fisheries, some are optimistic that a strategy should set:¹²⁴

an expectation that any fisheries-related plans, when created or revised, must specify how they will progress the objectives of the Oceans Strategic Action Plan and demonstrate progress against this in annual review reports.



A national level Oceans Policy could be a strategic instrument, going far beyond just an action plan for rolling out marine spatial plans. It could, with sufficient framing, be a form of “constitution” for the oceans.

A spotlight on Australia's Oceans Policy

Australia's oceans policy, released in 1998, consisted of two volumes. The first volume set out the overall vision and goals for the Policy, described the concept of ecosystem-based oceans planning and management, set out the institutional arrangements for implementation and identified some key initial actions to be carried out by the federal government. The second volume identified specific proposed actions for particular economic sectors and some other areas. The key implementing mechanism for the Policy was to be the development of what were called “regional marine plans”.¹²⁵

Four institutions were put in place to support implementation. A National Oceans Ministerial Board of Commonwealth Ministers was to have an oversight role and approve regional marine plans. Reporting to the Ministerial Board was an independent executive agency in the form of a National Oceans Office. A National Ocean Advisory Group was established to provide input from a range of sectors and it included members from industry, science, conservation and the community as well as government stakeholders. Regional Marine Plan Steering Committees were also established by the Ministerial Board. They comprised non-governmental and governmental regional stakeholders, and were to oversee the development of regional marine plans. An Oceans Policy Science Advisory Group was also subsequently established to provide science input.¹²⁶

However, it has been said that “the consensus is now that the Oceans Policy has failed to realise its full potential ... and Australian marine and coastal regulation continues to be described as highly fragmented, *ad hoc*, inconsistent and inefficient.”¹²⁷

A spotlight on Canada's Oceans Strategy

Under the Canadian Oceans Act 1997, the Minister of Fisheries and Oceans was given the task of leading and facilitating the development and implementation of a “national oceans management strategy”. The strategy was to be based on the principles of sustainable development, integrated management



Waves, Ōrere Point

and the precautionary approach. The Act emphasises the importance of ecosystem-based management, stating in the preamble that “conservation, based on an ecosystem approach, is of fundamental importance to maintaining biological diversity and productivity in the marine environment”.

A high-level strategy was released in 2002.¹²⁸ It was followed in 2005 by an Oceans Action Plan. This identified 18 specific initiatives to be undertaken across six federal departments.¹²⁹ A key focus of the Action Plan was the application of integrated management planning to large ocean management areas (a form of marine spatial planning). Planning for these areas was to be undertaken on a collaborative basis between the various management agencies, indigenous peoples and stakeholders. Implementation of the Oceans Action Plan was being overseen by a Deputy Minister’s Interdepartmental Committee on Oceans which consists of representatives of 19 federal departments and agencies involved in oceans management. However, the programme has struggled due to the marine plans lacking direct legal force.

In short, Canada’s experience has been one of trying to use a more integrated strategy, followed by an action plan, to coordinate multiple institutions and multiple legislative frameworks towards a common end. It has been about using an additional layer of measures to bring together what was already there (improve connections and align actions), rather than redesigning that system itself. It tries to act as the “glue”.

However, reviews have since identified impediments to the successful implementation of Canada’s Oceans Act. Specific problems include¹³⁰ no requirement for other federal departments to comply with or implement the Act and no specific provision to give integrated management plans legal force. Criticisms have continued in recent years, pointing out the failure to properly implement the legislation (with much discretion remaining with Ministers), the weakening of habitat protections through separate fisheries legislation, the lack of central leadership to protect threatened species, and inconsistency between the Strategy and the Action Plan expected to implement it.¹³¹ More targeted reforms (particularly to strengthen MPA deployment) have been progressed in the last few years, but have been achieved largely through direct legislative amendment rather than by strengthening the place of the strategy itself.¹³²

Reflecting on the Australian and Canadian examples, it would make sense for an oceans policy in Aotearoa New Zealand to be closely linked to a framework for marine spatial planning, just as national level policy under the RMA is closely linked to place-based planning for the coastal marine environment. Marine spatial planning would be one way – but not the only way – in which a policy would be implemented at place.

10.8 Do strategic tools need statutory backing?

When it comes to the creation of significant new tools, like marine spatial planning and an oceans policy, we need to consider whether such things require legal framing. The Sea Change Tai Timu Tai Pari marine spatial planning process was non-statutory, and provided many lessons. It was extremely valuable in bringing together various government agencies, Māori, councils and stakeholders to work together and consider how various tools could be deployed in a more integrated way. It provided a forum. However, implementation has been slow, including it taking over four years for the government to make a commitment to its implementation. And that has yet to become a legal commitment. So should we legislate for marine spatial planning? And for an oceans policy?

On the one hand, Sir Geoffrey Palmer and the Legislation Design and Advisory Committee remind us that we should not legislate unless there is good reason.¹³³ The statute book is already complex. On the other hand, inadequate legislative hooks have been partly blamed for the Australian oceans policy not being as effective as it could have been.¹³⁴ A review in Canada also identified the lack of legal force for oceans plans as a reason for them underperforming.¹³⁵ Arguably in that country, the difficulty in making progress with ocean planning resulted from ambiguity in the Oceans Act itself, and the voluntary nature of participation by partners and stakeholders. This suggests that the provisions of the Act could have been more prescriptive.¹³⁶

Another difficulty with making progress in implementing Canada’s Oceans Act was the fact that regulation of significant oceans activities, such as oil and gas exploitation, lay outside the jurisdiction of the Department of Fisheries and Oceans. Combined with the fact that oceans plans do not have regulatory effect, this institutional separation presented a formidable stumbling block when seeking implementation by other government departments. Perhaps the lesson is that an oceans policy or spatial planning does not necessarily need statutory backing, but if it does not have that, it does need a strong institutional champion.

Canada’s experience is not dissimilar to the implementation challenges that our first marine spatial plan in the Hauraki Gulf has faced (see

spotlight below), and which other areas face in even getting a marine spatial planning process off the ground. Other issues identified with the Canadian approach were changes in political priorities (and diversion of funds away from plan implementation), lack of operational guidance for integrated oceans management, and limited action plans to implement the oceans plans, given that they are high-level documents.¹³⁷ Overall, the review found that ocean planning had helped progress the development of an MPA network but had few other outcomes.¹³⁸

It is possible for marine spatial planning to be undertaken through non-statutory processes, with implementation achieved through formal avenues, following a plan's completion. However, without a statutory framework, such planning remains an ad hoc process that may or may not occur, and which remains outside the formal toolbox for marine management.

Even if a marine spatial planning project is initiated, there is a risk that the plan will not be completed, or will founder at the implementation phase. Non-statutory processes rely on strong political leadership and interest being maintained over multiple years. As demonstrated by the Sea Change Tai Timu Tai Pari project, this can be particularly difficult with Aotearoa New Zealand's three-year election cycle. There is also no clear path for implementation (eg through legal influence on lower level regulatory and funding decisions, or the ability for provisions to take direct effect).

Even with a high level of political interest and commitment to implementation, there are less than ideal procedural issues associated with relying on non-statutory plans. Processes such as iwi/hapū engagement, stakeholder participation, scientific input, and agency consideration of planning provisions would be uncertain (as there would be no legal requirement to undertake them) and would potentially be replicated in a subsequent statutory process to implement the plan. This would be costly in terms of agency resources and may lead to general consultation fatigue. In addition, implementation would occur under multiple pieces of legislation, with different purposes and overseen by different Ministers. As such, there is no guarantee that plan provisions will get implemented as an integrated package.

For a non-statutory plan to be successfully implemented, strong political leadership and interest needs to be present and maintained over multiple years. This is difficult with Aotearoa New Zealand's three-year political cycles.

Non-statutory marine spatial planning processes will usually require a duplication of public processes (eg consultation and hearings under various pieces of legislation). This may result in the plan losing its integration and coherence, as each of these statutory processes is only considering a part of the whole picture.



A statutory framework that both initiates marine spatial planning and provides agencies, mana whenua and stakeholders with guidance on principles and process steps could be provided to help with the implementation of marine spatial planning.



Careys Bay, Dunedin

Raewyn Peart

A spotlight on implementation of Sea Change Tai Timu Tai Pari: A spatial plan without a legislative basis

Implementation of the non-statutory marine spatial plan for the Hauraki Gulf has proved challenging. This was particularly due to the three-yearly electoral cycles and lack of an enduring champion to oversee the implementation of the plan.

A local government election was held just prior to the plan being adopted and several key members of the project steering group (the co-governance body), who were strong advocates for the plan, lost their seats. Once the plan was finalised and publicly released, both the project steering group (which provided administrative oversight) and the stakeholder working group (which developed the plan) were disestablished. After the 2016 local body election, the membership of the Hauraki Gulf Forum changed. Although the Forum had been the initiator of the project, a majority of its members no longer supported implementation of the plan. There was therefore no institutional champion for the plan's implementation as a coherent whole, and no formal process for broader public consultation on its provisions.¹³⁹

Auckland Council did establish a Political Reference Group, which first met in 2017, to provide oversight and guidance for council activities relevant to the plan as well as to integrate with the work programme of other agencies. Both Auckland Council and Waikato Regional Council evaluated the recommendations in the plan, identifying relevant actions and assessing them against current work programmes and budgets. Waikato Regional Council is currently reviewing its regional coastal plan, and this is being informed by the provisions in the Sea Change Tai Timu Tai Pari plan, as will subsequent land and water regional plans.¹⁴⁰

The Department of Conservation and Ministry for Primary Industries delayed any external activity to implement the plan, until after the national election in September 2017, and the confirmation of subsequent ministerial posts. But it was not until July 2019 that the Ministers of Conservation and Fisheries jointly appointed a Ministerial Advisory Committee to “help shape the Government’s response to the Conservation and fisheries proposals” in the plan.¹⁴¹ Officials were tasked with writing the government’s response with advice from the Committee.

A draft response document was completed just prior to the October 2020 general election, after which new Ministers of Conservation and Fisheries were appointed, and had to be brought up to speed. It took until June 2021 for the government to announce its commitment to implement proposals in the plan that are under its jurisdiction,¹⁴² and these have generally been received positively. However, these must still proceed through various avenues under other legislation, such as the Fisheries Act, and this has yet to happen. Because of the inadequacies of the underlying legal frameworks (especially for MPAs), more special legislation for the Gulf is anticipated to implement key elements.¹⁴³ Success still relies on the ongoing commitment of many parties.

As part of the 2000s oceans policy process, the Oceans Secretariat considered different theoretical approaches to dealing with the problem of integration, and the Canadian and Australian experiences of developing oceans policy. It concluded that both a strong legislative basis and a sound planning process were required to deliver the desired outcomes. The fragile nature of the implementation phase of the Sea Change process also suggests that some statutory framing may be useful, and it would be necessary if spatial plans were to have direct regulatory effect or a legally influential relationship with other implementation statutes (eg the RMA, Fisheries Act).

That said, excessive statutory prescription for marine spatial planning or an oceans policy may be counterproductive. In Victoria, a more flexible legislative background is provided in its Marine and Coastal Act 2018. There, a framework has been established, not just for spatial planning, but for a broader oceans policy (see spotlight below).



New Zealand fur seals, Doubtful Sound

Raewyn Peart

A spotlight on the Marine and Coastal Act 2018 (Victoria, Australia)

Victoria's Marine and Coastal Act establishes a tiered system of planning documents comprising a Marine and Coastal Policy, a Marine and Coastal Strategy, and a variety of regional and local plans. The Act also establishes a new Marine and Coastal Council and creates offences and enforcement mechanisms for unauthorised use or development of marine and coastal Crown land.¹⁴⁴

The Act *requires* development of a Marine and Coastal Policy¹⁴⁵ which was published in March 2020. It was developed by the Minister for Energy, Environment and Climate Change and sets out a vision for Victoria's marine area which "is for a healthy, dynamic and biodiverse marine and coastal environment that is valued in its own right, and that benefits the Victorian community, now and in the future".¹⁴⁶ A framework for marine spatial planning is a mandatory element of the policy under the Act, which is to establish "a process for achieving integrated and coordinated planning and management of the marine environment".¹⁴⁷ Victoria's marine environment extends three nautical miles offshore.

Unlike the case in the United Kingdom, the Marine and Coastal Act does not set out a process for developing marine plans; instead it is to be described in the Marine and Coastal Policy. This describes the planning process as "as a continuous, iterative process that will adapt according to new knowledge or needs" as opposed to a process that seeks to create a one-off "master plan". The first step is to determine marine planning areas, and prioritise when marine spatial planning will be undertaken for each, with such planning eventually to be undertaken on a state-wide basis.¹⁴⁸

The Minister is required to authorise a marine spatial planning process before it can commence. As part of the authorisation, the Minister will outline the scope of the process, who must be involved, the body which will coordinate and oversee the planning process and the implementation of its outcomes, and funding mechanisms.¹⁴⁹ This provides considerable flexibility to tailor the configuration of the marine spatial planning process to the particular area concerned. However it sets out some minimum requirements. Traditional Owners must be invited to "participate" in marine spatial planning with the method of participation to be determined by those groups.¹⁵⁰ Draft plans must be released for public comment.

The marine plan is required to "identify when, where, and how the goals and objectives for the planning area will be met", including identifying its scope; a vision, goals and objectives for the planning area; key issues; management approaches to address the key issues and achieve the goals and objectives (with a timeline for implementation); a zoning plan if required; and a monitoring, evaluation and reporting strategy for the plan. In addition, the plan should identify agencies or partners responsible for implementing specific actions within the plan.¹⁵¹

The Marine and Coastal Act enables the establishment of "regional and strategic partnerships" which have as their purpose "to respond to an identified regional issue relating to or affecting the marine and coastal environment" and to prepare a "product".¹⁵² This is a mechanism that the Victorian Government has indicated it may use to oversee the development of a marine spatial plan. The partnerships consist of two or more partner agencies, which can be government or non-government bodies that have an interest in or connection with the marine and coastal environment.¹⁵³ They can be established on direction by the Minister, or following a request to the Minister by the Victorian Marine and Coastal Council (an advisory body established by the Act) or a partner agency. They can only be established with the agreement of each partner agency and approval of the Minister.

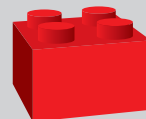
The instrument establishing a Partnership must identify a lead partner agency, the terms of reference, a statement of whether an implementation plan is necessary, and the reporting requirements. The lead agency is responsible for coordinating the preparation and implementation of the "product", in this case the marine spatial plan.¹⁵⁴ The Act also sets out consultation requirements with key stakeholders and the public, and includes a requirement for inviting and considering public submissions. The Minister formally approves the plan and publishes notice of the approval in the government *Gazette*.¹⁵⁵

Marine spatial plans, as products of regional and strategic partnerships, are statutory documents with legal effect under the Marine and Coastal Act. In determining consent applications on marine and coastal Crown land, the Minister must ensure that the consent is "consistent with" a marine spatial plan.¹⁵⁶ Crown land managers must take all reasonable steps to implement marine spatial plans in respect of the land managed on behalf of the

Crown.¹⁵⁷ However, there is no legal mechanism to achieve recognition or implementation of the plan by agencies operating under other statutes such as fisheries legislation.

The Victorian government has yet to embark on a marine spatial planning exercise under the new legislation and policy. The Department is currently undertaking a state-wide assessment of Victoria's marine environment to identify priority areas for marine spatial planning. It is developing guidelines which will provide instructions on how to undertake a planning process in a specific planning area. Following this, the Victorian government plans to scope and initiate the first marine spatial planning process in a priority area.¹⁵⁸

The Victorian government has adopted a flexible approach to marine spatial planning, providing a very broad structure for it within the legislation, and requiring a framework to be provided within a ministerially approved policy document. This enables more flexibility and learning by doing, where the framework can be regularly adjusted and updated as required. The policy is being implemented in a stepwise fashion, with priority areas being established first, before the first planning process is initiated. This makes sense and would be applicable to the Aotearoa New Zealand situation, where the process could be tailored to the regional context.



Marine spatial planning could have a broad and flexible legislative framing, allowing change and innovation to occur as practice and experience evolves.



Melbourne, Victoria

Credit

10.9 Concluding comments

In this chapter we have considered what we have called “strategic” and “integrative” tools. A future system could see existing tools become more future focused, or being used in a more strategic way, to drive change in the marine space. That could involve reimagining the purposes and principles of core legislation to be more about taking steps towards a different future than static management and protection. New tools could assist here, such as binding targets (and the accountability frameworks that could make them more robust, as we have for climate change) and mandatory and legally influential strategies (eg for the deployment of MPAs, or formalising the place of *Te mana o te taiao* – the *Aotearoa New Zealand biodiversity strategy 2020*). Of course, what that future should *be* is debatable (see Chapter 7). However, the point here is that a system could be oriented quite differently to pursue positive change of some kind. The RMA is already getting a strategic makeover, and changing tools under the Fisheries Act and Marine Reserves Act, could open up a world of other opportunities.

No matter how future focused a tool is, it does not exist in a silo. The system needs a mechanism by which tools are deployed, both in a

strategic (eg proactively rolling out AMAs or MPAs) and *coordinated* (eg considering where those things, and others, go relative to *each other*) way. Coordination needs to occur across both space and time, and within the context of achieving compatible goals.

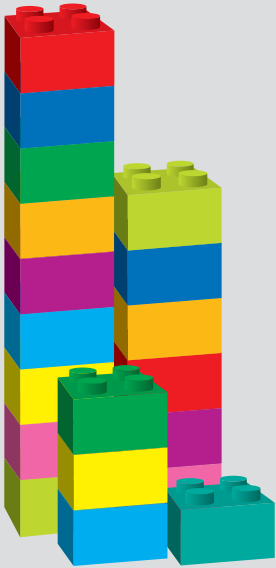
Some better integration of tools might be possible without too much upheaval, such as by better aligning the purposes of legislation, through creating cross-references or allowing one tool (eg the NZCPS) to influence another (eg fisheries plans). But integrative tools like marine spatial planning and an oceans policy have potential to provide stronger glue in what has become a fragmented system; one that does not reflect the interconnectedness of either the marine environment or the problems facing it.

More broadly, policy makers should think carefully about what the toolkit is there for. Is it a set of discretionary interventions that can be used by authorities when the urge strikes them or where politics makes it convenient? Or does it comprise measures that *need* to be proactively and strategically deployed in a coordinated manner, as a matter of law, to achieve clear statutory outcomes?

Raewyn Peart

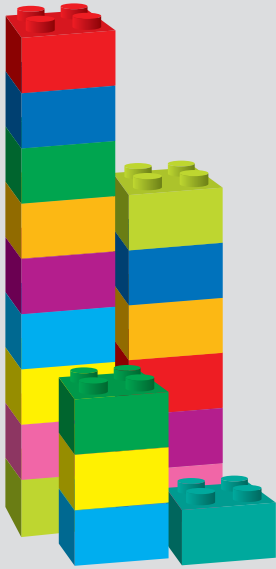


Ocean Beach, Hastings



Summary of options for reform: Strategic and integrative tools

- A future system could be made more strategic by recasting the purposes and principles of legislation to ones that drive towards a different future, rather than maintaining or protecting things or seeking static outcomes (eg wellbeing or sustainability).
- Mandatory targets could be used more systemically across a future system to drive positive change. Accountability mechanisms could be established around them to measure progress. Binding targets could cover many things, but may be particularly useful to return to a safe ecological space if limits have already been infringed.
- A future system could establish a more comprehensive range of trigger points that result in automatic or immediate management measures being taken. Here, the system would be more proactive in preparing for the future, providing greater agility when things change.
- Monitoring and reporting in a future system could be linked to obligations to conduct futures scanning exercises, to ensure that problems, opportunities and changes are pre-empted rather than leaving gaps in policy and regulatory frameworks to develop.
- Tools could be better coordinated in a future system by extending the responsibilities of institutions. If one institution has responsibilities for deploying (or engaging with) multiple tools, then they may be used in a more integrated way.
- There are a number of ways that connections could be improved between legislative frameworks, such as through cross-referencing, alignment of processes and the insertion of common principles.
- Strategies in a future system could be made mandatory (guided by revised and carefully crafted statutory purposes) and have strong legal effect on the tools needed to realise their objectives.
- An environmental research strategy, containing a specific part on marine research and information, could be made mandatory and have legal influence over how integrated research is created, funded and deployed to achieve clearer cross-cutting objectives for the marine environment. The strategy could provide for large “one-off” exercises such as a national coastal habitat mapping project.
- An Environmental Research Council or another independent agency such as an Oceans/Tikanga Commission could oversee the marine research and information system.
- An Environmental Research Council or another independent agency such as an Oceans/Tikanga Commission could oversee the marine research and information system.
- Funding for environmental research (or even marine components of it) could be ringfenced/ hypothecated using revenue from tools like resource rentals.



Summary of options for reform: Strategic and integrative tools *(continued)*

- A future system could provide for the mandatory creation of marine spatial plans to integrate or coordinate the use of other tools (regulatory and non-regulatory) in a particular place.
- Marine spatial plans could include environmental bottom lines, targets and outcomes for the marine area. They could be strategic only, and rely on implementation through other frameworks. Alternatively, they could include regulatory provisions themselves as an alternative pathway for things like environmental limits and MPAs.
- Marine spatial plans could be targeted to areas where there are particular issues or conflicts or they could cover all the country's marine areas. There could be specific trigger points specified for when a planning process was deemed necessary.
- A Marine Spatial Planning Strategy, developed by the Minister of Oceans, could provide a policy framework for marine spatial planning and set out a programme for developing marine spatial plans in targeted areas.
- A national level Oceans Policy could be a strategic instrument, going far beyond just an action plan for rolling out marine spatial plans. It could, with sufficient framing, be a form of "constitution" for the oceans.
- A statutory framework that both initiates marine spatial planning and provides agencies, mana whenua and stakeholders with guidance on principles and process steps could be provided to help with the implementation of marine spatial planning.
- Marine spatial planning could have a broad and flexible legislative framing, allowing change and innovation to occur as practice and experience evolves.



Kaikōura coast

Endnotes

- 1 Ministry for the Environment *Natural and Built Environments Bill: Parliamentary Paper on the Exposure Draft* (June 2021), cl 8.
- 2 Again, that resembles the more robust accountability mechanisms that now exist for climate change.
- 3 See Resource Management Review Panel *New Directions for Resource Management in New Zealand* (June 2020). There have been signals more recently that mandatory targets of some form may still be included in the legislation as per the Panel's recommendations.
- 4 Extending the reach of the Wildlife Act 1953. At present, the RMA and NZCPS are focused on avoiding harm.
- 5 Department of Conservation *Te Mana o Te Taiao - Aotearoa New Zealand Biodiversity Strategy 2020* (Department of Conservation, Wellington, August 2020).
- 6 The Guide moves beyond a single end goal – such as 30 percent marine protection by 2030 – instead focusing on progress. Its four stages of MPAs are: proposed/committed; designated; implemented; and actively managed. This recognises that that it may take years between an announcement to create an MPA to the time when in situ protection and management occurs, yet it is all part of the same process. The *levels* of protection are evaluated with reference to the biodiversity outcomes that different activities at different scales are expected to produce. Impact is determined by activity type, intensity, scale, duration and frequency relative to biodiversity conservation goals and is described as “none”, “low”, “moderate”, “high/large” or “incompatible with biodiversity conservation”. A future system could take a similarly nuanced approach to setting staged targets and measuring progress.
- 7 Oceans Act SC 1996 c 31, s 35(2).
- 8 For example, orcas, Right whales, bottlenose whales, belugas, leatherback turtles, abalone and seals.
- 9 See Raewyn Peart *Farming the sea* (Environmental Defence Society, Auckland, 2019).
- 10 That is conceptually similar to what is planned to happen with agricultural emissions, which will be subject to a pricing framework (within or alongside the ETS) if other measures are not achieved by a certain date.
- 11 See Jamie Morton “Back to the future: has NZ stopped looking ahead?” *The New Zealand Herald* (online ed, 13 November 2016) <https://www.nzherald.co.nz/nz/back-to-the-future-has-nz-stopped-looking-ahead/OW2X4VIAMG6YWWWTWDWRJYC6QM/?c_id=1&objectid=11745529>
- 12 See Well-being of Future Generations (Wales) Act 2015, s 4.
- 13 On the importance of locating current environmental challenges within anticipated future scenarios, see Parliamentary Commissioner for the Environment *The state of New Zealand's environment: Commentary by the Parliamentary Commissioner for the Environment on 'Environment Aotearoa 2015'* (June 2016) at 45.
- 14 See the spotlight on the *Motiti* decision in Chapter 11. We look at the risks of broadly focused institutions in Chapter 12.
- 15 Magnuson-Stevens Fishery Conservation and Management Act 16 USC §§ 1801-1891d
- 16 See Chapter 8 of the project's working paper on the institutional changes that have characterised fisheries management over recent times.
- 17 Which have not yet been created, although one is signalled to implement the Sea Change Tai Timu Tai Pari “Hauraki Gulf Marine Spatial Plan”.
- 18 Although this has not been without its issues – see Greg Severinsen *Reform of the Resource Management System: The urban context* (Environmental Defence Society, Auckland, August 2020).
- 19 Office of the Prime Minister's Chief Science Advisor *The Future of Commercial Fishing in Aotearoa New Zealand* (February 2021) at 18.
- 20 Under the Convention on Biological Diversity – see Chapter 3.
- 21 Parliamentary Commissioner for the Environment *A review of the funding and prioritisation of environmental research in New Zealand* (December 2020) at 35, 45 and following.
- 22 See Department of Conservation *The New Zealand Biodiversity Strategy* (Department of Conservation, Wellington, February 2000). Note this has been superseded by Department of Conservation *Te Mana o Te Taiao - Aotearoa New Zealand Biodiversity Strategy* (Department of Conservation, Wellington, August 2020).
- 23 At 67. For further background on the development and implementation of marine protection and MPA policy in Aotearoa New Zealand, see Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (Environmental Defence Society, Auckland, 2012).
- 24 Department of Conservation *The New Zealand Biodiversity Strategy* (Department of Conservation, Wellington, February 2000) at 67.
- 25 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (Environmental Defence Society, Auckland, 2012) at 134.
- 26 Department of Conservation and Ministry of Fisheries *Marine Protected Areas Policy and Implementation Plan* (Department of Conservation and Fisheries New Zealand, Wellington, 2005); Department of Conservation and Ministry of Fisheries *Marine Protected Areas: Classification, Protection Standard and Implementation Guidelines* (Department of Conservation and Fisheries New Zealand, Wellington, 2008).
- 27 At 3.
- 28 At 3.
- 29 At 6.
- 30 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening marine protection in New Zealand* (EDS, Auckland, 2012).
- 31 Department of Conservation *Te Mana o Te Taiao - Aotearoa New Zealand biodiversity Strategy 2020* (Department of Conservation, Wellington, August 2020) at 53-54: see objectives 10.3.3-10.5.3; objectives 10.6.1-10.6.3; and objectives 12.1.1-12.7.3.
- 32 Such as in statutory emissions reduction plans and in future urban development strategies required under the NPS on Urban Development.
- 33 See Greg Severinsen and Raewyn Peart *The Strategic Planning Act and Funding* (Environmental Defence Society, September 2021) available at <https://eds.org.nz/wp-content/uploads/2021/11/8.-The-Strategic-Planning-Act-and-Funding-FINAL.pdf>.
- 34 See generally Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening marine protection in New Zealand* (EDS, Auckland, 2012) from 178.
- 35 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (EDS, 2012) at 178.
- 36 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (EDS, 2012) at 178.
- 37 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (EDS, 2012) at 186.
- 38 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (EDS, 2012) at 180.
- 39 A Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (EDS, 2012) at 181; Marine Life Protection Act FGC § 2853(b).
- 40 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (EDS, 2012) at 182.
- 41 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (EDS, 2012) at 182.
- 42 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (EDS, 2012) at 189.
- 43 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (EDS, 2012) at 188-189.
- 44 Emily Saarman and others “The role of science in supporting marine protected area network planning and design in California” (2013) 74 *Ocean and Coastal Management* 45.
- 45 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (EDS, 2012) at 182.
- 46 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (EDS, 2012) at 182.
- 47 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding our Oceans: Strengthening Marine Protection in New Zealand* (EDS, 2012) at 183; Marine Life Protection Act FGC § 2857(c).
- 48 California Department of Fish and Wildlife “Regional MPA Statistics” <<https://wildlife.ca.gov/Conservation/Marine/MPAs/Statistics>>
- 49 Scripps Institution of Oceanography “Research highlight: Early results suggest California marine protected areas are a success” (5 September 2019) University of California <<https://scripps.ucsd.edu/news/research-highlight-early-results-suggest-california-marine-protected-areas-are-success>>
- 50 California Department of Fish and Wildlife “MPA Decadal Management Review” <<https://wildlife.ca.gov/Conservation/Marine/MPAs/Management/Decadal-Review>>
- 51 Parliamentary Commissioner for the Environment *A review of the funding and prioritisation of environmental research in New Zealand* (Office of the Parliamentary Commissioner for the Environment, Wellington, December 2020).
- 52 Parliamentary Commissioner for the Environment *A review of the funding and prioritisation of environmental research in New Zealand* (Office of the Parliamentary Commissioner for the

- Environment, Wellington, December 2020) at 64.
- 53 Noting that intellectual property concerns would need to be worked through when it comes to how mātauranga is presented and shared.
- 54 Prime Minister's Chief Science Advisor *The future of commercial fishing in Aotearoa New Zealand* (February 2021).
- 55 Prime Minister's Chief Science Advisor *The future of commercial fishing in Aotearoa New Zealand* (February 2021) at 2.
- 56 Prime Minister's Chief Science Advisor *The future of commercial fishing in Aotearoa New Zealand* (February 2021) at 3.
- 57 Prime Minister's Chief Science Advisor *The future of commercial fishing in Aotearoa New Zealand* (February 2021) at 23, 26.
- 58 UNESCO "MSP around the world" MSPglobal <<https://www.mspglobal2030.org/msp-roadmap/msp-around-the-world/>>
- 59 Directive 2014/89/EU of the European Parliament and of the Council of 23 July 2014 establishing a framework for maritime spatial planning [2014] OJ L257/135.
- 60 For a history of the development of marine spatial planning worldwide, including in-depth case studies, see Raewyn Peart *Turning the Tide: Integrated marine planning in New Zealand* (Environmental Defence Society, Auckland, 2018), ch 3 and Hauraki Gulf Forum *Spatial Planning for the Gulf: An international review of marine spatial planning initiatives and application to the Hauraki Gulf* (Hauraki Gulf Forum, Auckland, 2011).
- 61 Charles Ehler and Fanny Douvere *Marine spatial planning: A step-by-step approach toward ecosystem-based management* (UNESCO, Paris, 2009) at 18.
- 62 Directive 2014/89/EU of the European Parliament and of the Council of 23 July 2014 establishing a framework for maritime spatial planning [2014] OJ L257/135, art 3(2).
- 63 Melissa Foley and others "Guiding ecological principles for marine spatial planning" (2010) 34(5) *Marine Policy* 955.
- 64 Kelsey Serjeant and Raewyn Peart *Healthy Seas: Implementing marine spatial planning in New Zealand* (Environmental Defence Society, Auckland, 2019) at 4.
- 65 Hauraki Gulf Forum *State of our Gulf 2011* (Hauraki Gulf Forum, Auckland, August 2011) at 13.
- 66 Hauraki Gulf Forum *Spatial planning for the Gulf: An international review of marine spatial planning initiatives and application to the Hauraki Gulf* (Hauraki Gulf Forum, Auckland, 2011) at 40.
- 67 Sea Change Tai Timu Tai Pari Stakeholder Working Group: *Terms of reference* (Auckland Council, 2013) at 2-4, as cited in Environment Guide "Case study: Sea Change" (9 February 2018) <<https://www.environmentguide.org.nz/issues/marine/marine-spatial-planning/case-study-sea-change/>>
- 68 Seachange Stakeholder Working Group *Sea Change Tai Timu Tai Pari Hauraki Gulf Marine Spatial Plan* (Hauraki Gulf Forum in partnership with others, April 2017)
- 69 A copy of the plan and background documents can be accessed at <www.seachange.org.nz>
- 70 Auditor-General *Sea Change – Tai Timu Tai Pari: Creating a marine spatial plan for the Hauraki Gulf* (Office of the Auditor General, B.29[180], December 2018). This review found that in the working group, some interests were better represented than others; there was a lack of formal consultation on the proposals; and the 2017 Plan was not easily to implement for central and local government agencies. It needed more agency involvement, discussion with stakeholders, and communication with the public.
- 71 Department of Conservation, Fisheries New Zealand and Ministry for Primary Industries *Revitalising the Gulf: Government Action on the Sea Change Plan* (June 2021).
- 72 Elizabeth Macpherson and others "Hooks" and 'Anchors' for Relational Ecosystem-Based Marine Management" (2021) 130 *Marine Policy* 104561.
- 73 Elizabeth Macpherson and others "Hooks" and 'Anchors' for Relational Ecosystem-Based Marine Management" (2021) 130 *Marine Policy* 104561.
- 74 Intergovernmental Oceanographic Commission and Directorate General for Fisheries and Maritime Affairs *MSPglobal International Guide on Marine/Maritime Spatial Planning* (IOC-UNESCO, IOC/2021/MG/89, 2021) at 34.
- 75 See Mulcahy, Peart and Bull, above n 24, at 124-125.
- 76 Resource Management Review Panel *New Directions for Resource Management in New Zealand* (June 2020) at 144.
- 77 Kelsey Serjeant and Raewyn Peart *Healthy Seas: Implementing marine spatial planning in New Zealand* (Environmental Defence Society, Auckland, 2019) at 53.
- 78 Resource Management Review Panel *New Directions for Resource Management in New Zealand* (June 2020), 134-135.
- 79 Raewyn Peart and Brooke Cox *Governance of the Hauraki Gulf: A review of options* (Environmental Defence Society, Auckland, 2019) at 8.
- 80 Raewyn Peart *Turning the tide: Integrated marine planning in New Zealand* (Environmental Defence Society, Auckland, November 2018) at 36.
- 81 Ministry for the Environment *Natural and Built Environments Bill - Parliamentary Paper on the Exposure Draft* (June 2021) at [49].
- 82 Anne-Michelle Slater and Jim Claydon "Marine spatial planning in the UK: A review of the progress and effectiveness of the plans and their policies" (2020) 22(2) *Environmental Law Review* 85 at 86.
- 83 See Marine and Coastal Access Act 2009 (UK), pt 3.
- 84 Marine and Coastal Access Act 2009 (UK), s 44(1).
- 85 HM Government *UK Marine Policy Statement* (2011) at 10.
- 86 At 11.
- 87 At 7.
- 88 Marine and Coastal Access Act 2009 (UK), s 51(2).
- 89 Sections 51(3) and (5).
- 90 HM Government *UK Marine Policy Statement* (2011) at 10.
- 91 At 12.
- 92 Marine and Coastal Access Act 2009 (UK), s 55.
- 93 Scottish Government "Marine Planning Partnerships: Clyde and Shetland Isles" (5 December 2018) <<https://www.gov.scot/publications/marine-planning-partnerships-clyde-and-shetland-isles/>>.
- 94 See for example the membership of the Clyde Marine Planning Partnership: Clyde Planning Partnership "About the Clyde Marine Planning Partnership" <<https://www.clydemarineplan.scot/about-us/about-the-clyde-marine-planning-partnership/#membership>>.
- 95 Marine and Coastal Access Act 2009 (UK), sch 6, cl 1(1).
- 96 Schedule 6, cl 5(3).
- 97 Clause 5(5).
- 98 Clause 6.
- 99 Clause 7.
- 100 Clause 11.
- 101 Clause 10.
- 102 Clause 12.
- 103 Clause 13.
- 104 Clause 13.
- 105 Section 55.
- 106 Section 58.
- 107 Schedule 6, cl 58(2).
- 108 Section 61.
- 109 Anne-Michelle Slater and Jim Claydon "Marine spatial planning in the UK: A review of the progress and effectiveness of the plans and their policies" (2020) 22(2) *Environmental Law Review* 85 at 89.
- 110 Anne-Michelle Slater and Jim Claydon "Marine spatial planning in the UK: A review of the progress and effectiveness of the plans and their policies" (2020) 22(2) *Environmental Law Review* 85.
- 111 Lucy Greenhill, Tim Stojanovic and Paul Tett "Does marine planning enable progress towards adaptive governance in marine systems? Lessons from Scotland's regional marine planning process" (2020) 19(5) *Maritime Studies* 299 at 306.
- 112 Raewyn Peart *Turning the tide: Integrated marine planning in New Zealand* (Environmental Defence Society, Auckland, 2018) at 23.
- 113 Compare Office of the Prime Minister's Chief Science Advisor *The Future of Commercial Fishing in Aotearoa New Zealand* (February 2021).
- 114 See Minister for Oceans and Fisheries "Oceans and Fisheries portfolio – ensuring healthy ocean ecosystems" (2 July 2021).
- 115 For example, rainwater tanks that mitigate the need for alternative sources like desalination which could have consequential impacts on the marine environment.
- 116 Ministerial Advisory Committee on Oceans Policy *Healthy sea, healthy society: Towards an Oceans Policy for New Zealand* (Oceans Policy Secretariat, 30 September 2001) at 3.
- 117 Karen Scott "Does Aotearoa New Zealand Need an Oceans Policy for Modern Oceans Governance?" (2021) 35 *Ocean Yearbook*.
- 118 Such elements, with supporting institutions, could be given formal recognition in an Oceans

- Act, says Scott: at 38 and following. On an Oceans Act, see Chapter 6.
- 119 Karen Scott "Does Aotearoa New Zealand need an oceans policy for modern oceans governance?" (2021) 35 *Ocean Yearbook* at 38-43.
- 120 At 26-29.
- 121 Prime Minister's Chief Science Advisor *The future of commercial fishing in Aotearoa New Zealand* (February 2021) at 17.
- 122 At 17.
- 123 Elizabeth Macpherson and others "'Hooks' and 'Anchors' for Relational Ecosystem-Based Marine Management" (2021) 130 *Marine Policy* 104561.
- 124 Office of the Prime Minister's Chief Science Advisor *The Future of Commercial Fishing in Aotearoa New Zealand* (February 2021).
- 125 Raewyn Peart *Looking out to sea: New Zealand as a model for oceans governance* (Environmental Defence Society, Auckland, 2005) at 149-151.
- 126 At 150-152.
- 127 Elizabeth Macpherson and others "'Hooks' and 'Anchors' for Relational Ecosystem-Based Marine Management" (2021) 130 *Marine Policy* 104561.
- 128 Government of Canada *Canada's Oceans Strategy: Our Oceans, Our Future* (Fisheries and Oceans Canada, Ontario, 2002).
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- 131 Megan Bailey and others "Canada at a crossroad: The imperative for realigning ocean policy with ocean science" (2016) 63 *Marine Policy* 53.
- 132 Through mechanisms in Canada's Oceans Act see Oceans Act SC 1996 c 31.
- 133 Legislative Design and Advisory Committee *Legislation guidelines: 2018 edition* (March 2018) at 7. Compare Geoffrey Palmer "Law-making in New Zealand: Is there a better way?" (2014) 22 *Waikato LR* 1.
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- 135 Fisheries and Oceans Canada *Final evaluation report: Evaluation of the Ocean Management Program* (online ed, 17 January 2018). See generally Megan Bailey and others, "Canada at a crossroad: The imperative for realigning ocean policy with ocean science" (2016) 63 *Marine Policy* 53.
- 136 Similar to the Auckland Plan, which is a statutory document but has weak links to regulatory frameworks.
- 137 Fisheries and Oceans Canada *Final evaluation report: Evaluation of the Ocean Management Program* (online ed, 17 January 2018).
- 138 Fisheries and Oceans Canada *Final evaluation report: Evaluation of the Ocean Management Program* (online ed, 17 January 2018).
- 139 Raewyn Peart *Turning the tide: Integrated marine planning in New Zealand* (Environmental Defence Society, Auckland, 2018) at 28.
- 140 See Waikato Regional Council "Coastal policies and plans: Revitalising the Gulf" <<https://www.waikatoregion.govt.nz/council/policy-and-plans/coastal-policy/>>
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- 149 At 74.
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- 151 At 76.
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11 Legislative design



Parengarenga harbour

11.1 Introduction

It will be important to provide for a more integrated toolkit in a future system. Marine spatial plans and an overarching Oceans Policy provide potential ways forward. But in a first principles rethink, there is an opportunity to improve integration through deeper means, by restructuring our statutes. That could involve refining the boundaries between existing statutes, integrating them together, or splitting them along completely different lines. For example, it might be possible for a tool like marine spatial planning, discussed in Chapter 10, to be provided for within the RMA/NBA (eg through a regional policy statement), the proposed Strategic Planning Act (through regional spatial strategies), new MPA legislation, or a new, integrated Oceans Act. Fisheries tools like sustainability measures do not necessarily require a Fisheries Act. Nor do MPAs require an MPA Act.

In this chapter we consider why legislative design matters, how statutes could be layered quite differently to the status quo if we look through different “lenses”, and what that means when it comes to options for the future. We conclude by looking at what a radically different approach might look like – an Oceans Act. Overall, we are encouraged by the general direction in the Legislation Act 2012 to facilitate “the progressive and systematic revision of the New Zealand statute book so that . . . it is arranged more logically”.¹ The extensive fragmentation of the current oceans management system means it is a particularly good candidate for thinking about such things.

11.2 Why legislative design matters

It is by no means obvious what legislative arrangements would be “best”. It is not even clear whether what we have now is fundamentally broken. Most of the statutes we have at the moment could probably be justified in some way (for instance, the EEZ Act is a simpler framework than the RMA and responds to its different international legal framing)² and splitting them up differently risks replacing one issue with another.

Much may depend on whether our current approach is actually responsible for the biophysical and systemic problems discussed in Chapters 2 and 3. Even if it is, mechanisms other than legislative redesign (eg changes to the toolkit such as introducing marine spatial planning, or institutional measures such as providing for stronger oversight by an Oceans Ministry or Commission)³ might be equally or more effective at addressing problems with potentially less disruption. Some may see problems in the system as the product of poor implementation (eg a lack of political will to use tools like national direction, some sustainability measures and section 33 transfer of powers) or resourcing rather than the basic structure of our laws.

Others have suggested that fundamental legislative rearrangement does not get to the core of the issues we face, and more often is used as a way for policy makers to show something is being done. There is some truth here, in that:⁴

New Zealanders tend to exhibit an innocent and misplaced faith in the efficacy of legislation. We seem to be addicted to passing legislation for the sake of it. We seem to believe it will solve our innermost ills. The government must be seen to be acting or reacting. Passing a law is seen to be doing something.

Indeed, filling gaps like the lack of marine reserves in the EEZ, improving processes like planning under the RMA, and modernising the values underpinning legislation like the Wildlife Act would not necessarily require changes to statutory boundaries. Change could be driven through a series of targeted amendments or even, in some cases, subordinate instruments or non-statutory mechanisms. It would be reasonably straightforward, for example, to extend the application of the Marine Reserves Act to the EEZ, to change the purpose in the Wildlife Act, or to make the creation of an EEZ policy statement mandatory.

However, broader legislative boundaries do matter. Arguably many issues with the system stem from, or are exacerbated by, the manner in which we have designed them – where lines are drawn, what falls between the cracks, and how statutes interact with each other. If anything, the marine context has seen more carve-outs and workarounds than on land.⁵ In particular, the process of rationalising the management of different domains and sectors into one Act on land (in the RMA) never really occurred at sea (see Chapter 4).



Whale stranding, Coopers Beach

Raewyn Peart

Even with integrative tools like marine spatial planning, there are serious questions about whether this “glue” will be strong enough to give the overall system coherence and integration. Intra-statutory connections still tend to be tighter than those between statutes. A single act has a unifying purpose, processes that are designed to talk to each other, hierarchies between instruments that tend to be clearer, and (usually) a single agency responsible for oversight. In contrast, separate statutes can have conflicting purposes and different administering agencies. This can lead to regulatory impasses.

Fragmentation also risks processes that are disconnected. For example, sustainability measures under the Fisheries Act, plan reviews under the RMA, and the creation of protected areas under the Marine Reserves Act are seldom thought about in the same breath. Separation can produce conflicts that are either ignored or require litigation (eg to clarify the respective roles of the RMA and Fisheries Act in habitat protection). All these things are exacerbated when new statutes must interact with very old ones, and where the values underpinning them are either unclear or differ markedly.

The more statutes we have, the more interfaces are needed. This can create uncertainty, inefficiency and complexity. The words of the then Parliamentary Commissioner for the Environment two decades ago, are even more valid today, given the creation of yet more statutes, agencies and strategies since that time:⁶

we have an extraordinary plethora of legislation and agencies with marine responsibilities. There are 18 main statutes, 14 agencies and six government strategies for marine management. We have also signed up to at least 13 international conventions with marine implications. Inevitably there are gaps, overlaps and inefficiencies.

This does not necessarily make a single integrated statute the best option. But it suggests we should avoid having dozens of different statutes and creating new ones whenever a new intervention is needed. A single statute is also generally amended in a way that retains the coherence of the whole, whereas connections with others can be ignored or considered less closely. This can lead to frameworks growing apart over time, which is particularly noticeable with conservation laws that have been layered over many decades. Reforms to one statute might even leave a related one languishing in the values and assumptions of previous centuries. If the Marine Reserves Act had formed part of the reforms in 1991, it is highly doubtful that it would have the limited purpose it has today. The case of the Bryde’s whale shows how legislative design can matter in practice.

A spotlight on the Bryde’s whale: Why legislative fragmentation can matter

Bryde’s whales are nationally critically threatened. The Hauraki Gulf is favoured habitat for the whales, with up to 50 Bryde’s whales regularly using the Gulf at any one time, out of a larger national population of around 150. The Hauraki Gulf is also the location of the country’s busiest port (at Auckland). Geographically, the Bryde’s whales and commercial vessel traffic occupy overlapping water space. As a result of the spatial conflict, at least 17 whales have been killed by ship strike (with an historic average of around two deaths per year).⁷

In 2010, when the issue was publicised by scientists studying the whales, many of the ships coming and going from the Hauraki Gulf were travelling at well over 14 knots. International research indicated that there was a high chance of a whale dying if hit by a vessel travelling at that speed. But if the vessel’s speed was under 10 knots, an impacted whale had a good chance of surviving. For this reason, researchers concluded that vessels transiting the Hauraki Gulf needed to slow down (to 10 knots or less) in order to reduce the risk to the whales to an acceptable level, whilst maintaining safe ship navigation.⁸ It would seem a remarkably simple solution.

However, at least four potential mechanisms can be used to reduce vessel speed to protect whales, under existing legislation. The first is for the Minister of Conservation to create a marine mammal sanctuary under the Marine Mammals Protection Act with regulations setting a maximum speed within it. The second is for Auckland Council to change the coastal component of the Unitary Plan, under the RMA, to incorporate a new rule that makes operating a ship at speeds greater than 10 knots within the Hauraki Gulf a prohibited activity. Given Policy 11 of the NZCPS, which requires the *avoidance* of adverse effects on threatened species, there may well be a duty on the Council to act in this way. The third is for the Minister of Transport to recommend the creation of a Maritime Rule by Order-in-Council under the Maritime Transport Act to restrict ship speed in the Hauraki Gulf. The fourth potential mechanism is to present a proposal to the International Maritime Organization for a ship routing measure which could be voluntary or mandatory.

The advantage of such an international measure, is that the restriction would be noted on the relevant nautical charts, and would thereby be brought to the notice of overseas vessels entering the country.

This plethora of potential tools vividly illustrates the overlaps that exist between legislation. The targeted purpose of one Act – in this case, one would naturally gravitate towards the Marine Mammals Protection Act which has a protective purpose for the species in question – can in practice be achieved through tools provided under quite different legislation. This is not necessarily a bad thing, as it can be useful to have several regulatory tools in the toolbox when seeking to address an issue. However, overlap can lead to paralysis, because no agency is obviously in charge of addressing the issue. Agencies are often risk averse and would prefer, if possible, to leave politically contentious issues to someone else.

In the Bryde's whale example, none of the agencies with regulatory tools at their disposal took action to solve the issue. In this regulatory vacuum, fortunately a consortium of Auckland University scientists, the Hauraki Gulf Forum and EDS initiated a collaborative process which resulted in a voluntary agreement to reduce ship speed.⁹ This has undoubtedly saved many whale lives. Yet the underlying problem with the system remains.

While overlapping statutes can be a conscious design choice (two different statutes may deal with the same issue from different standpoints or using different tools, while overlapping spatially or in subject matter),¹⁰ they can, ironically, create gaps by giving a sense of complacency where a solution is assumed to belong elsewhere.

Yet it is a jurisdiction that has not really been exercised. Nor have habitat protection powers under the Fisheries Act been fully utilised, meaning that neither framework has been used to its full potential. Things fall between the cracks, and much remains unclear.

Overlaps between legislation can also cause complexity and multiple points of potential failure, where tools need to be used carefully in tandem to make a difference. This requires cooperation and joint processes that are not always easy to achieve under separate statutes. Difficulties in deploying large-scale MPA networks can be put down partly to the perception that they are just about the removal of existing “rights” rather than a broader discussion about what goes where. An MPA Strategy and marine spatial planning could assist (see Chapter 10) but we note that the root cause is we have separate statutes doing quite separate things.

This is very visible in the Hauraki Gulf, where the need for coordinated action across many frameworks and institutions (eg those addressing sedimentation, MPAs, fishing controls, Māori interests, ecological enhancement activities and other things) has resulted in a long non-statutory spatial planning process driven partly outside government. This has been followed by a protracted period of government consideration, before any marine protection tool is actually deployed in its service. Indeed, the waiting game continues at the time of writing.



Mussels, Ōtata

Raewyn Peart

The Bryde's whale is not an isolated example. It is characteristic of the system as a whole, which is split in ways that do not take an ecosystem approach to marine habitats or species. The most obvious example is the protection of marine habitats that are important to both fisheries and the broader biodiversity of the environment. This is dealt with under the RMA (and potentially EEZ Act), the Fisheries Act and the Marine Reserves Act. As explored later in this chapter, the relationship between the former two acts has recently been the subject of significant jurisprudence from the Court of Appeal, which concluded that regional councils retain substantial jurisdiction to regulate the impacts of fishing in the coastal marine area.¹¹

A spotlight on marine protected areas: the consequence of fragmentation

Fragmented legislation has meant that the existing system has struggled to progress the deployment of MPAs. Despite the existence of a non-statutory strategy (see Chapter 10), MPA creation has proceeded in an ad hoc and isolated manner.

There are notable gaps in the toolkit itself, of course (see Chapter 8), including within the EEZ. Yet statutory fragmentation is a big part of the picture too. The workaround for the EEZ gap has largely been through the use of Fisheries Act tools, meaning that such offshore “MPAs” are subject to the Fisheries Act purpose and principles rather than norms specifically designed for what we want MPAs to achieve. This has led to some existing MPAs within the EEZ being found wanting when it comes to protecting a representative range of biodiversity.¹²

They are also poorly integrated with other legislation. The protection of benthic areas from bottom trawling under the Fisheries Act does not protect them from non-fishing activities that can also damage seafloor biodiversity, such as deep sea mineral mining. This lacuna in protection was highlighted in the Chatham Rock Phosphate case.

Chatham Rock Phosphate Ltd obtained a mining permit for an area of the EEZ on the Chatham Rise under the Crown Minerals Act. The company then applied for a marine consent under the EEZ Act to physically remove phosphate nodules from the seabed. A complicating factor was that the nodules themselves supported rare cold-water coral communities, which would be destroyed by the mining activity, and would not re-establish once the phosphate was removed. Approximately half the proposed mining area overlapped with approximately 60 percent of a benthic protection area (the overlap being 5,236 square kilometres). While the decision-making committee turned down the application, partly because granting consent would be inconsistent with the benthic protection area, there was no legal requirement for it to do so on this basis.

Another key concern is that the expected outcomes of MPAs, even if they are established, may be undermined if other

impacts (in addition to fishing) are not managed effectively. Integration between land and sea is crucial for effective marine protection. Notable challenges are land-based sedimentation and pollution, which can impact on all types of MPAs (whether marine reserves intended to protect nature, or spatial controls intended to protect or recover fish stocks for utilisation).¹³ While the NZCPS has a sedimentation policy, it is weak (only requiring that activities not result in a *significant* increase in sedimentation) and has not been effective in addressing adverse impacts on MPAs in practice.

Sedimentation is a particular concern for the Auckland Long Bay Okura Marine Reserve, as earthworks for new urban development within the catchment, has created much greater sediment risk. Increasing housing development and intensification also causes substantial stormwater runoff with an associated cocktail of pollutants.¹⁴ Plastics, created on land, creep through the food chain and do not respect the invisible boundaries of a marine reserve. One can legitimately ponder what the point of preventing fishing, mining and other sea-based activities is, if the values being protected can be so drastically undermined by land-based stressors.

Additionally, creating MPAs can result in perverse environmental outcomes. They can displace fishing effort resulting in greater fishing pressure (and consequent depletion of marine life) in nearby areas. There is also the likelihood of increased fishing pressure around the boundary of MPAs themselves, reducing the effectiveness of the protective measures, as harvestable species moving outside the MPA boundary are caught.¹⁵ This means that MPA creation may need to be accompanied by other fisheries management measures, such as creating protective buffer areas around MPAs, and reducing the TAC for the broader area (and potentially buying back quota). The Sea Change Tai Timu Tai Pari Hauraki Gulf Marine Spatial Plan contained proposals along these lines, including the creation of Ahu Moana as a protective cloak around MPAs and measures to address any adverse effects of their creation on commercial fishers. As stated in the plan:¹⁶

Importantly, Ahu Moana are intended to be used as korowai (cloaks) to wrap around other types of MPA, buffering them from the edge pressures previously described.

The impacts of legislative fragmentation can be seen, not only in the impacts of land-based sedimentation on marine reserves, but also its effects on fishing. These impacts (eg the smothering of shellfish beds and destruction of habitats) are well established, but there is a disconnect between the interests that are affected (fishing) and the framework under which those impacts are managed (the RMA). There is no clear or directive mechanism by which the fishing aims of the Fisheries Act – maximising utilisation of stocks – can lead to controls on land-based activities generating sediment under the RMA (when, for example, a regional plan is being reviewed or urban growth is being considered under a district plan or regional policy statement). Sustainability measures under the Fisheries Act cannot be used to prevent activities further up catchments, only to control the impacts of fishing itself.¹⁷ As we pointed out previously:¹⁸

there is a serious lack of connection between management under the RMA and the protection of important fish habitat, with consequent impacts on the productivity of fish stocks, highlighted by the lax controls on forestry harvesting in the Marlborough Sounds with likely consequential effects on struggling blue cod and pāua stocks.

Conversely, some have questioned the extent to which controls under the Fisheries Act, in practice, take into account current and future land-based stressors when considering how much additional stress to put on stocks through fishing activity (eg when setting catch limits).

Where there are arbitrary spatial distinctions between frameworks this can also cause perverse incentives. The case of putting hard boundaries

around “protected” and “non-protected” areas like marine reserves has been mentioned above. Moreover, some may perceive the EEZ Act to be a “softer” framework than the RMA (partly due to the more robust participatory and notification provisions of the latter) and choose to conduct potentially harmful activities just on the seaward side of the 12 nautical mile boundary. For example, Ports of Auckland has an RMA permit from Auckland Council to dredge the seabed, but its marine dumping consent comes under the EEZ Act because the disposal site is just outside the boundary of the coastal marine area. This meant that over half the submissions lodged opposing the dredging permit were struck out, as they related to the impacts of dumping, which was deemed “out of scope” in RMA proceedings.¹⁹

Although legislative redesign is not a panacea for other problems in the system, a poorly designed suite of legislation can have real impacts. In particular, excessive fragmentation, or splitting statutes along inappropriate lines, can cause confusion, incoherence, inaccessibility and poorer environmental outcomes. Ironically, overlapping jurisdictions can create gaps in management, while spatial boundaries can create perverse incentives. The effects of fragmentation can be seen in the difficulties around protecting the Bryde’s whale and in creating a network of MPAs. These issues with statutory boundaries cannot be entirely addressed through integrative tools like marine spatial planning or creating better cross-referencing.



Sediment-generating earthworks, Beachlands, Auckland

11.3 Pragmatism and principle in legislative design

It is much easier to point out flaws in the existing system than to provide perfect alternatives. To some extent, boundaries will always be problematic. And the issues described above are not caused by fragmentation per se; instead, they may be a product of the specific *lines* along which statutes have been divided.

Thinking about design *principles* might help policy makers to evaluate pros and cons of different options and make considered judgements. Some substantive principles from Chapter 7 may be relevant here. For example, ecosystem-based management may tend to support options that integrate ecological concerns in a single legislative framework, including the ecological connections between land and sea, rather than having sectoral or spatial silos. But more targeted legislative design principles can also be considered. We refer readers to our previous work on resource management reform for more explanation of such principles.²⁰ In short, we should ensure that legislation and its boundaries are coherent, certain, accessible, durable, integrated, tailored to Aotearoa New Zealand's circumstances (including te Tiriti o Waitangi), and efficient.

In particular, a system is coherent if it has “clarity and intelligibility” or is “logically ordered”;²¹ and relationships between statutes need to be designed so it is clear how they operate together (even if they may be intentionally conflicting).²² A suite of legislation should be designed in a way that is intuitive and accessible to those who use it (avoiding extensive exceptions and carve-outs);²³ and people (including institutions) need to understand why statutes are arranged as they are, partly in order to determine easily whether they are affected by them.

Duplication and overlap should be avoided unless there is good reason.²⁴ Practically speaking, it may be that Te Tiriti settlement legislation needs to remain separate to avoid undermining its more targeted and carefully negotiated purpose. However, beyond that, a tikanga approach might also suggest a more integrated approach to the management of different sectors.

Legislative design principles can help in determining how we divide up the statute book and evaluate the benefits and costs of different options, although they do not provide hard and fast answers.

The principle of coherence tells us that close consideration should be given, not just to *how* statutes could be split up differently in the future, but also *why*. What is the reason for putting something in a separate statute?

One answer is that there is no real rationale. The statute book evolves organically to respond to issues as they emerge. It is a product of democratic discourse and historical circumstance, both of which are messy (see Chapter 4). The existing system can, to some extent, be explained in this way.

The role of pragmatism in legislative design

Some people speak of Aotearoa New Zealand's “statute book” as if it exists as a grand, coherent code of law. The reality is that legislation has emerged over many decades in a fairly ad hoc way. From time to time efforts have been made to create better integration (eg the reform process leading to the RMA or the creation of the umbrella Conservation Act), but such measures have been more than counteracted by a broader trend towards fragmentation, carve-outs and gap filling.²⁵

To some extent this is because it is easier to create a statute to respond to a particular issue than it is to reimagine or even amend wide ranging frameworks like the RMA. There is never the simplicity of a blank slate. For example, we have a lot of bespoke statutes like the Hauraki Gulf Marine Park Act and Fiordland (Te Moana o Atawhenua) Marine Management Act, which create more nuanced marine protected areas and integrative place-based mechanisms (including new institutional features) than are available under more general legislation like the Marine Reserves Act and RMA. It may have been simpler to create the EPA via a standalone act than add a new part to the Environment Act.

It is also interesting to consider the widespread view that we need to create a separate act for MPAs – an MPA Act – to replace the Marine Reserves Act. This is despite the potential for a range of spatial protections to be delivered through amendment to the purpose and machinery of broader frameworks like the RMA, EEZ Act, Wildlife Act, Fisheries Act and Marine Mammals Protection Act.²⁶ It can be simpler to just plug holes, create overlays and avoid the noise that inevitably comes from messing with the fabric of multiple other acts. The same phenomenon was seen when the EEZ Act was enacted; while policy makers considered more systemic options, they settled on something that just filled gaps.²⁷

Statutes also reflect the gradual development and layering of new areas of concern as they emerge over time. Conservation laws have developed iteratively, from the Wildlife Act, to the Marine Reserves Act, to the Marine Mammals Protection Act, to the Conservation Act and various bespoke statutes for particular places. They grapple with different issues as they have become more important, but reflect the concerns of the time they were enacted. Similarly, the need to manage fish stocks and prevent collapse pre-dated the realisation of the complexity of the connection between fisheries, habitat, land-based pressures and climate change. It is not surprising, then, that a well-developed and integrated fisheries framework – a lengthy, standalone act – has been in place for a long time, while conservation laws in the marine space are comparatively fragmented, simplistic and patchy.

However, some boundaries have been a conscious system design choice rather than just a pragmatic and reactive response to new issues. It is not just that it is too hard to reshuffle legislation; it is also, perhaps, because policy makers do not want to. Such choices partly reflect historical circumstances and assumptions about what the focus of management should be. For example, a sector-specific Fisheries Act has been deliberately separated from broader outcomes-focused frameworks like the RMA and EEZ Act, despite fish stocks being a natural resource that would otherwise fall within their ambit. The RMA was not simply layered on top of the 1983 Fisheries Act to respond to a particular issue as it emerged; instead, a conscious choice was made to continue to treat the frameworks separately. Fishing had always been a world apart with its own targeted institutions, management tools, stakeholders and norms.²⁸ And so it has remained.

Similarly, the rate of depletion of marine minerals was deliberately excluded from the RMA and EEZ Act, partly because minerals are finite resources that cannot be managed “sustainably” and the Crown has ownership rights, but also because a sectoral silo for mining had existed for a long time and was already well-developed.²⁹ Climate change has been largely carved out into a separate legal framework, despite coming within the purpose of the RMA.³⁰ The issue is too cross-cutting to deal with through changing the dozens of statutes that would otherwise need to be amended. It was simpler to place a statutory emissions reduction plan over the top of them and create an emissions trading scheme to the side. And it has been suggested that novel marine activities, such as carbon capture and storage

and offshore energy may end up with their own bespoke statutes and carve-outs, rather than being welcomed into the fold of broader acts like the RMA.³¹ These are practical responses.

Statutes can also be separated to give a focus to a particular area of concern. This could be a sector (eg shipping), an outcome (eg Covid-19 recovery), a resource (eg minerals) or a tool (eg an MPA). For example, submarine pipelines and cables are protected in bespoke legislation, even though their protection could conceivably be integrated into the RMA (which already protects areas from interference through designations and can spatially separate activities) or Maritime Transport Act (which already deals with the impacts of ships and international frameworks for liability). Moreover, a sectoral focus for marine oil spill preparation and response under the Maritime Transport Act highlights the value of focusing on the source of harm when objectives are clear (prevention), consequences are severe and direct (for health and marine life), and clear institutional responsibility is needed to deal with emergency situations (more akin to civil defence). Passing a targeted act can be a visible symbol that something *matters*.

None of this is to say such pragmatic approaches are necessarily *wrong*. As will be seen, there are some good reasons to split statutes up in these (and other) ways. Not least, the RMA would probably double in length and complexity if it tried to incorporate something like the emissions trading scheme or the QMS. Rather, we simply observe that the statute book is a bit like the universe: over time, it trends towards chaos.³² Carve-outs are created. Gaps are filled. And this can be as much the result of pragmatism, political compromise, and historical circumstance as it is about principle.

A future oceans management system could continue to take this approach, introducing targeted legislation to tackle new issues as they emerge. Area-specific MPA statutes – one for the Hauraki Gulf, one for the south-east of the South Island, and another for the Rangitāhua/Kermadec Islands ocean sanctuary – might be more politically achievable than tackling a blanket MPA Act. Such flexibility can have benefits when compared to grand legislative schemes, in that:³³

it needs to be reasonably clear from the purpose of a statute what is expected of [people] under it. There should not be so few statutes as to make it unclear what exactly each one does... the RMA itself is guilty of this, in its wide and contestable definition of sustainable management.

To some extent legislative design is, and must be, a product of pragmatism. It can be difficult to change large legislative schemes, and sometimes more effective and focused action can be achieved by enacting bespoke statutes as new issues arise.

A more principled explanation for dividing legislation is possible, however. The starting point is that every act has a different purpose.³⁴ If a purpose is distinct enough, it belongs in dedicated legislation.

For instance, fisheries and minerals legislation are focused on particular industries, but arguably the *reason* they have a separate existence is not because sectors require bespoke legislation per se. Rather it is because the purpose of managing those sectors (essentially, maximising long-term utilisation of fish stocks and maximising a financial return to New Zealanders) is quite different to the purpose of acts like the RMA. Nationally valuable resources like these are to be managed in an active, not passive, manner.³⁵ We do not have the same imperative with other resources (eg tidal energy), so we have no need for a separate “Tidal Energy Act” (yet). The same kind of purposive distinction can be seen in the division between the RMA and conservation legislation like the Marine Reserves Act and Wildlife Act. They are trying to achieve different things,³⁶ even though they are managing some of the same subjects (“environmental” impacts).³⁷

All this begs the question: when do two statutory purposes become distinct enough to warrant structural separation? For example, one could say that the “purposes” of two regional coastal plans under the RMA are quite different from each other, because they respond to different geographical contexts and have bespoke objectives. Yet that does not justify separate *statutes* for each.

Even more striking is the fact that water conservation orders under the RMA have their own more targeted purpose *within* one act.³⁸ Some statutes cobble many different purposes together in one place.³⁹ Yet the very similar purposes of the EEZ Act and RMA have been seen as distinct enough to warrant separate statutes.⁴⁰ In short, saying that legislative boundaries should be defined by different statutory purposes does not get us very far.

The most fundamental reason for putting something in a separate statute is that it has its own purpose. However, it is not always clear how purposes, and therefore statutory boundaries, should be divided up. Purposes can be highly specific or very broad, and they can overlap.



Danish seine net, Hauraki Gulf

Raewyn Peart

11.4 Statutory lenses

The concept of “lenses” can take us further. A lens is, essentially, about what our main concern is when we slice and dice legislative boundaries (and their purposes). We explained this in our work on resource management reform, where we invited the reader to:⁴¹

consider the RMA, which has an extremely wide scope. It deals with water and air quality, ecosystem health, noise pollution, the development of land, and many other things. It regulates activities based on the impacts they have on the environment, not the sector to which an activity belongs (eg agriculture), the kind of person doing the activity (eg the government), or the area in which something happens (eg cities).⁴² For convenience, we can call this kind of statute an

“outcomes-based” one ... Other statutes may be “sectoral”, in that they regulate a particular industry or activity (eg an act regulating mining or an act regulating transport).

There are many lenses that could be looked through. They could, for example, be spatial (we split up statutes based on their location), domain-based (concerned with the resource or subject being managed, like fish or marine wildlife), tool-based (an individual statute provides a home for one or more interventions, like the emissions trading scheme and QMS), or institutional (each statute covers some or all the statutory responsibilities of one or more particular institutions, like Maritime New Zealand). Every one of these lenses has, to some extent, been used to split statutes in the current system (see Figure 11.1). We explore some of these below, before considering potential reform options for the future.

Outcome	Domain	Sector	Space	Institutional and administrative	Tool
RMA	Climate Change Response Act	Fisheries Act	Some te Tiriti settlement legislation	Environmental Protection Authority Act	Marine Reserves Act
EEZ Act	Marine Mammals Protection Act	Fisheries Settlement Acts	Kaikōura (Te Tai o Marokura) Marine Management Act	Local Government Act	
Biosecurity Act	Wildlife Act	Crown Minerals Act	Sugar Loaf Islands Marine Protected Area Act	Local Government (Auckland Council) Act 2009	
Conservation Act	MACA Act	Continental Shelf Act	Fiordland (Te Moana o Atawhenua) Marine Management Act	Environment Act	
Hazardous Substances and New Organisms Act	Heritage New Zealand Pouhere Taonga Act	Maritime Transport Act	Hauraki Gulf Marine Park Act	Environmental Reporting Act	
Waste Minimisation Act		Submarine Cables and Pipeline Protection Act		Territorial Sea, Contiguous Zone and Exclusive Economic Zone Act	
Litter Act		Building Act			

Figure 11.1: The different lenses through which existing marine legislative boundaries have been created. Some statutes having an influence on te moana span multiple systems, including public policy areas like education, property and health and safety.

There are different lenses we can look through when deciding where to draw boundaries between statutes. For example, a statute's boundaries can be defined by its focus on (1) achieving a particular outcome, (2) regulating a particular sector, (3) managing a specific domain or resource, (4) dealing with a geographically defined space, (5) providing a particular tool, or (6) laying out the functions and powers of a particular institution.

An outcomes-based lens

As mentioned above, one type of division is an outcomes-based one. Here, statutes are split up because each is designed to achieve a particular type of outcome irrespective of space, sector, domain or institution. These often exist where achieving an outcome relies on managing all these things in an integrated way, suggesting that a single statute (and therefore a coordinated use of tools within it) is desirable.

However, it is not always straightforward to determine what a single outcome should encompass, and therefore what the boundaries of such statutes should be. For example, the RMA seeks an outcome of sustainable management. This sounds very broad (and it is) but it does not extend to other outcomes like the active utilisation of a resource (which is sought for sectors like fishing and mining), the permanent and strict protection of species or spaces (eg marine reserves and marine mammals), or the ownership of marine resources (eg the foreshore and seabed). An

alternative way to split up outcome-based statutes might be to focus one on environmental protection and another on economic development (which was one option promoted for RMA reform), although such outcomes would still have considerable overlap.

Focusing on a single cross-cutting outcome like sustainable management may sound integrative (and to some extent it is), but it can also have fragmentary effects by splitting things up in other ways. For example, other outcomes might need to be achieved in a particular space (eg the ocean), in the regulation of a particular sector (eg mining), or in the protection of a particular species (eg the fairy tern).

A statute can be defined by the type of outcome it seeks. These statutes cut across spaces, sectors, and domains. An example is the RMA, which seeks the outcome of “sustainable management” irrespective of who is conducting an activity, where it is⁴³ or why they are doing it.

A spatial lens

If one looks through a spatial lens, different statutes would apply to different places. This kind of division exists at the moment, for example, between the RMA and the EEZ Act, where there is a sharp jurisdictional line drawn at a 12 nautical mile limit. We can define their scope by drawing lines on a map. Other spatial boundaries can be seen between Acts focused on the marine environment, like the Marine Reserves Act and Submarine Cables and Pipelines Protection Act,⁴⁴ and statutes that span both land and sea, like the RMA, Biosecurity Act, Wildlife Act and (to some extent) the Fisheries Act.⁴⁵

Another manifestation of a spatial division is between legislation that applies across the whole of a marine area (like the Conservation Act), and that which applies only to very specific areas of the marine space (eg the Hauraki Gulf or Fiordland).⁴⁶ Often the reason for place-based legislation is not to create carve-outs, but rather to create an additional layer of management (ie *both* frameworks apply in a particular place) or to make sure the tools under other frameworks are being used in a coordinated and place based way (eg that MPAs, fishing restrictions and aquaculture space are mapped or planned together). In other words, fragmenting statutes across different geographical spaces can be one way of integrating the management of different sectors, resources or domains *within* that space.

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Feeding sea lion pup, Otago Peninsula

A statute can be defined by the space or location it applies to. This reflects that different areas may require different treatment or a more integrated approach to management. Many spatially defined statutes are observable in the current system. An example is the Hauraki Gulf Marine Park Act.

A sectoral lens

Another lens through which legislation can be split is sectoral. Here, different statutes apply to particular ways in which people use a resource (eg fishing, mining, shipping or tourism). The term “sector” is being used in a broad sense as a type of human activity (eg recreational boating, fishing and diving), not just where a use is commercial.

In the current oceans management system there is a notable division between outcomes-based acts like the RMA and EEZ Act and sectoral statutes like the Fisheries Act, the Crown Minerals Act and the Maritime Transport Act. The former set of laws is focused on the environment itself, whereas the latter are focused on the particular *ways* in which people use elements of the environment. Another way of describing this distinction might be statutes that are effects-based and those that are activity-based.⁴⁷

A statute can be defined by the sector (ie type of human activity) it applies to. There are a number of sectoral statutes in the current system, such as the Fisheries Act.

Sectoral boundaries can also be drawn to match the responsibilities or mandate of a specialist regulator or other institution. These two things sometimes go hand in hand. For example, under the Fisheries Act, Fisheries New Zealand (and the Minister) have the dominant role. The point of having a separate act can be partly to allow an entity to have close and integrated control over decision-making for a particular sector, where specialist expertise is required. Similarly, the Crown Minerals Act is largely the territory of the Minister of Energy and a branch of Ministry of Business, Innovation and Employment. And the Maritime Transport Act is a collection of matters designed to form a coherent set of responsibilities for Maritime New Zealand, as much as it is an act about shipping.⁴⁸

By contrast, outcome-based statutes like the RMA tend to involve a wider range of institutions (eg councils, courts, commissioners and multiple Ministers and agencies). While the Ministry for the Environment is responsible for overseeing the RMA, it is a stretch to say the statute is defined by the Ministry's role in the same way as the sectoral frameworks described above.⁴⁹

The boundaries of sectoral statutes (eg the Maritime Transport Act) can also be defined by the responsibilities of a particular regulator or public authority overseeing that sector (eg Maritime New Zealand).

A “domain” based lens

Statutes can be divided according to particular domains. The term “domain” defies universal definition. However, it is useful to think about a domain as a component of the marine environment that we value. An anthropocentric view might characterise these as “resources”. The key point is that when we speak of domains, we are concerned with managing the environment itself, not the particular ways in which a resource or resources are used by humans (these are *sectors* or *activities*).

Domains can cross spatial boundaries if they are broadly defined. For example, the climate cuts across land and sea,⁵⁰ as does biodiversity, soil and air. Even freshwater can be part of the marine environment, such as in estuaries.⁵¹ Domains themselves often overlap,⁵² due to the interconnection of everything in the natural world.

The “marine” environment is often treated as a domain in its own right when the broader resource management system is being considered (such as in environmental reporting).⁵³ But it can be broken down further into particular resources (eg seabed minerals), species or groups of species (eg fish) or features (eg heritage sites or seascapes). These are all about an aspect of the marine environment being managed, not the human activities affecting them. For example, while *fishing* is a sector, *fish* can instead be treated as a domain. A number of statutes in the current system can be regarded as domain-based ones, such as the Wildlife Act, Marine Mammals Protection Act, MACA Act and Climate Change Response Act. But our current Fisheries Act is really about *fishing* as a sector, not the management of fish per se.

Statutes can be divided up to manage different domains. A domain can be thought of as the thing that is valued, rather than the way in which it is used (a sector/activity). While the marine environment can be thought of as one large domain, it can also be broken down into a finer scale by statutes focused on particular resources (eg fish) or other aspects that are valued (eg wildlife). We have a number of domain-based acts in the current system, such as the Marine Mammals Protection Act.

technical exercise unfamiliar to the RMA or EEZ Act. The QMS as a whole falls into the same boat.

Statutes can be split up to provide a bespoke home or framing for complex tools, especially where they require specialist expertise to deploy or involve market rather than regulatory mechanisms.

Administrative and institutional lenses

Some legislative boundaries can be purely administrative or institutional in nature. For example, the Environment Act is a standalone statute that creates the Ministry for the Environment and Parliamentary Commissioner for the Environment, and the Environmental Protection Authority Act creates its eponymous Authority. The Environmental Reporting Act outlines an information gathering and reporting process that informs both New Zealanders and statutory decision-making. And the main concern of the Territorial Sea, Contiguous Zone and Exclusive Economic Zone Act is to recognise and delineate maritime zones in accordance with international law.⁵⁷

The logic behind these types of splits may sometimes be that a statute provides for processes or institutions that apply across multiple other statutes. The Environmental Reporting Act, for example, compiles information across many domains; it is not just about reporting on the outcomes achieved under the RMA, so would not be a good fit within it.

Administrative legislation (such as the Environmental Reporting Act) may be necessary where processes and institutions cut across other statutes.

The point of some administrative acts may also be to separate the backroom machinery of the system from the frameworks that contain the interventions that actually shape people's behaviours (and therefore) have to be highly visible to them. For example, a person seeking consent under the EEZ Act is unlikely to be concerned with provisions establishing the EPA or jurisdiction in the EEZ, and that Act is thus more user-friendly to applicants because of the location of such things elsewhere.⁵⁸

One way of thinking about this is that statutes can be split by grouping together all the provisions that would be most useful to those using each

A tools-based lens

Instead of focusing on a particular outcome, sector, location or domain, a statute can be a place for a specific tool or range of tools to be deployed. The Marine Reserves Act can be seen as this kind of statute. In the current system, we have relatively few standalone statutes that have the purpose of creating a particular tool.⁵⁴ Instead, legislation usually provides multiple tools to manage a particular domain (eg the Wildlife Act) or sector (eg the Fisheries Act) or to achieve an outcome (eg the RMA).

That said, the division between statutes can sometimes be explained by their use of quite different “families” of tools. For example, although climate change mitigation falls within the purview of the RMA, the emissions trading scheme is a highly complex market mechanism and forms the flagship⁵⁵ tool delivered under the Climate Change Response Act. So while this Act can be seen as a domain-based one (concerned with the climate), its separate existence can also be explained as a tool-based home for the emissions trading scheme. After all, it does not purport to manage everything to do with the climate; the RMA and other acts play an important role too.

The same kind of thing can be said for the tools under the Waste Minimisation Act (like product stewardship schemes) and the Biosecurity Act (eg pathway management plans). These pursue outcomes that are to a large extent within the broad framing of sustainable management under the RMA,⁵⁶ but they are not integrated within it, because they involve interventions not easily provided for in its “planning and consenting” ecosystem of tools. Instead, they have their own kinds of intervention around which a bespoke legislative frame is built. In other words, their separate existence is arguably defined more by their toolkit than by the outcomes they seek. The same logic can apply to the stock assessment process and setting of a TAC under the Fisheries Act, which not only has a distinct purpose (sustainable utilisation) but is also a highly complex and

act, and putting the remaining content elsewhere. After all, legislative design principles say that the statute book should be *accessible* as well as coherent. That might be about carving off administrative detail, making something like the RMA shorter and less “impenetrable”.⁵⁹ But it can also be about who the primary “audience” of a statute is (eg particular industries, decision-making institutions, environmentalists, or those being impacted). Existing legislation takes different approaches in different circumstances. For example:

- The fishing sector can be seen as the primary audience of the Fisheries Act and related legislation (rather than those concerned with fish *per se*), although it can equally be seen as a reasonably self-contained home for the responsibilities of Fisheries New Zealand as an institution.⁶⁰ Those concerned with “fish” more broadly need to engage with multiple acts (including the Wildlife Act and RMA).
- A lot of “backroom” machinery is contained in the extremely long Maritime Transport Act (including the establishment⁶¹ of Maritime New Zealand). That is unlikely to be useful to those operating ships (who might find it easier to instead locate the ship-related biosecurity requirements of the Biosecurity Act within the Maritime Transport Act) but may be more useful to Maritime New Zealand itself (its functions and objectives are firmly linked to most of the toolkit at its disposal).⁶²
- In contrast, the EPA has to juggle many fragments of legislation, including its generative legislation (the EPA Act) and those under which it has roles shared with others (eg the Hazardous Substances and New Organisms Act, RMA and EEZ Act). The EPA Act itself does not do all of these things.
- Similarly, the Minister (and Ministry) for the Environment are seen as more than capable of understanding their roles under multiple complex statutes (eg the EEZ Act, the Environmental Reporting Act, the Hazardous Substances and New Organisms Act, the Environment Act and the Climate Change Response Act). Legislative boundaries are not tailored to what would be simplest for the Ministry.

Statutes can be divided up so that most content of relevance to its primary audience is grouped together in one place. Other “administrative” material can be split off into other legislation.

Other lenses

Other lenses for splitting legislation may be possible, but less workable. One could be based on the role the system is performing. For example, it might be possible to have one act charged with setting environmental limits (applying to all domains and sectors), another having a planning and consenting framework for making trade-offs, one dealing with allocation, and another for enhancing the environment. While this is likely to run into many practical difficulties when the same tools are expected to perform multiple roles (what if allocative choices were made according to which proposal would best enhance the environment?), it does raise the interesting possibility of having a separate statute focused on imposing and defending environmental limits in the marine space.

Although splitting up statutes based on the different roles the system is performing is not a practical approach to legislative design (and not one reflected in the current system), one interesting possibility would be to separate a statute responsible for imposing and defending environmental limits.

Multiple lenses can be used in layers

There is no single lens through which current marine legislative frameworks have been split up. We have outcome-based frameworks like the RMA, sectoral ones like the Fisheries Act, administrative ones like the Environmental Reporting Act, tool-based ones like the Marine Reserves Act, domain-based ones like the MACA Act, and space-based ones like bespoke legislation for the Hauraki Gulf.

Some statutory boundaries can be explained by more than one lens. For example, it is not clear whether the Maritime Transport Act is focused on a sector (shipping) or defined by the various institutional roles performed by Maritime New Zealand. It might even be a spatial one – after all, we have separate *maritime* and *land* transport legislation. Similarly, the RMA and EEZ Act are outcomes-based statutes, but they can also be seen as spatial statutes (applying to defined areas). And the Continental Shelf Act is defined both by the sector it applies to (mining)⁶³ and the space it regulates (beyond the territorial sea). All this marine legislation fits together in a complex jigsaw puzzle involving both hard and overlapping spatial, sectoral and outcome-based jurisdictions. Sometimes interfaces between them remain quite unclear.

For a system to be coherent, it cannot chop and change lenses at will. For example, there would be little point in having a broad, outcome-based statute like the RMA, while also having separate statutes dealing with the environmental effects of fishing, mining, aquaculture, shipping and offshore wind farms. The more carve-outs there are, the less meaningful core statutes become.

However, this does not mean that one lens needs to be chosen as the only basis for splitting legislation. For example, a sectoral statute for fisheries could still co-exist with an outcomes-based statute like the RMA. Instead, we need to consider how multiple lenses might be *layered*, and how the resulting statutes might relate to each other. As we have pointed out previously (and giving it a marine flavour):⁶⁴

We can usefully think of lenses as existing in a hierarchy or sequence. We can start by choosing a primary lens (eg a sectoral one), which we apply across the whole system. For example, we could have (among others) a Mining Act, an [Aquaculture] Act, and a Transport Act. Each act [which would apply across land and sea] would deal with all issues relevant to the sector in question (such as managing the sector's environmental impacts, any funding decisions, and the allocation of resource use rights).

We can then consider what those statutes do not do, and apply a secondary lens (eg an institutional one) to fill those gaps across the whole system. For example, if institutions were needed to operate across multiple sectors, such as an Environment Court, [Oceans Commission] or regional councils, it would not be appropriate to include them in any sector-specific act. So we could enact specific statutes – an Environment Court Act, [an Oceans Commission Act] and a Local Government Act (among others). But we would not need to enact a separate statute establishing an institution concerned only, for example, with transport, like [Maritime New Zealand], because that would already fall firmly within the scope of a sectoral Transport Act.

Again, we could then consider what still remains to be done, and apply a tertiary lens (say, a location-based one) to fill any gaps. For example, if a location had a special character that could not be recognised through restrictions on particular sectors or the behaviour of a particular institution [eg a congested space like the Hauraki Gulf where marine spatial planning was required], we could enact a specific statute to do so.

Applying a hierarchy or sequence of lenses means that most of the content of the system is contained within those statutes created using a *primary* lens. Those statutes are the first cab off the rank, so to speak. Using sectors as a primary lens would result in sectoral statutes dealing with most things (the environmental impacts of each sector, the allocation of resources used by the sector, sector-specific institutions, and so forth). Statutes created using a secondary lens would therefore have less scope, because their role would be simply to fill any gaps that remain.

Most of the content of the system is contained in statutes that are created using a primary lens. Other lenses are then applied only to fill the gaps left by previous ones, not to overlap.

How lenses are layered in the current oceans management system

While they are not perfectly applied, the current system can be understood as *layers* of lenses (see Figure 11.2 below). We described this for the resource management system as a whole in previous work.⁶⁵ Yet how lenses are layered in the marine context is subtly different, partly owing to the existence of strong sectoral legislation.



Longlining, Hauraki Gulf

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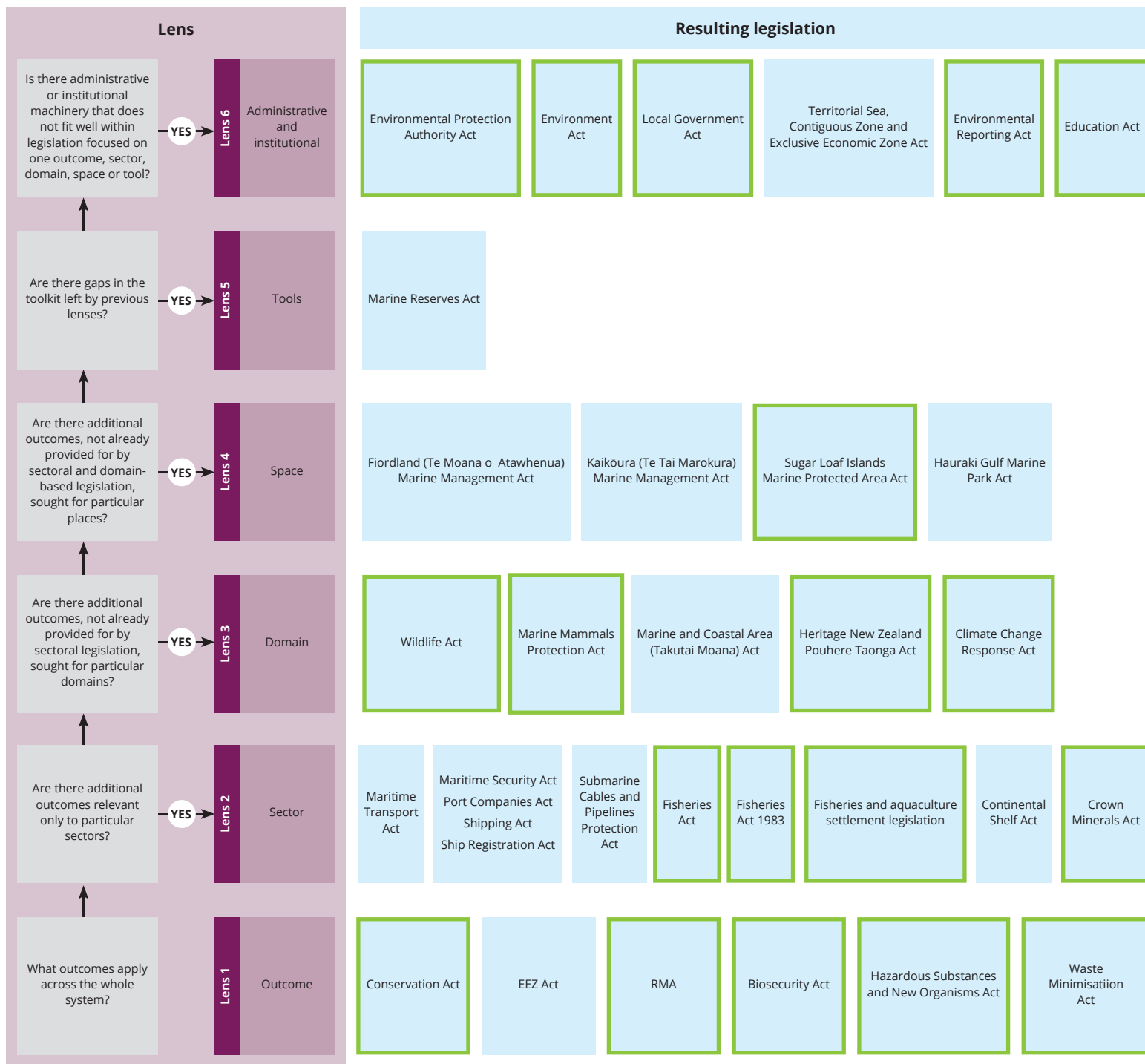


Figure 11.2: Layers of core legislation in the current oceans management system. Most of the system's content is found in outcome-based statutes, with gaps being filled by statutes created through other lenses (eg ones focusing on particular sectors, spaces or institutions). This figure is not to suggest that one statute is "dominant" or "subservient" to another at a different layer. Instead, it is about how the content of the system has been distributed between different types of statutes. Green framing indicates where statutes also apply to land and/or freshwater environments.

As on land, the primary lens that has been used is still an outcomes-based one. Statutes like the RMA, EEZ Act, Conservation Act, Waste Minimisation Act and Biosecurity Act seek general outcomes rather than being defined by a particular sector/activity, domain/resource, location,⁶⁶ tool or institution. Sustainable management applies to all industries and in all places.

A secondary, sectoral lens has been generally used to fill gaps in outcome-based frameworks. These gaps can be explained in two ways. First, they are because some sectors require the system to pursue *additional* outcomes that are specific to one activity and not others (eg fisheries⁶⁷ is about maximising yield as well as sustainability, shipping/boating is about complying with international law, safety requirements, and proactive planning for preventing and cleaning up oil spills,⁶⁸ and mining is about returning royalties to the Crown as a property owner). These would be an uneasy fit within even the broad purposes of the RMA, EEZ Act, or Conservation Act.⁶⁹

Secondly, gaps can be explained by the need for quite different *tools* to manage aspects of particular sectors. To some extent these tools follow naturally from a sector having an additional outcome being sought. For example, the RMA's planning and consenting framework is ill equipped to integrate a tool like a TAC or the QMS, which is designed to maximise sustainable yield. Similarly, internationally prescriptive design and health and safety codes for maritime transport would be an uneasy fit in the effects-based framework of the RMA,⁷⁰ as would a property allocation framework under the Crown Minerals Act (eg bids, block offers, royalties).

To some extent these sectoral statutes fill genuine gaps, rather than seeking to carve-out exceptions. For example, mining operations must comply with consents granted under the RMA and EEZ Act *as well* as tools like exploration permits under mining legislation. Commercial and recreational ships/boats must comply with RMA regulations on dumping in the coastal marine area.⁷¹ And although the Fisheries Act overlaps with the RMA, it can impose controls on the environmental impacts of fishing activity (at least in theory) so this is by no means a carve-out.⁷² Instead, a dual layer of controls are imposed for subtly different reasons (a point which is explored later in this chapter).⁷³ Similarly, although there is a lot of institutional cooperation,⁷⁴ compliance with RMA requirements for aquaculture does not remove the need to comply with Biosecurity Act requirements. In short, there is some coherence in the way boundaries have been split between outcome and sector-based legislation.

Where a broad outcome is sought across the whole system, it is generally contained within legislation built around achieving that outcome, and applies to all sectors and spaces. This leaves gaps, because some outcomes apply only to particular sectors, resulting in an additional layer of sectoral legislation such as the Fisheries Act.

Primary and secondary layers of legislation still leave some gaps, however. These are partly filled by domain-specific statutes. For example, while the Fisheries Act fills a gap in the RMA and EEZ Act relating to the proactive management of fish stocks, there is a whole category of fish that lies outside the concept of a “fishery”. These need to be protected and proactively enhanced (populations recovered) rather than utilised (eg great white sharks). This is the role of the domain-based Wildlife Act.

Similarly, the Wildlife Act and Marine Mammals Protection Act fill gaps in the RMA, EEZ Act and (perhaps surprisingly) Conservation Act that are not about managing fishing (and are therefore not dealt with already by the Fisheries Act). That includes impacts on protected marine life from other activities (eg parasitic threats, intentional hunting, harassment and other direct interactions like shark cage diving, whale watching, and interference with seabird and shorebird habitats).⁷⁵ These have much more directive purposes than the RMA and EEZ Act.

Moreover, while sectoral statutes determine who owns seabed minerals and fishing rights (which are core to those sectors), it is left to the broader domain-based MACA Act to resolve the question of ownership of the foreshore and seabed itself (and the customary rights and title that go along with that). Ownership is a significant gap left by outcomes-based statutes like the RMA, potentially because it is a marine-specific issue, or because it is not necessary to resolve property rights to achieve sustainable management.⁷⁶ Domain-based legislation for heritage (including marine heritage) and climate change fill other gaps left by outcomes-based statutes, but largely because they use quite different tools, not because they seek to achieve fundamentally different things from the RMA.⁷⁷

Some domains are managed to achieve more specific things than are provided for in outcomes-based or sectoral statutes, or by using an additional range of tools. This results in a layer of domain-based legislation that fills gaps, such as the Marine Mammals Protection Act.

A fourth layer of legislation fills gaps that still remain. These are created using a spatial lens. General outcome-based acts, sector-focused acts and domain or resource-focused acts can still fail to recognise the special characteristics of particular places. This partly⁷⁸ explains the existence of a number of geographically specific statutes for Kaikōura, the Hauraki Gulf, the Sugar Loaf Islands and Fiordland. Generally, these do not seek carve-outs from other frameworks created using earlier lenses, but instead seek to coordinate tools used under them in a way that reflects the values of the particular location. Because iwi and hapū are linked strongly to place, whenua and moana, te Tiriti settlement legislation is often location specific.

The Marine Reserves Act, which seeks to preserve particular areas in their natural state as the habitat of marine life for scientific study, can be seen as a location-specific statute. However, it can also be seen as a “tool” based statute akin to the Reserves Act on land. It is about creating new protections, not just managing the values of places that have already been identified as special. While other spatial protection tools are available under outcome-based, sectoral and domain-based statutes (eg RMA zoning, Fisheries Act taiāpure and marine mammal sanctuaries), it fills a gap that remains in their combined toolkit: a strong, no-take protected area designation protecting a space from all potentially harmful activities.

A layer of location-specific legislation fills remaining gaps to the extent that previous lenses do not reflect the particular needs or values of a place. A further layer fills gaps in the toolkit, notably for marine reserves.

A final layer of legislation in the current system can be described as institutional or administrative. Where institutions and processes relate squarely to the outcomes, sectors, domains and spaces in previous layers of statutes, that is where relevant provisions are generally found. For example, it is logical that the Department of Conservation is established under the Conservation Act and that Maritime New Zealand's existence is continued under the Maritime Transport Act. However, some may not be a comfortable fit in any of these previous layers, given their cross-cutting nature. For example, the Ministry for the Environment, the Parliamentary Commissioner for the Environment and the EPA have roles that span many other statutes, so their creation in separate acts makes sense.⁷⁹

A final layer of administrative and institutional legislation fills gaps where such things are not a comfortable fit under legislation created under previous lenses.

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Kaikōura coast